arXiv:2608.27346v4 [math.CV] 21 Sep 2026

THE COMPLETE CROUZEIX CONJECTURE IN DIMENSION THREE AND THE CLOUÂTRE–OSTERMANN–RANSFORD CONJECTURE

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Abstract. We settle the complete Crouzeix conjecture for matrices of order at most three and prove the Q-algebra case of the completely bounded Clouâtre–Ostermann–Ransford (COR) conjecture for homomorphisms into matrices of order at most three, via sharp abstract column and row estimates. Our approach also establishes the scalar COR conjecture in a stronger form, for homomorphisms with commutative range on Banach algebras with unity satisfying von Neumann's inequality. Under contractivity of the symmetrized map, this result holds on arbitrary Hilbert spaces without initial boundedness assumptions on the homomorphism or the antilinear map. For arbitrary operator algebras, we disprove the complete COR conjecture by an exact three-dimensional example with target matrix order two. We also prove the sharp complete bound for every matrix subalgebra containing the diagonal, in arbitrary matrix order and on arbitrary target Hilbert spaces. We also obtain column and row square-function inequalities with sharp norm bounds, strict scalar bounds for operators similar to normal operators, sharp complete spectral constants for scaled q-numerical ranges in dimensions two and three, and rigidity, stability, and representing-measure results.

1. Introduction

In 2004, Crouzeix [13] conjectured that for $A \in M_n(\mathbb{C})$

$$\|p(A)\| \le 2 \max_{z\in W(A)} |p(z)| \tag{1.1}$$

for every complex polynomial p, where

$$W(A) = \{\langle Ax, x\rangle : \|x\| = 1\}$$

is the numerical range of A. As evidence for his conjecture, the paper included a proof of the 2×2 case. It is easy to see that the constant 2 is sharp already in dimension two.

In 2007, Crouzeix [14] obtained the first dimension-free bound, replacing 2 in (1.1) by 11.08. In the same paper, he introduced the completely bounded version: For matrix-valued polynomials $F = (f_{ij})$, with $F(A) = (f_{ij}(A))$, he conjectured that

$$\|F(A)\| \le 2 \max_{z\in W(A)} \|F(z)\|, \tag{1.2}$$

and proved the corresponding inequality with 2 replaced by 11.08. We refer to (1.2) as the *complete Crouzeix conjecture*, which is equivalent to asking whether $W(A)$ is a complete 2-spectral set for A.

The spectral-set approach to numerical ranges was initiated by Delyon and Delyon [20], and subsequently used by Crouzeix and Palencia [19] to lower the bound in (1.2) to $1+\sqrt{2}$. The constant $1+\sqrt{2}$ is optimal for the abstract lemma used in the Crouzeix–Palencia argument [34], although for every fixed numerical-range domain a strictly smaller scalar bound is available [31, Theorems 2 and 3]. Recently, Jin [27] and Lorist and Schwenninger [30] independently proved the scalar conjecture. Both papers explicitly note that their methods do not directly yield the completely bounded inequality. Indeed, in the Lorist–Schwenninger argument, the obstruction is a commutativity hypothesis that is not available after matrix amplification. For (1.2), $1+\sqrt{2}$ therefore remains the best universal bound.

Our first main result settles the complete conjecture for matrices of order at most three.

Theorem 1.1. *Let $A \in M_n(\mathbb{C})$ with $n \le 3$. Then*

$$\|F(A)\| \le 2 \max_{z\in W(A)} \|F(z)\| \tag{1.3}$$

for every matrix-valued polynomial F of arbitrary size. Thus $W(A)$ is a complete 2-spectral set for A.

More generally, if (1.2) fails for some $A \in M_n(\mathbb{C})$, then it already fails for an $s\times r$ matrix-valued polynomial with

$$r, s \ge 2, \qquad r+s \le n; \tag{1.4}$$

in particular $n \ge 4$, and in dimension four the only possible matrix level is 2×2.

The case $n = 2$ of the complete inequality was already known [3, Section 5.2] while the new result is the case $n = 3$.

The proof of Theorem 1.1 combines new square-function inequalities, which control scalar, column, and row matrix levels, with a finite-dimensional reduction that restricts the size of a possible counterexample.

The same finite-dimensional reduction, combined with abstract analogs of the column and row estimates, also yields a complete abstract result. Clouâtre, Ostermann, and Ransford formulated abstract scalar and completely bounded versions of Crouzeix's conjecture for symmetrized homomorphisms [12]. We prove their complete conjecture through dimension three for Q-algebras, that is, quotients of uniform algebras equipped with the quotient matrix norms; see Definition 7.6. In the scalar setting, Theorem 7.1 extends the uniform-algebra theorem of Badea, O'Loughlin, and Virtanen [5, Theorem 2.1] to homomorphisms with commutative range on Banach algebras with unity satisfying von Neumann's inequality. Under contractivity of the symmetrized map, it removes the initial boundedness assumptions on the homomorphism and the antilinear map.

Theorem 1.2. *Let $\mathcal{B}$ be a Q-algebra with unity, let $\mathcal{H}$ be a Hilbert space, let $\theta : \mathcal{B} \to B(\mathcal{H})$ be a completely bounded homomorphism preserving the identity, and let $\alpha : \mathcal{B} \to \mathcal{B}$ be an antilinear map preserving the identity. Define*

$$\Phi_{\theta,\alpha}(b) = \frac{1}{2}\big(\theta(b) + \theta(\alpha(b))^*\big), \qquad b \in \mathcal{B}.$$

Suppose that $\Phi_{\theta,\alpha}$ is completely contractive. If $b_1, \dots, b_q \in \mathcal{B}$ satisfy

$$\big\|(b_1, \dots, b_q)^{\mathsf{T}}\big\| \le 1,$$

then

$$\left\| \begin{pmatrix} \theta(b_1) \\ \vdots \\ \theta(b_q) \end{pmatrix} \right\| \le 2, \qquad \big\| \begin{pmatrix} \theta(b_1) & \cdots & \theta(b_q) \end{pmatrix} \big\| \le 2. \tag{1.5}$$

Consequently, if $\mathcal{H} = \mathbb{C}^d$ and $d \le 3$, then

$$\|\theta\|_{\mathrm{cb}} \le 2.$$

In this finite-dimensional assertion, θ need not initially be assumed bounded. The constant 2 is best possible.

Notice that no boundedness assumption on α is needed in Theorem 1.2. The complete COR conjecture was formulated for arbitrary operator algebras, with α completely contractive [12, Conjecture 6.1]. In this generality it is false. We use the same notation $\Phi_{\theta,\alpha}$ for the symmetrized map on an arbitrary operator algebra, and write T_d for the upper triangular subalgebra of $M_d(\mathbb{C})$. Complete contractivity of an antilinear map refers to its entrywise amplifications.

Theorem 1.3. *There are a three-dimensional noncommutative operator algebra $\mathcal{A} \subset M_3(\mathbb{C})$, an injective completely bounded homomorphism $\theta : \mathcal{A} \to M_2(\mathbb{C})$ preserving the identity, and an antilinear map $\alpha : \mathcal{A} \to \mathcal{A}$ preserving the identity such that*

$$\|\alpha\|_{\mathrm{cb}} = \|\Phi_{\theta,\alpha}\|_{\mathrm{cb}} = 1,$$

but

$$2 < \frac{2(48 + \sqrt{2665})}{19(3 + \sqrt{5})} \le \|\theta\| = \|\theta\|_{\mathrm{cb}} \le \frac{8\sqrt{29}}{19}. \tag{1.6}$$

Moreover, $\theta(\mathcal{A}) = T_2$ and $\mathcal{A} \cap \mathcal{A}^* = \mathbb{C}I_3$.

The target matrix order and the dimension of the domain algebra are the smallest possible, by Corollary 8.5. The homomorphism in Theorem 1.3 has noncommutative range T_2, whereas the numerical-range functional calculus of a single matrix has commutative range.

The complete bound nevertheless holds for every matrix subalgebra containing the diagonal. These are the digraph algebras [22, p. 78].

Theorem 1.4. *Let $\mathcal{B} \subset M_n(\mathbb{C})$ be a subalgebra containing every diagonal matrix. Let $\theta : \mathcal{B} \to B(\mathcal{H})$ be a homomorphism preserving the identity, and let $\alpha : \mathcal{B} \to \mathcal{B}$ be an antilinear contraction preserving the identity. If $\Phi_{\theta,\alpha}$ is completely contractive, then*

$$\|\theta\|_{\mathrm{cb}} \le 2.$$

The constant 2 is best possible.

There is no restriction on n or on the target Hilbert space, and only ordinary contractivity of α is assumed. The theorem includes upper triangular matrix algebras, block triangular matrix algebras, and all finite block incidence algebras. In particular, the bound holds for T_2 with its usual matrix norms, but fails for the algebra in Theorem 1.3, which is algebraically isomorphic to T_2. Thus the bound depends on the concrete operator-algebra structure, not only on the algebraic isomorphism class.

We next state the square-function inequalities underlying Theorem 1.1. Let $K_A = \overline{W(A)}$, and denote by $\mathcal{A}(K_A)$ the functions continuous on K_A and holomorphic in its interior. For $k \geq 1$, let $\mathcal{W}_k = \{1, \dots, q\}^k$ and $\mathcal{W}_0 = \{\emptyset\}$. Given $f_1, \dots, f_q \in \mathcal{A}(K_A)$ and $\alpha = (\alpha_1, \dots, \alpha_k) \in \mathcal{W}_k$, we set $f_\alpha = f_{\alpha_1} \cdots f_{\alpha_k}$ and $f_\emptyset = 1$. For self-adjoint operators X and Y, we write $X \preceq Y$ when $Y - X$ is positive.

Theorem 1.5. *Let $A \in \mathcal{B}(H)$ and let $f_1, \dots, f_q \in \mathcal{A}(K_A)$ satisfy*

$$\sum_{j=1}^{q} |f_j(z)|^2 \leq 1 \qquad (z \in K_A). \tag{1.7}$$

For every $0 < \tau < 1$, the series

$$P_\tau(f, A) = \sum_{k=0}^{\infty} \tau^{2k} \sum_{\alpha \in \mathcal{W}_k} f_\alpha(A)^* f_\alpha(A), \tag{1.8}$$

$$Q_\tau(f, A) = \sum_{k=0}^{\infty} \tau^{2k} \sum_{\alpha \in \mathcal{W}_k} f_\alpha(A) f_\alpha(A)^* \tag{1.9}$$

converge in operator norm and satisfy

$$I \preceq P_\tau(f, A) \preceq \frac{1}{1-\tau} I, \qquad I \preceq Q_\tau(f, A) \preceq \frac{1}{1-\tau} I. \tag{1.10}$$

In particular,

$$\big\| (f_1(A), \dots, f_q(A))^{\mathsf{T}} \big\| \leq 2, \qquad \| (f_1(A), \dots, f_q(A)) \| \leq 2. \tag{1.11}$$

The constant 2 is best possible.

Choosing $q = 1$ and $\tau = 1/2$, the preceding theorem recovers and extends (1.1) to every $f \in \mathcal{A}(K_A)$. The terms with $k \geq 2$ and the other values of τ will be used later to obtain rigidity and stability results.

The proof of Theorem 1.5 combines the positive operator-valued double-layer calculus of Delyon–Delyon and Crouzeix–Palencia [19, 20, 35] with a multivariable version of Jin's sampling argument [27]. Jin's proof produces the scalar Gramian estimate by sampling a matrix-valued Herglotz kernel on the disk. We carry out the sampling on the unit ball of $\mathbb{C}^q$, which produces the two tuple square functions above.

The passage from Theorem 1.5 to Theorem 1.1 is finite-dimensional. A rectangular form of the finite-level argument associated with Smith's lemma [36], together with a variational lemma of Hartz and McCarthy [25], shows that if the complete functional-calculus norm for $A \in M_d(\mathbb{C})$ exceeds 2, then this is already detected at an $s \times r$ matrix level with $r, s \geq 2$ and $r + s \leq d$. The inequalities $r, s \geq 2$ follow from (1.11). Thus no such pair exists when $d \leq 3$, while for $d = 4$ only $r = s = 2$ remains. This dimension count is the reason that the tuple square functions settle the complete conjecture through dimension three.

For the remaining statements, define

$$\psi(A) = \sup\{\|f(A)\| : f \in \mathcal{A}(K_A),\ \max_{z \in K_A} |f(z)| \leq 1\}$$

and let $\psi_{\mathrm{cb}}(A)$ be the corresponding supremum over matrix-valued functions of arbitrary size. Thus, $\psi(A) \leq \psi_{\mathrm{cb}}(A)$, and Theorem 1.1 gives $\psi_{\mathrm{cb}}(A) \leq 2$ when $d \leq 3$.

For $d \geq 1$, put

$$Q_{\mathrm{cb}}(d) = \sup\{\psi_{\mathrm{cb}}(A) : A \in M_d(\mathbb{C})\}.$$

Thus $Q_{\mathrm{cb}}(d)$ is the least constant in the complete numerical-range inequality for all matrices of order d.

The higher-order terms in the square-function estimates yield the following sharpening of Theorem 1.1.

Theorem 1.6. *Let $A \in M_d(\mathbb{C})$ with $d \leq 3$. Then*

$$\psi_{\mathrm{cb}}(A) = 2 \iff \psi(A) = 2.$$

The square-function estimates also improve the elementary bound obtained from a diagonalization. For an invertible matrix V, write $\operatorname{cond}(V) = \|V\| \left\|V^{-1}\right\|$, and set

$$\Phi(\kappa) = \left(\frac{8\kappa^2}{\kappa^2 + 1 + \sqrt{(\kappa^2-1)^2 + 4}} \right)^{1/2}, \qquad \kappa \geq 1. \tag{1.12}$$

Theorem 1.7. *Let $d \leq 3$ and let $A \in M_d(\mathbb{C})$ be diagonalizable, say $A = V\Lambda V^{-1}$ with Λ diagonal. Then*

$$\psi_{\mathrm{cb}}(A) \leq \min\{\operatorname{cond}(V), \Phi(\operatorname{cond}(V))\} < 2. \tag{1.13}$$

The scalar counterpart holds in every dimension whenever A is similar to a normal operator. Indeed, every diagonalizable matrix satisfies the scalar Crouzeix inequality strictly; conversely, if $A = V\Lambda V^{-1}$ and $\psi(A) = s < 2$, then

$$\operatorname{cond}(V)^4 > \frac{s^4}{4(4 - s^2)}.$$

Thus, if $A_j = V_j \Lambda_j V_j^{-1}$ and $\psi(A_j) \to 2$, then $\operatorname{cond}(V_j) \to \infty$ for every choice of eigenvector matrices V_j.

This condition-number viewpoint is closely related to recent work of Aouichaoui and O'Loughlin [1], which characterizes scaled q-numerical ranges as unions of numerical ranges under similarities with bounded condition number and determines their sharp scalar spectral constants. We obtain a completely bounded analog of their similarity-amplification result; see Theorem 5.5. For $A \in M_d(\mathbb{C})$ with $d \geq 2$ and $0 < |q| \leq 1$, let

$$W_q(A) = \{\langle Ax, y\rangle : \|x\| = \|y\| = 1,\ \langle x, y\rangle = q\}, \qquad \Omega_q(A) = q^{-1} W_q(A).$$

Theorem 9.3 shows that, for $d = 2, 3$, the sharp universal complete spectral constant for $\Omega_q(A)$ is $\max\{1, 2|q| / (1 + (1 - |q|^2)^{1/2})\}$. Thus, $\Omega_q(A)$ is a complete spectral set for every matrix of order two or three if and only if $|q| \leq 4/5$; in every dimension at least two, this holds whenever $|q| \leq 1/\sqrt{2}$.

The freedom in choosing τ and the higher-order terms in Theorem 1.5 also give information at and near the sharp constant. If an operator T satisfies the two square-function estimates at $\tau = 1/2$ and $\|T\| = 2$ is attained, then T has a two-dimensional reducing subspace on which it is unitarily equivalent to $\left(\begin{smallmatrix} 0 & 0 \\ 2 & 0 \end{smallmatrix}\right)$. There is a quantitative version: Let $A \in M_n(\mathbb{C})$. If $f \in \mathcal{A}(K_A)$ satisfies

$$\max_{z \in K_A} |f(z)| = 1, \qquad \|f(A)\| \geq 2 - \varepsilon, \qquad 0 \leq \varepsilon \leq 2 - \frac{\sqrt{15}}{2},$$

then $f(A)$ is within $8\sqrt{\varepsilon}$ of an operator having this block as a reducing summand. Thus near equality forces a concrete approximate nilpotent structure.

There is also a function-theoretic form of this rigidity. If f maximizes $\|f(A)\|$ among functions bounded by one on a convex domain Ω and x is a unit vector with $\|f(A)x\| = \|f(A)\|$, then the associated representing probability measure μ on $\partial\Omega$ is characterized by

$$\langle h(A)x, x\rangle = \int_{\partial\Omega} h \, d\mu \qquad (h \in \mathcal{A}(\overline{\Omega}));$$

for its existence and uniqueness, see [7]. We show that near equality, all nonzero Fourier coefficients of the pushforward measure $f_*\mu$ under f are quantitatively small. At equality, $f_*\mu$ is normalized arclength measure on the unit circle, and along sequences approaching equality these measures converge weakly to it.

Jin's scalar rigidity theorem shows that $\psi(A) = 2$ forces $W(A)$ to be a nondegenerate disk [28, Theorem 1]. Together with Theorem 1.6 and Crouzeix's disk classification [16, Lemma 10.2], this gives the full complete equality classification in dimension three; see Corollary 12.5. In particular, a nondisk numerical range has complete constant strictly less than two in dimensions at most three. We also classify scalar polynomial equality in every finite dimension in Theorem 12.2. After translating the disk centre to zero, every such polynomial is a monomial, and the matrix has the corresponding reducing Crabb–Choi–Crouzeix block. Consequently, a polynomial of degree at least d with $p(A) \neq 0$ satisfies the scalar inequality strictly for $A \in M_d(\mathbb{C})$.

Finally, we apply the general theory to the family

$$A_{\mu,r} = \begin{pmatrix} i\mu & 0 & ir \\ 0 & -i\mu & ir \\ ir & ir & -1 \end{pmatrix}, \qquad \mu > 0, \qquad r \geq 0. \tag{1.14}$$

Up to multiplication by -1 and unitary equivalence, this is a real family studied numerically by Crouzeix [16] while related numerical work appears in [17, 24]. Since

$$\operatorname{Re} A_{\mu,r} = \operatorname{diag}(0, 0, -1),$$

the boundary of its numerical range contains $[-i\mu, i\mu]$. For $r > 0$, the matrix is unitarily irreducible. Since $\mu > 0$, the nontrivial boundary segment $[-i\mu, i\mu]$ also shows that its numerical range is not a disk.

Corollary 12.4 therefore gives $\psi_{\mathrm{cb}}(A_{\mu,r}) < 2$ for every $\mu > 0$ and $r \geq 0$. We retain the quantitative bound obtained from an eigenvector matrix at diagonalizable parameters. The nondiagonalizable parameters form the curve

$$\mu = t\sqrt{1-2t}, \qquad r = (1-t)\sqrt{t}, \qquad 0 < t < \frac{1}{2}.$$

At $t = 1/3$, the matrix has Jordan form $J_3(-1/3)$ and is affinely equivalent to the 3×3 Kac–Murdock–Szegő matrix. We retain an independent proof of strictness there using the rational-domain construction of Crouzeix, Greenbaum, and Li [17], with containment certified in exact rational arithmetic by Sturm sequences [6]. Although every point of the defective curve satisfies strictness, $\psi_{\mathrm{cb}}(A_{\mu(t),r(t)}) \to 2$ as $t \to 1/2^-$. Thus the strict bounds cannot be uniform along this curve.

The paper is organized as follows. Section 2 sets up the quotient operator algebra of the numerical-range functional calculus. Section 3 proves the tuple square-function inequalities. Sections 4 and 5 develop the quantitative similarity estimates and the rectangular extremal argument. Section 6 proves Theorems 1.1, 1.6, and 1.7. Section 7 proves the scalar abstract estimates and the Q-algebra extension. Section 8 gives the counterexample to the complete COR conjecture and proves the bound for digraph algebras. Section 9 contains the scaled q-numerical-range results. Sections 10 and 11 treat equality, stability, and representing measures, and Section 12 classifies scalar polynomial equality and complete equality in dimension three. Finally, Section 13 studies the family (1.14).

2. The numerical-range functional calculus

For $K \subset \mathbb{C}$ compact and convex, give $M_{s,r}(\mathcal{A}(K))$ the matrix norm

$$\|F\|_K = \max_{z \in K} \|F(z)\|.$$

These matrix norms make $\mathcal{A}(K)$ an operator algebra with unity. Let $A \in M_d(\mathbb{C})$ and put $K = W(A)$. Polynomials are dense in $\mathcal{A}(K)$ by Mergelyan's theorem, including when K has empty interior. The Crouzeix–Palencia inequality [19], applied to differences of polynomials, therefore defines $f(A)$ by continuous extension of the polynomial calculus. This agrees with the holomorphic functional calculus whenever the latter applies.

The functional calculus is the homomorphism preserving the identity

$$\rho_A : \mathcal{A}(K) \longrightarrow M_d(\mathbb{C}), \qquad \rho_A(f) = f(A).$$

Let

$$\mathcal{I}_A = \ker \rho_A, \qquad \mathfrak{A}_A = \mathcal{A}(K)/\mathcal{I}_A.$$

We equip $\mathfrak{A}_A$ with the quotient matrix norms. Thus, for $F \in M_{s,r}(\mathcal{A}(K))$,

$$\|F + M_{s,r}(\mathcal{I}_A)\| = \inf\{\|F + G\|_K : G \in M_{s,r}(\mathcal{I}_A)\}. \tag{2.1}$$

The Crouzeix–Palencia inequality [19] shows that ρ_A is bounded. Since $\rho_A(1) = I_d$, its kernel $\mathcal{I}_A$ is a proper closed two-sided ideal. By [10, Corollary 3.2], the quotient $\mathfrak{A}_A$ is an operator algebra with unity. When a concrete representation is needed, we use an identity-preserving completely isometric representation given by [10, Theorem 3.1].

The homomorphism ρ_A induces an injective homomorphism preserving the identity

$$\theta_A : \mathfrak{A}_A \longrightarrow M_d(\mathbb{C}), \qquad \theta_A(f + \mathcal{I}_A) = f(A).$$

Proposition 2.1. *Let $A \in M_d(\mathbb{C})$. Then*

$$\theta_A(\mathfrak{A}_A) = \operatorname{alg}(A), \qquad \dim \mathfrak{A}_A = \deg m_A \leq d, \tag{2.2}$$

where m_A is the minimal polynomial of A. Moreover,

$$\|\theta_A\| = \psi(A), \qquad \|\theta_A\|_{\mathrm{cb}} = \psi_{\mathrm{cb}}(A). \tag{2.3}$$

The same equality holds at every fixed rectangular level.

Proof. The image of the polynomial calculus is $\mathrm{alg}(A)$. Since this algebra is finite-dimensional, it is closed. Polynomial approximation on K therefore gives $\rho_A(\mathcal{A}(K)) = \mathrm{alg}(A)$. The first isomorphism theorem gives the first equality in (2.2). A monogenic matrix algebra has basis $I, A, \dots, A^{m-1}$, where $m = \deg m_A$. This proves the dimension formula.

Fix r, s and let $c_{s,r}(A)$ be the norm of the rectangular functional calculus on $M_{s,r}(\mathcal{A}(K))$. For $X = F + M_{s,r}(\mathcal{I}_A)$, the definition of the quotient norm gives

$$\left\|\theta_A^{(s,r)}(X)\right\| = \|F(A)\| \le c_{s,r}(A)\,\|F+G\|_K \qquad (G \in M_{s,r}(\mathcal{I}_A)).$$

Taking the infimum over G shows $\left\|\theta_A^{(s,r)}\right\| \le c_{s,r}(A)$. Conversely,

$$\|F + M_{s,r}(\mathcal{I}_A)\| \le \|F\|_K\,,$$

so $c_{s,r}(A) \le \left\|\theta_A^{(s,r)}\right\|$. Thus, equality holds at each rectangular level. Taking $r = s = 1$ and then the supremum over all levels proves (2.3). It is immaterial whether the suprema are taken over functions in $\mathcal{A}(K)$ or over polynomials, because entrywise polynomial approximation is uniform at every fixed matrix level. □

We shall repeatedly use approximate lifts from the quotient.

Lemma 2.2. *Let $X \in M_{s,r}(\mathfrak{A}_A)$ and suppose that $\|X\| \le 1$. For every $\varepsilon > 0$, there is $F \in M_{s,r}(\mathcal{A}(W(A)))$ such that*

$$F + M_{s,r}(\mathcal{I}_A) = X, \qquad \|F\|_{W(A)} \le 1 + \varepsilon. \tag{2.4}$$

Proof. This is the definition of the quotient matrix norm in (2.1). □

The scalar and complete ratios are unchanged by taking adjoints.

Lemma 2.3. *For $A \in M_d(\mathbb{C})$,*

$$\psi(A^*) = \psi(A), \qquad \psi_{\mathrm{cb}}(A^*) = \psi_{\mathrm{cb}}(A).$$

Proof. One has $W(A^*) = \{\overline{z} : z \in W(A)\}$. For a scalar function f, set $f^{\#}(z) = \overline{f(\overline{z})}$. For a matrix-valued function $F = (f_{ij})$, set

$$F^{\#}(z) = \left(\overline{f_{ji}(\overline{z})}\right).$$

Then $F^{\#}(A^*) = F(A)^*$ and $\max_{W(A^*)} \left\|F^{\#}\right\| = \max_{W(A)} \|F\|$. The assertions follow in both directions. □

3. Positive kernels and iterated square functions

For $q \ge 1$, let

$$\mathbb{B}_q = \{w \in \mathbb{C}^q : \|w\| < 1\}.$$

We write $w^{\mathsf{T}} a = \sum_{j=1}^q w_j a_j$. This expression is bilinear.

Definition 3.1. A scalar kernel K on $\mathbb{B}_q$ is positive if

$$[K(w_i, w_j)]_{i,j=1}^N \succeq 0$$

for every finite family $w_1, \dots, w_N \in \mathbb{B}_q$. An $M_d(\mathbb{C})$-valued kernel is positive when the corresponding block matrices are positive.

Lemma 3.2. *Let $a \in \overline{\mathbb{B}_q}$ and put*

$$c_a(w) = \frac{1 + w^{\mathsf{T}} a}{1 - w^{\mathsf{T}} a}.$$

Then

$$K_a(w, z) = \frac{c_a(w) + \overline{c_a(z)}}{1 - z^* w} \tag{3.1}$$

is a positive kernel on $\mathbb{B}_q$.

Proof. A direct calculation gives

$$K_a(w, z) = \frac{2}{(1 - w^{\mathsf{T}} a)(1 - \overline{z^{\mathsf{T}} a})} \left(1 + \frac{z^* \{I - \overline{a}(\overline{a})^*\} w}{1 - z^* w}\right). \tag{3.2}$$

The matrix $I - \overline{a}(\overline{a})^*$ is positive. If $M \succeq 0$, then

$$\frac{z^*Mw}{1 - z^*w} = \sum_{k=0}^{\infty} (z^*Mw)(z^*w)^k$$

is a positive kernel. Hence the expression in parentheses in (3.2) is positive. Multiplication by a scalar function of w and its conjugate at z preserves positivity. □

We next give the double-layer formula in the form used below. This is the positive operator-valued double-layer calculus of Delyon–Delyon and Crouzeix–Palencia [19,20]. A detailed recent account is given in [35].

Lemma 3.3. *Let $B \in M_d(\mathbb{C})$, and let Ω be a bounded convex domain with smooth boundary such that $W(B) \subset \Omega$. There is a positive map preserving the identity*

$$\Phi : C(\partial\Omega) \longrightarrow M_d(\mathbb{C})$$

with the following property. If h is holomorphic on a neighborhood of $\overline{\Omega}$, then

$$2\Phi(h) = h(B) + C_h^*, \qquad C_h \in \mathrm{alg}(B). \tag{3.3}$$

Proof. Let $\nu(\zeta)$ be the outward unit normal to $\partial\Omega$, viewed as a complex number, and let s be arclength. Set

$$\mu(\zeta, B) = \frac{1}{2\pi}\left(\nu(\zeta)(\zeta I - B)^{-1} + \overline{\nu(\zeta)}(\overline{\zeta} I - B^*)^{-1}\right). \tag{3.4}$$

Put $Z = \zeta I - B$ and $R = Z^{-1}$. Convexity gives

$$\nu Z^* + \overline{\nu} Z \succeq 0.$$

Indeed, its quadratic form at a unit vector x is

$$2\,\mathrm{Re}\,\{\overline{\nu}(\zeta - x^*Bx)\} \geq 0.$$

Congruence by R gives $\nu R + \overline{\nu} R^* \succeq 0$. Thus $\mu(\zeta, B) \succeq 0$.

For the positive orientation of $\partial\Omega$, one has $\nu\, ds = -i\, d\zeta$. The Cauchy formula and its adjoint give

$$\int_{\partial\Omega} \mu(\zeta, B)\, ds = 2I.$$

Define

$$\Phi(g) = \frac{1}{2}\int_{\partial\Omega} g(\zeta)\mu(\zeta, B)\, ds.$$

Then Φ is positive and $\Phi(1) = I$. Splitting the two terms in (3.4) gives (3.3), where

$$C_h = \frac{1}{2\pi i}\int_{\partial\Omega} \overline{h(\zeta)}(\zeta I - B)^{-1}\, d\zeta.$$

Every resolvent of B belongs to $\mathrm{alg}(B)$. Hence $C_h \in \mathrm{alg}(B)$. □

Let $F = (f_1, \ldots, f_q)^{\mathsf{T}}$, where the f_j are holomorphic near $\overline{\Omega}$ and

$$\sum_{j=1}^{q} |f_j(z)|^2 \leq 1 \qquad (z \in \overline{\Omega}). \tag{3.5}$$

For $w \in \mathbb{B}_q$, define

$$h_w(\zeta) = \frac{1 + w^{\mathsf{T}}F(\zeta)}{1 - w^{\mathsf{T}}F(\zeta)}, \qquad H(w) = \Phi(h_w). \tag{3.6}$$

The denominator does not vanish. The function H is holomorphic and $H(0) = I$.

Lemma 3.4. *The kernel*

$$\mathcal{L}_H(w, z) = \frac{H(w) + H(z)^*}{1 - z^*w} \tag{3.7}$$

is positive on $\mathbb{B}_q$.

Proof. For $\zeta \in \partial\Omega$, put $a = F(\zeta)$. By (3.5), $a \in \overline{\mathbb{B}_q}$. For finite points $w_1, \ldots, w_N$, Lemma 3.2 and (3.4) show that

$$[K_{F(\zeta)}(w_i, w_j)\mu(\zeta, B)]_{i,j=1}^N \succeq 0.$$

Integration gives

$$[\mathcal{L}_H(w_i, w_j)]_{i,j=1}^N = \frac{1}{2}\int_{\partial\Omega} [K_{F(\zeta)}(w_i, w_j)\mu(\zeta, B)]_{i,j=1}^N\, ds.$$

□

Put

$$T_j = f_j(B), \qquad T(w) = \sum_{j=1}^{q} w_j T_j.$$

The double-layer formula applied to h_w gives

$$H(w) - (I - T(w))^{-1} \in \mathrm{alg}(B^*). \tag{3.8}$$

Indeed,

$$h_w(B) = (I + T(w))(I - T(w))^{-1} = 2(I - T(w))^{-1} - I,$$

and (3.8) follows from (3.3). The sample points and the additional point at the origin used below are the multivariable form of the one-variable sampling argument in [27].

Proposition 3.5. *Let $B \in M_d(\mathbb{C})$ have simple spectrum. Let Ω be a bounded smooth convex domain with $W(B) \subset \Omega$. Suppose that $f_1, \dots, f_q$ are holomorphic near $\overline{\Omega}$ and satisfy (3.5). Then, for every $0 < \tau < 1$,*

$$\sum_{k=0}^{\infty} \tau^{2k} \sum_{\alpha \in \mathcal{W}_k} f_\alpha(B)^* f_\alpha(B) \preceq \frac{1}{1-\tau} I. \tag{3.9}$$

Proof. Choose a diagonalization

$$B = S \,\mathrm{diag}(\beta_1, \dots, \beta_d) S^{-1}$$

and put $G = S^*S$. For $1 \le i \le d$, set

$$\lambda_i = \begin{pmatrix} f_1(\beta_i) \\ \vdots \\ f_q(\beta_i) \end{pmatrix} \in \overline{\mathbb{B}_q}.$$

For $1 \le j \le q$, let

$$\Lambda_j = \mathrm{diag}(f_j(\beta_1), \dots, f_j(\beta_d)).$$

Thus $T_j = S\Lambda_j S^{-1}$. Put

$$D(w) = \sum_{j=1}^{q} w_j \Lambda_j = \mathrm{diag}(w^{\mathsf{T}}\lambda_1, \dots, w^{\mathsf{T}}\lambda_d).$$

Since the eigenvalues β_i are distinct, polynomial interpolation gives

$$\mathrm{alg}(B^*) = \{(S^{-1})^* \Delta S^* : \Delta \text{ is diagonal}\}.$$

It follows from (3.8) that there is a diagonal holomorphic matrix $\Psi(w)$, with $\Psi(0) = 0$, such that

$$S^* H(w) S = G(I - D(w))^{-1} + \Psi(w) G. \tag{3.10}$$

Fix $0 < \tau < 1$. Set

$$c_{ij} = \lambda_i^* \lambda_j, \qquad w_i = \tau \overline{\lambda_i}.$$

Some of the points w_i may coincide. Repeated sampling points are allowed in the positivity condition. We have

$$w_j^* w_i = \tau^2 c_{ij}, \qquad w_i^{\mathsf{T}} \lambda_j = \tau c_{ij}.$$

Define Hermitian matrices P, Q, Y by

$$P_{ij} = \frac{G_{ij}}{1 - \tau^2 c_{ij}}, \qquad Q_{ij} = \frac{G_{ij}}{1 - \tau c_{ij}}, \qquad Y = Q - P. \tag{3.11}$$

Let $e_1, \dots, e_d$ be the standard basis of $\mathbb{C}^d$. Fix $u = (u_1, \dots, u_d)^{\mathsf{T}} \in \mathbb{C}^d$. In the positive-kernel inequality for (3.7), first take the congruence by S in each block. Use the points w_i and the vectors $u_i e_i$. Add the point 0 with the vector

$$v = -G^{-1} P u. \tag{3.12}$$

Write ψ_i for the ith diagonal entry of Ψ. The contribution of Ψ to the resulting quadratic form is

$$2\,\mathrm{Re} \sum_{i=1}^{d} \overline{u_i}\, \psi_i(w_i)(Gv + Pu)_i,$$

which is zero by (3.12).

The contribution of $G(I - D(w))^{-1}$ from the nonzero sample points is the matrix M with entries

$$M_{ij} = \frac{2G_{ij}}{(1 - \tau c_{ij})(1 - \tau^2 c_{ij})}.$$

The scalar identity

$$\frac{2}{(1-\tau x)(1-\tau^2 x)} = \frac{2}{1-\tau^2 x} + \frac{2}{1-\tau}\left(\frac{1}{1-\tau x} - \frac{1}{1-\tau^2 x}\right)$$

gives

$$M = 2P + \frac{2}{1-\tau}Y. \tag{3.13}$$

The block between the nonzero sample points and the origin is $G+Q$, and the origin block is $2G$. Positivity and (3.12) give

$$0 \preceq M - (G+Q)G^{-1}P - PG^{-1}(G+Q) + 2PG^{-1}P.$$

Using (3.13) and $Q = P + Y$, we obtain

$$\frac{2}{1-\tau}Y - YG^{-1}P - PG^{-1}Y \succeq 0. \tag{3.14}$$

Define

$$\widetilde{T}_j = G^{1/2}\Lambda_j G^{-1/2}, \qquad \widehat{P} = G^{-1/2}PG^{-1/2}, \qquad \widehat{Q} = G^{-1/2}QG^{-1/2}, \qquad \widehat{Y} = \widehat{Q} - \widehat{P}.$$

For $\alpha = (\alpha_1, \dots, \alpha_k)$, put

$$\Lambda_\alpha = \Lambda_{\alpha_1} \cdots \Lambda_{\alpha_k}, \qquad \widetilde{T}_\alpha = \widetilde{T}_{\alpha_1} \cdots \widetilde{T}_{\alpha_k},$$

with $\Lambda_\varnothing = \widetilde{T}_\varnothing = I$. For every $k \geq 0$,

$$\sum_{\alpha \in \mathcal{W}_k} (\Lambda_\alpha^* G \Lambda_\alpha)_{ij} = G_{ij} c_{ij}^k.$$

Expanding the denominators in (3.11) gives norm-convergent series

$$\widehat{P} = \sum_{k=0}^{\infty} \tau^{2k} \sum_{\alpha \in \mathcal{W}_k} \widetilde{T}_\alpha^* \widetilde{T}_\alpha, \tag{3.15}$$

$$\widehat{Q} = \sum_{k=0}^{\infty} \tau^{k} \sum_{\alpha \in \mathcal{W}_k} \widetilde{T}_\alpha^* \widetilde{T}_\alpha. \tag{3.16}$$

Since

$$\tau^k - \tau^{2k} \geq \frac{1-\tau}{\tau}\tau^{2k} \qquad (k \geq 1),$$

we have $\widehat{P} \succeq I$ and

$$\widehat{Y} \succeq \frac{1-\tau}{\tau}(\widehat{P} - I). \tag{3.17}$$

Congruence of (3.14) by $G^{-1/2}$ gives

$$\frac{2}{1-\tau}\widehat{Y} - \widehat{Y}\widehat{P} - \widehat{P}\widehat{Y} \succeq 0. \tag{3.18}$$

Let e be a unit eigenvector of $\widehat{P}$, with $\widehat{P}e = \gamma e$, and put $y = e^*\widehat{Y}e$. Then $\gamma \geq 1$ and

$$0 \leq 2\left(\frac{1}{1-\tau} - \gamma\right) y.$$

If $\gamma > 1/(1-\tau)$, then (3.17) gives $y > 0$, which is impossible. Hence

$$\widehat{P} \preceq \frac{1}{1-\tau} I. \tag{3.19}$$

The polar decomposition $S = UG^{1/2}$ gives $T_j = U\widetilde{T}_j U^*$. Thus (3.15) and (3.19) are unitarily equivalent to (3.9). □

We use two elementary approximation facts.

Lemma 3.6. *Let $K \subset \mathbb{C}$ be compact and convex, and let V be an open set containing K. There is a bounded convex domain Ω with smooth boundary such that*

$$K \subset \Omega, \qquad \overline{\Omega} \subset V.$$

Moreover, matrices with simple spectrum are dense in $M_d(\mathbb{C})$, and

$$d_{\mathrm{H}}(W(A), W(B)) \leq \|A - B\|. \tag{3.20}$$

Proof. Choose $\delta > 0$ such that $K + \delta\overline{\mathbb{D}} \subset V$. Let d_K be the distance to K. This is a convex 1-Lipschitz function. Let v_η be a smooth convolution of d_K at scale η, and put

$$u_\eta(z) = v_\eta(z) + \eta\,|z|^2\,.$$

The Lipschitz estimate gives $|v_\eta - d_K| \le \eta$. Choose R with $K \subset R\overline{\mathbb{D}}$, and take $\eta > 0$ so small that

$$\eta(1+R^2) < \delta/4, \qquad \delta/2 - \eta > \delta/4.$$

Then the set

$$\Omega = \{z : u_\eta(z) < \delta/4\}$$

contains K. If $d_K(z) \ge \delta/2$, then $v_\eta(z) \ge d_K(z) - \eta > \delta/4$, so $\Omega \subset K + (\delta/2)\mathbb{D}$. The function u_η is smooth, proper, and strictly convex. It has at most one critical point, which is its global minimum. The minimum is less than $\delta/4$, so the gradient does not vanish on the displayed level set. Thus Ω has smooth boundary.

Simple-spectrum matrices are dense because the discriminant of the characteristic polynomial is a nonzero polynomial in the matrix entries. Finally, for every unit vector x,

$$|x^*(A-B)x| \le \|A-B\|\,.$$

Applying this with A and B interchanged proves (3.20). □

Proof of Theorem 1.5 for matrices. First suppose that every f_j is holomorphic on an open neighborhood of $W(A)$. Fix $\varepsilon > 0$. Lemma 3.6 gives a bounded smooth convex domain Ω such that $W(A) \subset \Omega$, every f_j is holomorphic near $\overline{\Omega}$, and

$$\sup_{z\in\overline{\Omega}} \sum_{j=1}^{q} |f_j(z)|^2 \le 1 + \varepsilon.$$

Put $\widetilde{f}_j = (1+\varepsilon)^{-1/2} f_j$. Choose simple-spectrum matrices $B_n \to A$. By (3.20), $W(B_n) \subset \Omega$ for all large n. Proposition 3.5 applies to B_n and the functions $\widetilde{f}_j$.

Fix N. Retain the terms with $k \le N$ and let $n \to \infty$. Continuity of the holomorphic functional calculus gives

$$\sum_{k=0}^{N} \tau^{2k} \sum_{\alpha\in\mathcal{W}_k} \widetilde{f}_\alpha(A)^*\widetilde{f}_\alpha(A) \preceq \frac{1}{1-\tau} I.$$

Letting $\varepsilon \to 0^+$ gives the same inequality with f_α in place of $\widetilde{f}_\alpha$. The partial sums are increasing and bounded. In finite dimensions their positive increments have summable traces, and therefore summable norms. This proves norm convergence and the column estimate.

For general $f_j \in \mathcal{A}(W(A))$, choose polynomial tuples converging uniformly to $(f_1,\dots,f_q)$. After multiplication by a scalar converging to one, the polynomial tuples satisfy the pointwise column bound. Applying the preceding result to each finite partial sum and passing to the limit proves the column estimate. The row estimate follows by applying the column estimate to A^* and to

$$f_j^{\#}(z) = \overline{f_j(\overline{z})}.$$

The functions of A commute, so reversal of a word does not change its product. □

We now pass to bounded operators. The following argument is the finite-dimensional Krylov compression used in numerical-range functional-calculus estimates [14].

Completion of the proof of Theorem 1.5. Suppose first that $f_1,\dots,f_q$ are polynomials, each of degree at most m. Fix a unit vector $x \in H$ and an integer $N \ge 0$. Let

$$\mathcal{K}_{N,x} = \operatorname{span}\{x, Ax, \dots, A^{mN}x\}, \qquad B = P_{\mathcal{K}_{N,x}} A|_{\mathcal{K}_{N,x}}.$$

The numerical range of the compression B is contained in K_A. For every word α with $|\alpha| \le N$,

$$f_\alpha(B)x = f_\alpha(A)x. \tag{3.21}$$

Indeed, the degree of f_α is at most mN, and $B^j x = A^j x$ for $0 \le j \le mN$. Applying the matrix case to B and using (3.21) gives

$$\sum_{k=0}^{N} \tau^{2k} \sum_{\alpha\in\mathcal{W}_k} \|f_\alpha(A)x\|^2 \le \frac{1}{1-\tau}.$$

Since this holds for every unit vector x, the corresponding operator partial sum is bounded by $(1-\tau)^{-1}I$.

Now let $f_1, \dots, f_q \in \mathcal{A}(K_A)$ satisfy (1.7). For each n, choose polynomials $p_{j,n}$ converging uniformly to f_j on K_A. There are numbers $\varepsilon_n \to 0^+$ such that

$$\sum_{j=1}^{q} |p_{j,n}(z)|^2 \le (1+\varepsilon_n)^2 \qquad (z \in K_A).$$

Apply the polynomial result to $p_{j,n}/(1+\varepsilon_n)$. For each fixed N, uniform convergence and continuity of the functional calculus permit passage to the limit. Thus every column partial sum is bounded by $(1-\tau)^{-1}I$. The increasing sequence of partial sums therefore converges strongly to an operator satisfying the same bound.

In fact, the convergence is in operator norm. Fix $\tau < \sigma < 1$ and put

$$S_k = \sum_{\alpha \in \mathcal{W}_k} f_\alpha(A)^* f_\alpha(A).$$

For $M > N$, the bound already proved at σ gives

$$\begin{aligned} 0 &\preceq \sum_{k=N+1}^{M} \tau^{2k} S_k \\ &\preceq \left(\frac{\tau}{\sigma}\right)^{2(N+1)} \sum_{k=N+1}^{M} \sigma^{2k} S_k \\ &\preceq \frac{(\tau/\sigma)^{2(N+1)}}{1-\sigma} I. \end{aligned}$$

Thus the partial sums are Cauchy in operator norm.

Apply the column estimate to A^* and the reflected functions to obtain the row estimate. Keeping only the terms of degrees zero and one gives

$$\sum_{j=1}^{q} f_j(A)^* f_j(A) \preceq \frac{1}{\tau(1-\tau)} I,$$

and the corresponding row inequality. The minimum of the scalar factor occurs at $\tau = 1/2$, which proves (1.11). For sharpness, take $q = 1$, $f_1(z) = z$, and

$$A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}.$$

Then $W(A) = \overline{\mathbb{D}}$ and $\|A\| = 2$. □

Corollary 3.7. *Under the hypotheses of Theorem 1.5, let P_τ and Q_τ denote its column and row square functions. For $0 < \tau \le 1/2$,*

$$I \preceq P_\tau, Q_\tau \preceq (1+4\tau^2) I.$$

The upper bound is sharp for every τ in this interval.

Proof. For $k \ge 1$ and $0 < \tau \le 1/2$, one has $\tau^{2k} \le 4\tau^2 4^{-k}$. Positivity of each term gives

$$P_\tau - I \preceq 4\tau^2 (P_{1/2} - I) \preceq 4\tau^2 I,$$

and the same argument applies to Q_τ. For sharpness, take $A = \left(\begin{smallmatrix} 0 & 0 \\ 2 & 0 \end{smallmatrix}\right)$ and $f(z) = z$. Then $W(A) = \overline{\mathbb{D}}$, $A^2 = 0$, and both square functions have norm $1 + 4\tau^2$. □

4. Stein equations and normal similarities

Let $T = (T_1, \dots, T_q)$ be a commuting tuple of bounded operators. For $X \in \mathcal{B}(H)$, put

$$\mathcal{L}_T(X) = \sum_{j=1}^{q} T_j^* X T_j, \qquad \mathcal{R}_T(X) = \sum_{j=1}^{q} T_j X T_j^*.$$

The two square functions in Theorem 1.5 are the resolvent series

$$P_\tau = \sum_{k=0}^{\infty} \tau^{2k} \mathcal{L}_T^k(I), \qquad Q_\tau = \sum_{k=0}^{\infty} \tau^{2k} \mathcal{R}_T^k(I).$$

Proposition 4.1. *Under the assumptions of Theorem 1.5, with $T_j = f_j(A)$, one has*

$$P_\tau = I + \tau^2 \sum_{j=1}^{q} T_j^* P_\tau T_j, \tag{4.1}$$

$$Q_\tau = I + \tau^2 \sum_{j=1}^{q} T_j Q_\tau T_j^*. \tag{4.2}$$

Moreover,

$$I \preceq P_\tau, Q_\tau \preceq \frac{1}{1-\tau} I.$$

Proof. Let $P_{\tau,N}$ and $Q_{\tau,N}$ be the partial sums. They converge strongly. Multiplication on the left and right by a fixed bounded operator preserves strong convergence on bounded sets. Shifting the two series gives (4.1) and (4.2). The bounds are contained in Theorem 1.5. □

The two Stein equations give, in general, two different similarities. One transforms the tuple into a column contraction and the other into a row contraction. Both similarities depend on the tuple.

Corollary 4.2. *Under the assumptions of Proposition 4.1, put*

$$C_j = \sqrt{\tau}\, P_\tau^{1/2} T_j P_\tau^{-1/2}, \qquad D_j = \sqrt{\tau}\, Q_\tau^{-1/2} T_j Q_\tau^{1/2}.$$

Then

$$\sum_{j=1}^{q} C_j^* C_j \preceq I, \qquad \sum_{j=1}^{q} D_j D_j^* \preceq I. \tag{4.3}$$

Furthermore,

$$\operatorname{cond}(P_\tau^{1/2}),\ \operatorname{cond}(Q_\tau^{1/2}) \le (1-\tau)^{-1/2}.$$

For $0 < \tau \le 1/2$, the condition numbers are at most $\sqrt{1+4\tau^2}$, and the same similarities satisfy

$$\sum_j C_j^* C_j \preceq \frac{4\tau}{1+4\tau^2} I, \qquad \sum_j D_j D_j^* \preceq \frac{4\tau}{1+4\tau^2} I.$$

Proof. Equation (4.1) gives

$$\sum_{j=1}^{q} C_j^* C_j = \tau P_\tau^{-1/2} \left(\sum_{j=1}^{q} T_j^* P_\tau T_j \right) P_\tau^{-1/2} = \frac{1}{\tau}(I - P_\tau^{-1}) \preceq I.$$

The last inequality follows from $P_\tau^{-1} \succeq (1-\tau)I$. The row calculation is identical. The condition-number estimates follow from $I \preceq P_\tau, Q_\tau \preceq (1-\tau)^{-1} I$. For the additional estimates, use Corollary 3.7 and the identities $\sum_j C_j^* C_j = \tau^{-1}(I - P_\tau^{-1})$ and $\sum_j D_j D_j^* = \tau^{-1}(I - Q_\tau^{-1})$.

□

We next prove a quantitative estimate for tuples similar to commuting normal tuples. The following moment inequality is the required form of Hölder's inequality.

Lemma 4.3. *Let $N_1, \dots, N_q$ be commuting normal operators, and assume*

$$\sum_{j=1}^{q} N_j^* N_j \preceq I.$$

For a word α, put $N_\alpha = N_{\alpha_1} \cdots N_{\alpha_k}$. Then, for every vector z and every $k \ge 1$,

$$\left(\sum_{|\alpha|=k} \|N_\alpha z\|^2 \right) \|z\|^{2(k-1)} \ge \left(\sum_{j=1}^{q} \|N_j z\|^2 \right)^k. \tag{4.4}$$

Proof. By the joint spectral theorem, there is a projection-valued measure E and bounded scalar functions n_j such that $N_j = \int n_j \, dE$. Put

$$a = \sum_{j=1}^{q} |n_j|^2, \qquad \omega(\Gamma) = \langle E(\Gamma) z, z \rangle .$$

The hypothesis gives $0 \le a \le 1$, and

$$\sum_{|\alpha|=k} \|N_\alpha z\|^2 = \int a^k \, d\omega, \qquad \sum_{j=1}^{q} \|N_j z\|^2 = \int a \, d\omega.$$

Hölder's inequality gives (4.4). □

For $0 < s < 2$ and $0 < \tau < 1$, put

$$L_\tau(s) = \frac{\tau^3(1-\tau)s^4}{1-\tau(1-\tau)s^2}, \qquad L(s) = \sup_{0<\tau<1} L_\tau(s). \tag{4.5}$$

The denominator is positive because $s < 2$.

Theorem 4.4. *Let $T_1, \dots, T_q \in \mathcal{B}(H)$ satisfy the column square-function estimate in Theorem 1.5. Suppose that*

$$T_j = SN_jS^{-1}, \qquad 1 \le j \le q,$$

where S is invertible, the operators N_j are commuting and normal, and $\sum_j N_j^* N_j \preceq I$. Put

$$s = \left\| (T_1, \dots, T_q)^{\mathsf{T}} \right\|, \qquad \kappa = \operatorname{cond}(S).$$

Then, for every $0 < \tau < 1$,

$$\tau^2 s^2 + \frac{\tau^4 s^4}{\kappa^4(1 - \tau^2 s^2/\kappa^2)} \le \frac{\tau}{1-\tau}. \tag{4.6}$$

Consequently,

$$s \le \min\{\kappa, \Phi(\kappa)\} < 2, \tag{4.7}$$

where Φ is given by (1.12). If $0 < s < 2$, then

$$\kappa^4 \ge L(s) > \frac{s^4}{4(4-s^2)}. \tag{4.8}$$

The same assertions hold for a row tuple if the row square-function estimate is assumed.

Proof. The column operator associated with $(N_1, \dots, N_q)$ is a contraction, so $s \le \kappa$. Choose unit vectors x_m such that

$$\sum_{j=1}^{q} \|T_j x_m\|^2 \longrightarrow s^2,$$

and put $z_m = S^{-1}x_m$. For $k \ge 1$, Lemma 4.3 gives

$$\begin{aligned}
\sum_{|\alpha|=k} \|T_\alpha x_m\|^2 &\ge \frac{1}{\|S^{-1}\|^2} \sum_{|\alpha|=k} \|N_\alpha z_m\|^2 \\
&\ge \frac{1}{\|S^{-1}\|^2} \frac{\left(\sum_j \|N_j z_m\|^2\right)^k}{\|z_m\|^{2(k-1)}} \\
&\ge \frac{1}{\kappa^{2k}} \left(\sum_j \|T_j x_m\|^2 \right)^k .
\end{aligned}$$

In the column square function, keep the exact term of degree one and use the last lower bound from degree two onward. First let $m \to \infty$ in each finite partial sum and then let the number of terms tend to infinity. We obtain

$$1 + \tau^2 s^2 + \sum_{k=2}^{\infty} \frac{\tau^{2k} s^{2k}}{\kappa^{2k}} \le \frac{1}{1-\tau}.$$

The geometric series converges because $s \le \kappa$ and $\tau < 1$. This is (4.6).

If $\kappa = 1$, then $s \le 1$. Suppose that $\kappa > 1$. At $\tau = 1/2$, put $u = s^2/4$. Inequality (4.6) becomes

$$u + \frac{u^2}{\kappa^4(1 - u/\kappa^2)} \le 1.$$

Equivalently,

$$(\kappa^2 - 1)u^2 - \kappa^2(\kappa^2 + 1)u + \kappa^4 \ge 0.$$

The smaller zero is

$$u_\kappa = \frac{2\kappa^2}{\kappa^2 + 1 + \sqrt{(\kappa^2-1)^2 + 4}}.$$

The first two terms of the square function give $0 \le u \le 1$. The quadratic is positive at 0 and negative at 1, so $u \le u_\kappa$. This proves (4.7).

If $0 < s < 2$, retaining only the terms of degrees zero, one, and two in the square function gives

$$1 + \tau^2 s^2 + \frac{\tau^4 s^4}{\kappa^4} \le \frac{1}{1-\tau}.$$

Rearrangement gives $\kappa^4 \ge L_\tau(s)$ for every $0 < \tau < 1$, and hence $\kappa^4 \ge L(s)$. At $\tau = 1/2$,

$$L_{1/2}(s) = \frac{s^4}{4(4-s^2)}.$$

A direct differentiation gives

$$\frac{\partial}{\partial \tau} L_\tau(s)\Big|_{\tau=1/2} = \frac{s^4}{4-s^2} > 0,$$

which proves the strict inequality in (4.8).

For a row tuple, apply the column assertion to $(T_1^*, \dots, T_q^*)$. The similarity operator is $(S^{-1})^*$ and has the same condition number. The operators T_j commute, so the word sums agree after reversal. □

In the next corollary, the conclusion that $\psi(A) < 2$ for every diagonalizable matrix A also follows from Theorem 10.1; see Remark 10.2.

Corollary 4.5. *Let $A \in \mathcal{B}(H)$ and suppose that*

$$A = VNV^{-1},$$

where V is invertible and N is normal. Put $\kappa = \operatorname{cond}(V)$. For every nonzero $f \in \mathcal{A}(K_A)$, set

$$s_f = \frac{\|f(A)\|}{\|f\|_{K_A}}.$$

Then

$$s_f \le \min\{\kappa, \Phi(\kappa)\} < 2. \tag{4.9}$$

If $0 < s_f < 2$, then

$$\kappa^4 \ge L(s_f) > \frac{s_f^4}{4(4-s_f^2)}. \tag{4.10}$$

Consequently, every diagonalizable matrix, of arbitrary order, has scalar Crouzeix ratio strictly smaller than 2.

Proof. Normalize f so that $\|f\|_{K_A} = 1$. Then

$$f(A) = Vf(N)V^{-1}.$$

The operator $f(N)$ is normal, and $\|f(N)\| \le 1$ because $\sigma(N) = \sigma(A) \subset K_A$. Apply Theorem 4.4 with $q = 1$. □

5. Rectangular completely bounded extremals

Let $\mathfrak{A}$ be a finite-dimensional operator algebra with unity and let

$$\theta : \mathfrak{A} \longrightarrow M_d(\mathbb{C})$$

be a homomorphism preserving the identity. For $F \in M_{s,r}(\mathfrak{A})$, the amplification $\theta^{(s,r)}(F)$ acts from $\mathbb{C}^r \otimes \mathbb{C}^d$ to $\mathbb{C}^s \otimes \mathbb{C}^d$.

For $x \in \mathbb{C}^r \otimes \mathbb{C}^d$, let E_x be the smallest subspace of $\mathbb{C}^r$ such that $x \in E_x \otimes \mathbb{C}^d$. Its dimension is the Schmidt rank of x. The compression below is the rectangular form of the finite-level argument associated with Smith's lemma [36].

Lemma 5.1. *One has*

$$\|\theta\|_{\mathrm{cb}} = \max_{1 \le r,s \le d} \left\|\theta^{(s,r)}\right\|. \tag{5.1}$$

At every fixed rectangular level, the norm is attained on the closed unit ball of $M_{s,r}(\mathfrak{A})$.

Proof. Let $F \in M_n(\mathfrak{A})$ with $\|F\| \le 1$, and choose unit vectors $x, y \in \mathbb{C}^n \otimes \mathbb{C}^d$ such that

$$\left|\left\langle \theta^{(n)}(F)x, y\right\rangle\right| = \left\|\theta^{(n)}(F)\right\|.$$

Let $E_x, E_y \subset \mathbb{C}^n$ be the coefficient subspaces of x and y, and put $r = \dim E_x$, $s = \dim E_y$. Then $r, s \le d$. The compression

$$P_{E_y} F|_{E_x} : E_x \longrightarrow E_y$$

does not increase the matrix norm and preserves the displayed matrix coefficient. This proves that the left side of (5.1) is bounded by the right side. The reverse inequality follows by adjoining zero rows and columns.

The space $M_{s,r}(\mathfrak{A})$ is finite-dimensional. Its closed unit ball is compact, and the map $F \mapsto \left\|\theta^{(s,r)}(F)\right\|$ is continuous. Hence the norm is attained. □

The next lemma uses the variational argument of Hartz and McCarthy [25, Lemma 4.1]. We allow perturbations in a C^*-subalgebra on which the homomorphism preserves adjoints. The scalar-matrix case gives the rectangular reduction below, and the more general form will be used for digraph algebras.

Lemma 5.2. *Let $\mathcal{A}$ be an operator algebra with unity, and let $\theta : \mathcal{A} \to M_d(\mathbb{C})$ be a homomorphism preserving the identity. Suppose that $F \in M_n(\mathcal{A})$ satisfies $\|F\| \le 1$ and*

$$C = \left\|\theta^{(n)}\right\| > 1, \qquad R = \theta^{(n)}(F), \qquad \|Rx\| = C$$

for some unit vector $x \in \mathbb{C}^n \otimes \mathbb{C}^d$. Then

$$\langle Rx, (D \otimes I)x\rangle = 0 \qquad (D \in M_n(\mathbb{C})). \tag{5.2}$$

More generally, if $\mathcal{D} \subset \mathcal{A}$ is a C^*-subalgebra with the same identity and $\theta|_{\mathcal{D}}$ is a $*$-homomorphism, then

$$\left\langle Rx, \theta^{(n)}(B)x\right\rangle = 0 \qquad (B \in M_n(\mathcal{D})). \tag{5.3}$$

Proof. We prove the more general assertion. By [10, Theorem 3.1], we may represent $\mathcal{A}$ completely isometrically as a subalgebra of $B(\mathcal{L})$ containing the identity. This representation preserves adjoints on $\mathcal{D}$ by [2, Proposition 1.2.8]. For $\|E\| < 1$, the transformation

$$m_E(X) = (I - EE^*)^{-1/2}(X + E)(I + E^*X)^{-1}(I - E^*E)^{1/2}$$

maps the closed operator ball into itself. Indeed, put $\Delta_E = (I - E^*E)^{1/2}$. For $\|X\| \le 1$, direct calculation gives

$$I - m_E(X)^* m_E(X) = \Delta_E (I + X^*E)^{-1}(I - X^*X)(I + E^*X)^{-1}\Delta_E \ge 0.$$

Fix $B \in M_n(\mathcal{D})$ and put $K = \theta^{(n)}(B)$. For all sufficiently small $t \in \mathbb{C}$, the two square-root factors in $m_{tB}(F)$ belong to $M_n(\mathcal{D})$, and the remaining inverse is given by a norm-convergent Neumann series. Hence $m_{tB}(F) \in M_n(\mathcal{A})$ and $\|m_{tB}(F)\| \le 1$. Since $C = \|Rx\| < \infty$, the map $\theta^{(n)}$ is bounded. Applying it to the Neumann and binomial series, and using $\theta^{(n)}(B^*) = K^*$, gives

$$\theta^{(n)}(m_{tB}(F)) = m_{tK}(R), \qquad \|m_{tK}(R)\| \le C.$$

The target inverse exists for all sufficiently small t, since $|t|\,\|K\|\,\|R\| < 1$.

Since $\|R\| \le C$ and $\|Rx\| = C$, one has $R^*Rx = C^2x$. Expansion at $t = 0$ gives

$$m_{tK}(R)x = Rx + tKx - \bar{t}\,RK^*Rx + O(|t|^2).$$

With inner products linear in the first variable, it follows that

$$\|m_{tK}(R)x\|^2 = C^2 + 2(1 - C^2)\,\mathrm{Re}\big(\bar{t}\,\langle Rx, Kx\rangle\big) + O(|t|^2).$$

The left side is at most C^2. Varying the argument of t and then letting $t \to 0$ proves (5.3). Taking $\mathcal{D} = \mathrm{span}\{1\}$ proves (5.2). □

Theorem 5.3. *Suppose that $\mathfrak{A}$ is finite-dimensional and $\|\theta\|_{\mathrm{cb}} > 1$. There are positive integers r, s and an element $F \in M_{s,r}(\mathfrak{A})$ such that*

$$r + s \le d, \qquad \|F\| \le 1, \qquad \left\|\theta^{(s,r)}(F)\right\| = \|\theta\|_{\mathrm{cb}}. \tag{5.4}$$

If $r + s$ is minimal, norming input and output vectors have Schmidt ranks r and s, and their Schmidt-vector families in $\mathbb{C}^d$ are mutually orthogonal.

Proof. Put $C = \|\theta\|_{\mathrm{cb}} > 1$. By Lemma 5.1, choose a pair (r,s) with $r+s$ minimal among the pairs at which C is attained, and choose an extremal $F \in M_{s,r}(\mathfrak{A})$. There are unit vectors

$$x \in \mathbb{C}^r \otimes \mathbb{C}^d, \qquad y \in \mathbb{C}^s \otimes \mathbb{C}^d$$

such that

$$\theta^{(s,r)}(F)x = Cy, \qquad \theta^{(s,r)}(F)^* y = Cx. \tag{5.5}$$

Minimality implies that the Schmidt ranks of x and y are r and s. Otherwise, the compression in Lemma 5.1 would give an extremal with a smaller sum.

On $\mathbb{C}^s \oplus \mathbb{C}^r$, define

$$\widehat{F} = \begin{pmatrix} 0 & F \\ 0 & 0 \end{pmatrix}, \qquad \widehat{x} = \begin{pmatrix} 0 \\ x \end{pmatrix}, \qquad \widehat{y} = \begin{pmatrix} y \\ 0 \end{pmatrix}.$$

Then $\left\|\widehat{F}\right\| \le 1$ and $\theta^{(r+s)}(\widehat{F})\widehat{x} = C\widehat{y}$. Hence $\left\|\theta^{(r+s)}\right\| = C$. Apply Lemma 5.2 to $\widehat{F}$ and $\widehat{x}$. Using (5.5), we obtain

$$\langle \widehat{y}, (D \otimes I)\widehat{x} \rangle = 0 \qquad (D \in M_{r+s}(\mathbb{C})).$$

Taking

$$D = \begin{pmatrix} 0 & D_0 \\ 0 & 0 \end{pmatrix}, \qquad D_0 \in M_{s,r}(\mathbb{C}),$$

gives

$$\langle y, (D_0 \otimes I)x \rangle = 0 \qquad (D_0 \in M_{s,r}(\mathbb{C})). \tag{5.6}$$

Take Schmidt decompositions

$$x = \sum_{j=1}^{r} a_j e_j \otimes \xi_j, \qquad y = \sum_{i=1}^{s} b_i f_i \otimes \eta_i,$$

where $a_j, b_i > 0$, and where (ξ_j) and (η_i) are orthonormal families in $\mathbb{C}^d$. Taking D_0 to be a matrix unit in (5.6) gives

$$\langle \eta_i, \xi_j \rangle = 0 \qquad (1 \le i \le s,\ 1 \le j \le r).$$

Thus the two Schmidt-vector families are mutually orthogonal, and $r+s \le d$. □

The first variation also gives simplicity of the largest singular value at a minimal rectangular extremal in the following cases.

Proposition 5.4. *Let $F \in M_{s,r}(\mathfrak{A})$ satisfy (5.4), with $r+s$ minimal, and put $C = \|\theta\|_{\mathrm{cb}} > 1$. If $r+s \ge d-1$, then the largest singular value of $R = \theta^{(s,r)}(F)$ is simple.*

Proof. Let $\mathcal{K} = \ker(R^*R - C^2 I)$. Every nonzero vector in $\mathcal{K}$ norms R after normalization. Applying Lemma 5.2 to the square matrix obtained by placing F in an upper-right corner, as in the preceding proof, gives

$$\langle Rx, (D \otimes I)x \rangle = 0 \qquad (x \in \mathcal{K},\ D \in M_{s,r}(\mathbb{C})).$$

For each fixed D, polarization on the complex vector space $\mathcal{K}$ therefore gives

$$\langle Ru, (D \otimes I)v \rangle = 0 \qquad (u, v \in \mathcal{K}).$$

Let $E, G \subset \mathbb{C}^d$ be the smallest subspaces for which

$$\mathcal{K} \subset \mathbb{C}^r \otimes E, \qquad R\mathcal{K} \subset \mathbb{C}^s \otimes G.$$

Taking D to be a matrix unit shows that $E \perp G$. Minimality of $r+s$ implies that every nonzero $x \in \mathcal{K}$ and Rx have Schmidt ranks r and s, respectively. Hence

$$\dim E \ge r, \qquad \dim G \ge s.$$

Since $E \perp G$,

$$0 \le (\dim E - r) + (\dim G - s) \le d - r - s \le 1.$$

Consequently, either $\dim E = r$ or $\dim G = s$. In the first case, identify $\mathbb{C}^r \otimes E$ with $M_r(\mathbb{C})$. Every nonzero element of $\mathcal{K}$ is invertible. If X, Y were linearly independent elements of $\mathcal{K}$, then $\det(X + tY)$ would have degree r with nonzero leading coefficient. A complex root would give a nonzero singular element of $\mathcal{K}$. Thus $\dim \mathcal{K} = 1$. In the second case, apply the same argument to $R\mathcal{K} \subset \mathbb{C}^s \otimes G \simeq M_s(\mathbb{C})$. Since R is injective on $\mathcal{K}$, again $\dim \mathcal{K} = 1$.

□

The corresponding scalar functional-calculus statement is the exact similarity amplification theorem of Aouichaoui and O'Loughlin [1, Theorem 3.2].

Theorem 5.5. *Let $\mathfrak{A}$ be an operator algebra with unity and let $\theta : \mathfrak{A} \to M_d(\mathbb{C})$ be a bounded homomorphism preserving the identity with*

$$C = \|\theta\|_{\mathrm{cb}} > 1.$$

For every $\gamma \geq 1$, there is a positive invertible matrix $S_\gamma \in M_d(\mathbb{C})$ such that

$$\mathrm{cond}(S_\gamma) = \gamma, \qquad \left\|\mathrm{Ad}_{S_\gamma^{-1}} \circ \theta\right\|_{\mathrm{cb}} = \gamma C, \tag{5.7}$$

where $\mathrm{Ad}_{S^{-1}}(X) = S^{-1}XS$.

Proof. Every bounded map into $M_d(\mathbb{C})$ is completely bounded by the coefficient-space compression argument in Lemma 5.1. The ideal $J = \ker\theta$ is closed, and $\mathfrak{A}/J$, with its quotient matrix norms, is finite-dimensional, since it is linearly isomorphic to $\theta(\mathfrak{A})$. Let $\overline{\theta}$ be the induced homomorphism. At each rectangular level,

$$\left\|\overline{\theta}^{(s,r)}\right\| = \left\|\theta^{(s,r)}\right\|.$$

Indeed, the quotient map is completely contractive, and a quotient matrix of norm at most one has, for every $\varepsilon > 0$, a lift of norm at most $1 + \varepsilon$. Applying θ to such lifts and letting $\varepsilon \to 0$ gives the reverse inequality. It is therefore enough to carry out the following construction for $\overline{\theta}$ on $\mathfrak{A}/J$.

Choose a minimal rectangular extremal as in Theorem 5.3. Thus there are $F \in M_{s,r}(\mathfrak{A}/J)$ and unit vectors $x \in \mathbb{C}^r \otimes \mathbb{C}^d$, $y \in \mathbb{C}^s \otimes \mathbb{C}^d$ such that

$$\|F\| \leq 1, \qquad \overline{\theta}^{(s,r)}(F)x = Cy.$$

Let E and G be the subspaces of $\mathbb{C}^d$ spanned by the Schmidt vectors of x and y, respectively. Theorem 5.3 gives $E \perp G$. Define

$$S_\gamma = \sqrt{\gamma}\, P_E + \frac{1}{\sqrt{\gamma}} P_G + P_{(E\oplus G)^\perp}.$$

Then S_γ is positive and $\mathrm{cond}(S_\gamma) = \gamma$. Moreover,

$$(I_r \otimes S_\gamma)x = \sqrt{\gamma}\, x, \qquad (I_s \otimes S_\gamma^{-1})y = \sqrt{\gamma}\, y.$$

Therefore

$$\begin{aligned}&\left(\mathrm{Ad}_{S_\gamma^{-1}} \circ \overline{\theta}\right)^{(s,r)}(F)x \\ &\quad = (I_s \otimes S_\gamma^{-1})\overline{\theta}^{(s,r)}(F)(I_r \otimes S_\gamma)x = \gamma C y.\end{aligned}$$

This proves the lower bound in (5.7). The reverse bound follows from

$$\left\|\mathrm{Ad}_{S_\gamma^{-1}} \circ \overline{\theta}\right\|_{\mathrm{cb}} \leq \left\|S_\gamma^{-1}\right\| \|S_\gamma\| \left\|\overline{\theta}\right\|_{\mathrm{cb}} = \gamma C.$$

The same approximate-lift argument applies to $\mathrm{Ad}_{S_\gamma^{-1}} \circ \theta$. It transfers the computed norm γC from the quotient to $\mathfrak{A}$.

□

6. Complete bounds and equality in dimension three

We first transfer the square functions from the uniform algebra to its quotient.

Lemma 6.1. *Let $A \in M_d(\mathbb{C})$. Suppose that $X = (x_1, \ldots, x_q)^{\mathsf{T}} \in M_{q,1}(\mathfrak{A}_A)$ satisfies $\|X\| \leq 1$, and put $T_j = \theta_A(x_j)$. Then, for every $0 < \tau < 1$,*

$$\sum_{k=0}^{\infty} \tau^{2k} \sum_{\alpha \in \mathcal{W}_k} T_\alpha^* T_\alpha \preceq \frac{1}{1-\tau} I. \tag{6.1}$$

The analogous row estimate holds for every $X \in M_{1,q}(\mathfrak{A}_A)$ with $\|X\| \leq 1$.

Proof. Fix $\varepsilon > 0$. Lemma 2.2 gives a lift $F = (f_1, \ldots, f_q)^{\mathsf{T}}$ with

$$\max_{z \in W(A)} \left(\sum_{j=1}^{q} |f_j(z)|^2\right)^{1/2} \leq 1 + \varepsilon.$$

The operators $f_j(A)$ are the fixed operators T_j. Apply Theorem 1.5 to $f_j/(1+\varepsilon)$. For every finite partial sum, let $\varepsilon \to 0^+$. This gives (6.1). The row assertion is proved in the same way. □

Proof of Theorem 1.1. Let $A \in M_3(\mathbb{C})$ and put $C = \psi_{\mathrm{cb}}(A)$. By Proposition 2.1, $C = \|\theta_A\|_{\mathrm{cb}}$. If $C \le 1$, there is nothing to prove. Suppose that $C > 1$. Theorem 5.3 gives positive integers r, s and $X \in M_{s,r}(\mathfrak{A}_A)$ such that

$$r + s \le 3, \qquad \|X\| \le 1, \qquad \left\|\theta_A^{(s,r)}(X)\right\| = C.$$

The possible pairs are

$$(r,s) = (1,1), \qquad (1,2), \qquad (2,1).$$

They are respectively a scalar, a two-component column, and a two-component row. Lemma 6.1, at $\tau = 1/2$ and after retaining the terms of degrees zero and one, gives $C \le 2$ in each case. This proves (1.3).

The same proof applies in dimension two. For the lower estimate, let

$$J = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}.$$

Then $W(J) = \overline{\mathbb{D}}$ and $\|J\| = 2$. Direct sums with a zero block give the same lower estimate in dimension three. Hence $Q_{\mathrm{cb}}(2) = Q_{\mathrm{cb}}(3) = 2$.

More generally, if $\psi_{\mathrm{cb}}(A) > 2$, a minimal extremal cannot have $r = 1$ or $s = 1$, by Lemma 6.1. Thus $r, s \ge 2$ and $r + s \le d$. When $d = 4$, this leaves only $(r,s) = (2,2)$. □

We now identify equality in dimension three.

Proof of Theorem 1.6. The implication from right to left follows from

$$\psi(A) \le \psi_{\mathrm{cb}}(A) \le 2.$$

Suppose that $\psi_{\mathrm{cb}}(A) = 2$ and $\psi(A) < 2$. Choose a minimal rectangular extremal as in Theorem 5.3. The scalar pair $(1,1)$ cannot attain the completely bounded norm. Hence the pair is either $(1,2)$ or $(2,1)$.

Suppose first that the pair is $(1,2)$. Write

$$X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \qquad T_j = \theta_A(x_j).$$

There is a unit vector $\xi \in \mathbb{C}^3$ such that

$$\|T_1\xi\|^2 + \|T_2\xi\|^2 = 4. \tag{6.2}$$

At $\tau = 1/2$, Lemma 6.1 gives

$$I + \frac{1}{4}\sum_{j=1}^{2} T_j^* T_j + \frac{1}{16}\sum_{i,j=1}^{2} (T_iT_j)^*(T_iT_j) + \cdots \preceq 2I.$$

Taking the quadratic form at ξ and using (6.2), the first two displayed terms already have value 2. Hence

$$T_iT_j\xi = 0 \qquad (1 \le i, j \le 2). \tag{6.3}$$

Let

$$\eta = \frac{1}{2}\begin{pmatrix} T_1\xi \\ T_2\xi \end{pmatrix} \in \mathbb{C}^2 \otimes \mathbb{C}^3.$$

The minimality assertion in Theorem 5.3 says that η has Schmidt rank two. Thus $T_1\xi$ and $T_2\xi$ are linearly independent. The orthogonality relation (5.6) gives

$$\xi \perp T_1\xi, \qquad \xi \perp T_2\xi. \tag{6.4}$$

Since $\dim \mathbb{C}^3 = 3$, the two vectors $T_1\xi, T_2\xi$ span $\xi^\perp$. It follows from (6.3) that each T_i vanishes on $\xi^\perp$. Therefore

$$T_iT_j = 0 \qquad (1 \le i, j \le 2). \tag{6.5}$$

The space $L = \operatorname{span}\{T_1, T_2\}$ is two-dimensional and $L^2 = \{0\}$. It is contained in $\operatorname{alg}(A)$. Since $I \notin L$ and $\dim \operatorname{alg}(A) \le 3$, one has

$$\operatorname{alg}(A) = \mathbb{C}I \oplus L.$$

Thus $A = aI + N$ for some $N \in L$. By (6.5), $N^2 = 0$, and consequently

$$\operatorname{alg}(A) = \operatorname{span}\{I, N\}.$$

This contradicts $\dim(\mathbb{C}I \oplus L) = 3$.

If the minimal pair is $(2,1)$, apply the preceding argument to A^* and the adjoint column. Lemma 2.3 preserves both the scalar and completely bounded ratios. This completes the proof. □

We next prove the quantitative estimate for diagonalizable matrices.

Proof of Theorem 1.7. Put $\kappa = \mathrm{cond}(V)$ and $C = \psi_{\mathrm{cb}}(A)$. If $C = 1$, the conclusion is immediate. Otherwise, choose a minimal rectangular extremal $X \in M_{s,r}(\mathfrak{A}_A)$. As before, (r, s) is one of $(1,1), (1,2), (2,1)$.

Consider first a column $X = (x_1, \dots, x_q)^{\mathsf{T}}$, where $q = 1$ or $q = 2$, and put $T_j = \theta_A(x_j)$. The class x_j has well-defined values on the eigenvalues of A. Indeed, if $g \in \mathcal{I}_A$, then

$$0 = V^{-1} g(A) V = g(\Lambda),$$

so g vanishes at every diagonal entry of Λ. Let

$$N_j = V^{-1} T_j V.$$

The matrices N_j are diagonal. Lemma 2.2 gives, for every $\varepsilon > 0$, a column lift whose pointwise norm is at most $1 + \varepsilon$. Evaluation at the eigenvalues and passage to the limit give

$$\sum_{j=1}^{q} N_j^* N_j \preceq I.$$

Lemma 6.1 gives the required column square function for the tuple (T_j). Theorem 4.4 therefore gives

$$\big\| (T_1, \dots, T_q)^{\mathsf{T}} \big\| \le \min\{\kappa, \Phi(\kappa)\}.$$

The row case follows by taking adjoints. Since the chosen extremal has norm C, the displayed estimate proves (1.13). The formula (1.12) gives $\Phi(\kappa) < 2$ for every finite κ. □

7. The abstract Crouzeix conjecture

We first consider the scalar estimate. Badea, O'Loughlin, and Virtanen proved the scalar Clouâtre–Ostermann–Ransford conjecture for bounded homomorphisms on uniform algebras, without requiring the antilinear map to be contractive [5, Theorem 2.1]. The following theorem removes the initial boundedness assumptions and allows more general Banach algebras, under an explicit commutation hypothesis.

A complex Banach algebra $\mathcal{B}$ with unity satisfies von Neumann's inequality if

$$\|p(b)\| \le \max_{|z| \le 1} |p(z)| \qquad (\|b\| \le 1)$$

for every polynomial p. Uniform algebras and operator algebras with unity have this property. We use the disk algebra $A(\mathbb{D})$ with the uniform norm on $\overline{\mathbb{D}}$.

Theorem 7.1. *Let $\mathcal{B}$ be a complex Banach algebra with unity satisfying von Neumann's inequality, let $\mathcal{H}$ be a nonzero Hilbert space, and let $\theta : \mathcal{B} \to B(\mathcal{H})$ be a complex-linear homomorphism preserving the identity, not assumed to be bounded. Suppose that $\Psi : \mathcal{B} \to B(\mathcal{H})$ is a complex-linear contraction preserving the identity and*

$$[\theta(b), 2\Psi(b^n)^* - \theta(b^n)^*] = 0 \qquad (b \in \mathcal{B},\ n \ge 1). \tag{7.1}$$

Then θ is bounded and $\|\theta\| \le 2$.

In particular, let $\alpha : \mathcal{B} \to \mathcal{B}$ be an antilinear map preserving the identity and put

$$\Phi_{\theta,\alpha}(b) = \frac{1}{2}\big(\theta(b) + \theta(\alpha(b))^*\big).$$

If $\Phi_{\theta,\alpha}$ is contractive and

$$[\theta(b), \theta(\alpha(b^n))] = 0 \qquad (b \in \mathcal{B},\ n \ge 1),$$

then θ is bounded and $\|\theta\| \le 2$. The commutation condition holds whenever $\theta(\mathcal{B})$ is commutative. No boundedness assumption on α is needed.

Proof. Fix $b \in \mathcal{B}$ with $\|b\| \le 1$ and put $T = \theta(b)$. Since an algebra homomorphism preserving the identity preserves inverses,

$$\sigma(T) \subseteq \sigma_{\mathcal{B}}(b), \qquad r(T) \le 1,$$

where $r(T)$ is the spectral radius. This uses only the algebraic properties of θ. By von Neumann's inequality, polynomial evaluation at b extends to a contractive homomorphism $C_b : A(\mathbb{D}) \to \mathcal{B}$ preserving the identity. Thus $L = \Psi \circ C_b$ is a contraction preserving the identity.

We recall why ordinary contractivity of L gives a unitary dilation. The formula

$$\widetilde{L}(f + \overline{g}) = L(f) + L(g)^*, \qquad f, g \in A(\mathbb{D}),$$

is well defined on $A(\mathbb{D}) + A(\mathbb{D})^*$, because the intersection of the disk algebra and its conjugate consists of the constants. For a unit vector $x \in \mathcal{H}$, the functional $f \mapsto \langle L(f)x, x\rangle$ has norm and value at 1 equal to one. Its norm-preserving Hahn–Banach extension to $C(\mathbb{T})$ is a state. Testing against these

extensions shows that $\widetilde{L}$ is positive. Positivity and the identity $\widetilde{L}(1) = I$ bound $\widetilde{L}$ on self-adjoint elements, and hence on the whole space. Since the trigonometric polynomials are dense in $C(\mathbb{T})$, it extends to a positive map preserving the identity on $C(\mathbb{T})$. This map is completely positive because $C(\mathbb{T})$ is commutative [37, Theorem 4]. Stinespring's representation theorem, with the normalization in [37, Theorem 1 and Section 3], therefore gives a Hilbert space $\mathcal{K}$, an isometry $V : \mathcal{H} \to \mathcal{K}$, and a unitary $U \in B(\mathcal{K})$ such that

$$\Psi(b^n) = V^* U^n V \qquad (n \geq 1).$$

Set

$$E_n = 2V^* U^{*n} V - T^{*n} = 2\Psi(b^n)^* - \theta(b^n)^*.$$

These operators commute with T by (7.1). For $0 < t < 1$, the operators tT and tU have the corresponding perturbations $t^n E_n$. They are uniformly bounded in n, since

$$\|t^n E_n\| \leq 2 + \|(tT)^n\| \quad \text{and} \quad r(tT) < 1.$$

Lemma 1 of Lorist and Schwenninger [30] therefore gives $t\,\|T\| \leq 2$. Letting $t \uparrow 1$ proves $\|\theta(b)\| \leq 2$, and taking the supremum over b proves boundedness and the asserted norm estimate.

For the final assertion, $\Phi_{\theta,\alpha}$ is complex-linear, satisfies $\Phi_{\theta,\alpha}(1) = I$, and

$$2\Phi_{\theta,\alpha}(b^n)^* - \theta(b^n)^* = \theta(\alpha(b^n)).$$

Apply the first part with $\Psi = \Phi_{\theta,\alpha}$. □

The commutation hypothesis in Theorem 7.1 cannot be omitted in general, even if both α and $\Phi_{\theta,\alpha}$ are completely contractive. Theorem 1.3 gives such an example, and (8.19) displays the failure of commutation.

Remark 7.2. The dilation in the proof of Theorem 7.1 represents the contractive map to which it is applied. Its compressed coordinate need not agree with the homomorphism θ.

For example, take $\mathcal{B} = A(\mathbb{D})$ and define

$$\theta(f) = f(1/4), \qquad \alpha(f)(\zeta) = \overline{f(-\bar{\zeta})}.$$

Then $\theta : \mathcal{B} \to \mathbb{C}$ is a completely contractive homomorphism preserving the identity. The map α is antilinear and multiplicative, with $\alpha(1) = 1$ and $\alpha^2 = I$. For $F = [f_{ij}] \in M_n(A(\mathbb{D}))$,

$$[\alpha(f_{ij})](\zeta) = \overline{F(-\bar{\zeta})}, \qquad \|[\alpha(f_{ij})]\| = \|F\|,$$

where the bar denotes entrywise complex conjugation. Thus α is a complete isometry. The symmetrized map is

$$\Phi(f) := \Phi_{\theta,\alpha}(f) = \frac{1}{2}\big(f(1/4) + f(-1/4)\big).$$

It is completely contractive, since

$$\left\|\Phi^{(n)}(F)\right\| \leq \frac{1}{2}\big(\|F(1/4)\| + \|F(-1/4)\|\big) \leq \|F\|_{M_n(A(\mathbb{D}))}.$$

Let z denote the coordinate function. Every unitary moment dilation of Φ, with isometry V and unitary U, satisfies

$$V^* U V = \Phi(z) = 0, \qquad \theta(z) = \frac{1}{4}.$$

Hence it cannot also satisfy $V^* U V = \theta(z)$. The example satisfies the hypotheses of the complete abstract problem and $\|\theta\|_{\mathrm{cb}} = 1$. It therefore excludes this additional moment requirement, without contradicting the abstract Crouzeix bound.

For an operator algebra with unity, a complex-linear map P is called real positive if $\operatorname{Re} b \geq 0$ implies $\operatorname{Re} P(b) \geq 0$, where $\operatorname{Re} b = (b + b^*)/2$. The next consequence extends [5, Theorem 3.3] from uniform algebras to operator algebras with unity and removes the initial boundedness assumptions on both maps.

Corollary 7.3. *Let $\mathcal{B}$ be an operator algebra with unity, let $\mathcal{H}$ be a nonzero Hilbert space, and let $\theta : \mathcal{B} \to B(\mathcal{H})$ be a complex-linear homomorphism preserving the identity. Suppose that $P : \mathcal{B} \to B(\mathcal{H})$ is complex-linear and real positive, $P(1) = 2I$, and*

$$[P(b^n) - \theta(b^n), \theta(b)^*] = 0 \qquad (b \in \mathcal{B},\ n \geq 1).$$

Then P and θ are bounded, $\|P\| = 2$, and $\|\theta\| \leq 2$.

Proof. By [9, Lemma 2.1], a real-positive map between operator algebras with unity is bounded and has norm equal to the norm of its value at the identity. Hence $\|P\| = 2$ and $\Psi = P/2$ is a contraction preserving the identity. Taking adjoints in the commutation condition gives (7.1). Theorem 7.1 now applies. □

We next give the bounds for operator radii, using the exact formula of Badea, Crouzeix, and Klaja [4, Theorem 3.2]. For $\rho \geq 1$, let $\mathcal{C}_\rho$ consist of the operators $S \in B(\mathcal{H})$ for which there are a Hilbert space $\mathcal{K} \supset \mathcal{H}$ and a unitary $U \in B(\mathcal{K})$ satisfying

$$S^n = \rho P_{\mathcal{H}} U^n|_{\mathcal{H}} \qquad (n \geq 1).$$

Define

$$w_\rho(S) = \inf\{t > 0 : t^{-1}S \in \mathcal{C}_\rho\}, \qquad \|\theta\|_{w_\rho} = \sup_{\|b\|\leq 1} w_\rho(\theta(b)).$$

Thus $w_1(S) = \|S\|$ and $w_2(S) = w(S)$, where $w(S) = \sup_{\|x\|=1} |\langle Sx, x\rangle|$ is the numerical radius. For $K \geq 1$, put

$$\kappa_\rho(K) = \frac{K^2 + 1 + \sqrt{(K^2+1)^2 - 4\rho(2-\rho)K^2}}{2\rho K},$$

$$c_\rho = \kappa_\rho(2) = \frac{5 + \sqrt{9 + 16(\rho-1)^2}}{4\rho}.$$

Corollary 7.4. *Under the hypotheses of Theorem 7.1 or Corollary 7.3, let $K = \|\theta\|$. Then*

$$\|\theta\|_{w_\rho} = \kappa_\rho(K) \leq c_\rho \qquad (\rho \geq 1).$$

Each constant c_ρ is best possible. In particular, $\sup_{\|b\|\leq 1} w(\theta(b)) \leq 5/4$, and $5/4$ is best possible.

Proof. The preceding results give $1 \leq K \leq 2$. The equality is [4, Theorem 3.2(a)]. To obtain the bound, write

$$x_K = \frac{1}{2}(K + K^{-1}), \qquad \kappa_\rho(K) = \frac{x_K + \sqrt{x_K^2 - \rho(2-\rho)}}{\rho}.$$

Both expressions are increasing for $K \geq 1$, so $\kappa_\rho(K) \leq \kappa_\rho(2)$. Sharpness follows from the example below. □

For a completely bounded map on an operator algebra, write

$$\|\theta\|_{w_\rho,\mathrm{cb}} = \sup_{m\geq 1} \sup_{\|F\|_{M_m(\mathcal{B})}\leq 1} w_\rho(\theta^{(m)}(F)).$$

The following consequence uses only ordinary contractivity of the symmetrized map, even when the domain algebra is noncommutative.

Corollary 7.5. *Let $\mathcal{B}$ be an operator algebra with unity and let $\theta : \mathcal{B} \to B(\mathcal{H})$ be a complex-linear homomorphism preserving the identity, where $\mathcal{H} \neq \{0\}$ and $\dim \theta(\mathcal{B}) \leq 2$. Let $\alpha : \mathcal{B} \to \mathcal{B}$ be an antilinear map preserving the identity. If $\Phi_{\theta,\alpha}$ is contractive, then*

$$\|\theta\|_{\mathrm{cb}} = \|\theta\| \leq 2,$$

$$\|\theta\|_{w_\rho,\mathrm{cb}} = \kappa_\rho(\|\theta\|) \leq c_\rho \qquad (\rho \geq 1).$$

These conclusions apply, in particular, to homomorphisms into $M_2(\mathbb{C})$ with commutative range. The constants 2 and c_ρ are best possible.

Proof. An algebra of dimension at most two with unity is commutative, so Theorem 7.1 gives boundedness and $\|\theta\| \leq 2$. Meyer proved that a bounded homomorphism of linear rank at most two preserving the identity satisfies $\|\theta\|_{\mathrm{cb}} = \|\theta\|$ [32, Corollary 3.6]. The radius formula now follows from [4, Theorem 3.2(b)]. A commutative subalgebra of $M_2(\mathbb{C})$ containing I_2 has dimension at most two: if it contains a nonscalar matrix X, it is contained in the commutant $\mathrm{span}\{I, X\}$ of X. The scalar case is immediate. □

To see sharpness of all the preceding constants, take $\mathcal{B} = A(\mathbb{D})$ and define

$$J = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \qquad \theta(f) = f(0)I + f'(0)J, \qquad \alpha(f)(z) = \overline{f(0)}.$$

Since $J^2 = 0$, the map θ preserves the identity and is a homomorphism with two-dimensional range, and

$$\Phi_{\theta,\alpha}(f) = f(J/2).$$

Since $J/2$ is a contraction, this map is completely contractive. The similarity $J = D(J/2)D^{-1}$ with $D = \mathrm{diag}(2, 1)$ gives $\|\theta\|_{\mathrm{cb}} \leq 2$, while $\|\theta(z)\| = 2$. Thus $\|\theta\| = \|\theta\|_{\mathrm{cb}} = 2$, and the radius formulas give

equality with c_ρ. The map $P = 2\Phi_{\theta,\alpha}$ also satisfies Corollary 7.3: the contraction $\Phi_{\theta,\alpha}$ preserves the identity and is real positive by [9, Corollary 2.2], so P is real positive as well. The required commutation follows from the commutativity of $\theta(A(\mathbb{D}))$.

We next prove the column and row estimates in Theorem 1.2 for quotients of uniform algebras.

Definition 7.6. A Q-algebra with unity is an operator algebra $\mathcal{B}$ with unity that is completely isometrically isomorphic to $\mathcal{A}/I$, where $\mathcal{A} \subset C(X)$ is a uniformly closed algebra containing the constant function 1, where X is a compact Hausdorff space and I is a proper closed ideal. The matrix norms are the quotient matrix norms. Thus, with $q_I : \mathcal{A} \to \mathcal{A}/I$,

$$\|[b_{ij}]\|_{M_{s,r}(\mathcal{A}/I)} = \inf\left\{\sup_{x\in X} \|[f_{ij}(x)]\| : q_I(f_{ij}) = b_{ij}\right\}.$$

This is the terminology of Meyer [32, Section 1]. It includes uniform algebras, with $I = 0$, and quotients such as $A(\mathbb{D})/z^2A(\mathbb{D})$. The latter contains a nonzero nilpotent element and hence is not a uniform algebra.

For $q \geq 1$, let $A(\mathbb{B}_q)$ denote the ball algebra on the closed unit ball of $\mathbb{C}^q$.

Lemma 7.7. *Let $\Psi : A(\mathbb{B}_q) \to B(\mathcal{H})$ be a complete contraction preserving the identity. There are a Hilbert space $\mathcal{K}$, an isometry $V : \mathcal{H} \to \mathcal{K}$, and commuting normal operators $Q_1, \dots, Q_q \in B(\mathcal{K})$ such that*

$$\sum_{j=1}^{q} Q_j^* Q_j = \sum_{j=1}^{q} Q_j Q_j^* = I$$

and

$$\Psi(h) = V^* h(Q_1, \dots, Q_q)V, \qquad h \in A(\mathbb{B}_q).$$

Proof. Regard the ball algebra as a subalgebra of $C(\partial\mathbb{B}_q)$. Arveson's theorem [2, Theorem 1.2.9] gives a completely positive extension $\widetilde{\Psi} : C(\partial\mathbb{B}_q) \to B(\mathcal{H})$ preserving the identity. By [37, Theorem 1 and Section 3], there are a $*$-representation preserving the identity

$$\pi : C(\partial\mathbb{B}_q) \longrightarrow B(\mathcal{K})$$

and an isometry $V : \mathcal{H} \to \mathcal{K}$ such that $\Psi(h) = V^*\pi(h)V$ for $h \in A(\mathbb{B}_q)$.

Put $Q_j = \pi(z_j)$, where z_j is the jth coordinate function. The operators $Q_1, \dots, Q_q$ commute and are normal. Since $\sum_j |z_j|^2 = 1$ on $\partial\mathbb{B}_q$, they satisfy both spherical identities. The compression formula holds first for polynomials and then for every $h \in A(\mathbb{B}_q)$ by uniform approximation. □

Let $\mathcal{W}_n$ be the set of words of length n in $\{1, \dots, q\}$. For $\omega = (\omega_1, \dots, \omega_n) \in \mathcal{W}_n$, put

$$T_\omega = T_{\omega_1} \cdots T_{\omega_n}, \qquad Q_\omega = Q_{\omega_1} \cdots Q_{\omega_n}.$$

If $j \in \{1, \dots, q\}$, then ωj denotes the word obtained by adjoining j to ω.

Lemma 7.8. *Let $T_1, \dots, T_q \in B(\mathcal{H})$ be commuting operators. Let $Q_1, \dots, Q_q \in B(\mathcal{K})$ be a commuting normal tuple satisfying*

$$\sum_{j=1}^{q} Q_j^* Q_j = I,$$

let $V : \mathcal{H} \to \mathcal{K}$ be an isometry, and put

$$E_\omega = 2V^* Q_\omega^* V - T_\omega^*.$$

Assume that every E_ω commutes with every T_j. For $n \geq 1$, define

$$\mathcal{T}_n : \mathcal{H} \longrightarrow \mathcal{H}^{\mathcal{W}_n}, \qquad \mathcal{T}_n x = (T_\omega x)_{\omega\in\mathcal{W}_n},$$

and

$$\mathcal{E}_n : \mathcal{H}^{\mathcal{W}_n} \longrightarrow \mathcal{H}, \qquad \mathcal{E}_n(x_\omega)_{\omega\in\mathcal{W}_n} = \sum_{\omega\in\mathcal{W}_n} E_\omega x_\omega.$$

If

$$\sup_{n\geq 1} \|\mathcal{E}_n\| < \infty,$$

then

$$\left\| \begin{pmatrix} T_1 \\ \vdots \\ T_q \end{pmatrix} \right\| \leq 2.$$

Proof. Let $\mathcal{A}_n$ be the row operator with entries $V^*Q_\omega^*V$, $\omega \in \mathcal{W}_n$. The spherical identity gives

$$\sum_{\omega\in\mathcal{W}_n} Q_\omega^* Q_\omega = I.$$

Thus $\mathcal{A}_n$, being a compression of the corresponding row, has norm at most one. Since

$$\mathcal{E}_n = 2\mathcal{A}_n - \mathcal{T}_n^*,$$

one has $\|\mathcal{T}_n\| \le 2 + \|\mathcal{E}_n\|$. The uniform bound for $\mathcal{T}_n$ therefore follows. Conversely, $\|\mathcal{E}_n\| \le 2 + \|\mathcal{T}_n\|$, so either one of these bounds implies the other.

Put

$$\kappa = \left\| \begin{pmatrix} T_1 \\ \vdots \\ T_q \end{pmatrix} \right\|$$

and suppose that $\kappa > 2$. If the column operator does not attain its norm, choose unit vectors x_m on which its norm tends to κ and pass to Hilbert-space ultrapowers with respect to a free ultrafilter. The class of (x_m) is a unit vector at which the ultrapower of $\mathcal{T}_1$ attains the norm κ. The defining relations and the uniform bounds for $\mathcal{T}_n$ and $\mathcal{E}_n$ pass to the ultrapowers. Since $\mathcal{T}_1^*\mathcal{T}_1 \preceq \kappa^2 I$, equality of its quadratic form at this vector gives the equation below. It is therefore enough to treat the case in which there is a unit vector $x \in \mathcal{H}$ such that

$$\sum_{j=1}^{q} T_j^* T_j x = \kappa^2 x.$$

For $n \ge 1$, define

$$m_n = \sum_{\omega\in\mathcal{W}_n} \operatorname{Re}\langle E_\omega T_\omega x, x\rangle .$$

The sequence (m_n) is bounded because

$$|m_n| \le \|\mathcal{E}_n\|\,\|\mathcal{T}_n\| .$$

Set

$$y = \sum_{j=1}^{q} Q_j^* V T_j x, \qquad u = y - \kappa V x.$$

The row $(Q_1^*, \dots, Q_q^*)$ is a contraction, so $\|y\| \le \kappa$. Moreover,

$$2\operatorname{Re}\langle y, Vx\rangle = \kappa^2 + m_1.$$

Consequently,

$$\|u\|^2 \le 2\kappa^2 - \kappa m_1 - \kappa^3. \tag{7.2}$$

For every $\omega \in \mathcal{W}_n$, commutativity of the T_j^* gives

$$T_{\omega j}^* T_j x = T_\omega^* T_j^* T_j x.$$

It follows that

$$\sum_{j=1}^{q} E_{\omega j} T_j x = 2V^* Q_\omega^* y - \kappa^2 T_\omega^* x.$$

Using $u = y - \kappa V x$, we obtain

$$\left(\kappa E_\omega - \sum_{j=1}^{q} E_{\omega j} T_j\right) x = \kappa(\kappa-1) T_\omega^* x - 2V^* Q_\omega^* u.$$

Take real parts of the inner products with $T_\omega^* x$ and sum over $\omega \in \mathcal{W}_n$. Since every E_ω commutes with every T_j, the two terms on the left sum respectively to κm_n and m_{n+1}. Completing the square termwise and using that V^* is a contraction gives

$$\begin{aligned} \kappa m_n - m_{n+1} &\ge -\frac{1}{\kappa(\kappa-1)} \sum_{\omega\in\mathcal{W}_n} \|V^* Q_\omega^* u\|^2 \\ &\ge -\frac{1}{\kappa(\kappa-1)} \sum_{\omega\in\mathcal{W}_n} \|Q_\omega^* u\|^2 \\ &= -\frac{\|u\|^2}{\kappa(\kappa-1)}. \end{aligned} \tag{7.3}$$

A commuting family of normal operators is doubly commuting. Hence

$$\sum_{\omega\in\mathcal{W}_n} Q_\omega Q_\omega^* = \left(\sum_{j=1}^{q} Q_j Q_j^*\right)^n = I.$$

After division of (7.3) by κ^n and summation from $n=1$ to N, we get

$$m_1 - \frac{m_{N+1}}{\kappa^N} \geq -\frac{\|u\|^2}{\kappa(\kappa-1)} \sum_{n=1}^{N} \frac{1}{\kappa^n}.$$

Letting $N\to\infty$ and using boundedness of (m_n) gives

$$\kappa m_1 \geq -\frac{\|u\|^2}{(\kappa-1)^2}.$$

Together with (7.2), this yields

$$\|u\|^2 \left(1 - \frac{1}{(\kappa-1)^2}\right) \leq \kappa^2(2-\kappa),$$

which is impossible when $\kappa > 2$. □

Remark 7.9. For $q=1$, Lemma 7.8 reduces to the unitary-dilation case of the perturbation lemma of Lorist and Schwenninger [30, Lemma 1]. The proof above adds the summation over words needed for tuples.

Lemma 7.10. *Let $\mathcal{B}$ be a Q-algebra with unity. Transposition is an isometry from $M_{s,r}(\mathcal{B})$ onto $M_{r,s}(\mathcal{B})$ for all $r,s\geq 1$. If $b_1,\dots,b_q\in\mathcal{B}$ satisfy*

$$\left\|(b_1,\dots,b_q)^{\mathsf{T}}\right\| \leq 1,$$

then the polynomial calculus extends to a complete contraction preserving the identity

$$C_b : A(\mathbb{B}_q) \longrightarrow \mathcal{B}, \qquad C_b(h) = h(b_1,\dots,b_q).$$

For every $n\geq 1$, the row and column formed by $b_\omega = b_{\omega_1}\cdots b_{\omega_n}$, $\omega\in\mathcal{W}_n$, both have norm at most one. In particular, $\mathcal{B}$ satisfies von Neumann's inequality.

Proof. The transpose assertion is [32, Corollary 5.3]; its rectangular form follows by adjoining zero rows and columns. Identify $\mathcal{B}$ with $\mathcal{A}/I$. For every $\varepsilon>0$, the quotient column norm gives simultaneous lifts $f_j\in\mathcal{A}$ with

$$q_I(f_j) = b_j, \qquad \sup_{x\in X} \sum_{j=1}^{q} |f_j(x)|^2 \leq (1+\varepsilon)^2.$$

For any matrix-valued polynomial P in q commuting variables, complete contractivity of q_I gives

$$\|P(b_1,\dots,b_q)\| \leq \sup_{\|z\|\leq 1+\varepsilon} \|P(z)\|.$$

Letting $\varepsilon\to 0^+$ and using polynomial density in $A(\mathbb{B}_q)$ proves the asserted extension at every matrix level. Finally,

$$\sum_{\omega\in\mathcal{W}_n} |z_\omega|^2 = \left(\sum_{j=1}^{q} |z_j|^2\right)^n \leq 1 \qquad (z\in\overline{\mathbb{B}_q}).$$

Apply the rectangular amplifications of C_b to the monomial row and column. The case $q=1$ gives von Neumann's inequality. □

Theorem 7.11. *Let $\mathcal{B}$ be a Q-algebra with unity, let $\theta:\mathcal{B}\to B(\mathcal{H})$ be a completely bounded homomorphism preserving the identity, and let $\alpha:\mathcal{B}\to\mathcal{B}$ be an antilinear map preserving the identity. Suppose that $\Phi_{\theta,\alpha}$ is completely contractive. If $b_1,\dots,b_q\in\mathcal{B}$ satisfy*

$$\left\|(b_1,\dots,b_q)^{\mathsf{T}}\right\| \leq 1,$$

then

$$\left\| \begin{pmatrix} \theta(b_1) \\ \vdots \\ \theta(b_q) \end{pmatrix} \right\| \leq 2, \qquad \left\| \begin{pmatrix} \theta(b_1) & \cdots & \theta(b_q) \end{pmatrix} \right\| \leq 2. \tag{7.4}$$

No boundedness assumption on α is required.

Proof. By Lemma 7.10, $\Psi = \Phi_{\theta,\alpha} \circ C_b$ is a complete contraction preserving the identity. Let $V : \mathcal{H} \to \mathcal{K}$ and $(Q_1, \dots, Q_q)$ be the normal spherical dilation from Lemma 7.7.

Set $T_j = \theta(b_j)$ and $L = 2\Phi_{\theta,\alpha} - \theta$. Then L is completely bounded, $\|L\|_{\mathrm{cb}} \le 2 + \|\theta\|_{\mathrm{cb}}$, and

$$L(b) = \theta(\alpha(b))^*.$$

For $\omega \in \mathcal{W}_n$, write $b_\omega = b_{\omega_1} \cdots b_{\omega_n}$. Since $V^* Q_\omega V = \Phi_{\theta,\alpha}(b_\omega)$, we have

$$2V^* Q_\omega^* V - T_\omega^* = L(b_\omega)^* = \theta(\alpha(b_\omega)).$$

These operators commute with every T_j because $\mathcal{B}$ is commutative. Lemma 7.10 gives

$$\left\|(T_\omega)^{\mathsf{T}}_{\omega\in\mathcal{W}_n}\right\| \le \|\theta\|_{\mathrm{cb}}, \qquad \|(L(b_\omega)^*)_{\omega\in\mathcal{W}_n}\| = \left\|(L(b_\omega))^{\mathsf{T}}_{\omega\in\mathcal{W}_n}\right\| \le \|L\|_{\mathrm{cb}}.$$

Lemma 7.8 therefore provides the column estimate.

For the row estimate, apply the same lemma to $S_j = T_j^*$ and $R_j = Q_j^*$. Both tuples are commuting, and $(R_1, \dots, R_q)$ is again a normal spherical tuple. Since the T_j commute, reversal of the indexed products does not change the word operators. The corresponding correction term is

$$2V^* R_\omega^* V - S_\omega^* = 2V^* Q_\omega V - T_\omega = L(b_\omega).$$

It commutes with every S_j. By Lemma 7.10, the word row and column over $\mathcal{B}$ both have norm at most one. Hence

$$\left\|(S_\omega)^{\mathsf{T}}_{\omega\in\mathcal{W}_n}\right\| = \|(T_\omega)_{\omega\in\mathcal{W}_n}\| \le \|\theta\|_{\mathrm{cb}},$$

while

$$\|(L(b_\omega))_{\omega\in\mathcal{W}_n}\| \le \|L\|_{\mathrm{cb}}.$$

The perturbation lemma now gives the row estimate. □

The column and row estimates also control arbitrary rectangular levels with a dimension-dependent constant.

Lemma 7.12. *Under the hypotheses of Theorem 7.11, for all positive integers r, s,*

$$\left\|\theta^{(s,r)}\right\| \le 2\sqrt{\min\{r,s\}}.$$

The same estimate holds for the homomorphism induced by θ on a quotient $\mathcal{B}/J$, where J is a closed ideal contained in $\ker\theta$ and the quotient is equipped with the operator-algebra matrix norms.

Proof. Let $F = [f_{ij}] \in M_{s,r}(\mathcal{B})$ satisfy $\|F\| \le 1$, and put $Z = \theta^{(s,r)}(F)$. Each column of F has norm at most one. Theorem 7.11 therefore gives column norm at most two for its image. For $x = \sum_{j=1}^r e_j \otimes x_j \in \mathbb{C}^r \otimes \mathcal{H}$,

$$\|Zx\| \le 2\sum_{j=1}^r \|x_j\| \le 2\sqrt{r}\,\|x\|.$$

Thus $\|Z\| \le 2\sqrt{r}$. Each row of F also has norm at most one. By Lemma 7.10, its transpose is a column contraction. Applying the row estimate to the s rows gives

$$\|Zx\|^2 \le 4s\,\|x\|^2,$$

so $\|Z\| \le 2\sqrt{s}$.

For the quotient assertion, lift a rectangular matrix of quotient norm at most one to matrices over $\mathcal{B}$ of norm at most $1+\varepsilon$, apply the preceding estimate after division by $1+\varepsilon$, and let $\varepsilon \to 0^+$. □

Theorem 7.13. *Let $\mathcal{B}$ be a Q-algebra with unity, let $\theta : \mathcal{B} \to M_d(\mathbb{C})$ be a homomorphism preserving the identity, and let $\alpha : \mathcal{B} \to \mathcal{B}$ be an antilinear map preserving the identity. Suppose that $\Phi_{\theta,\alpha}$ is completely contractive. If $d \le 3$, then*

$$\|\theta\|_{\mathrm{cb}} \le 2. \tag{7.5}$$

More generally, for $d \ge 2$,

$$\|\theta\|_{\mathrm{cb}} \le 2\sqrt{\lfloor d/2 \rfloor}. \tag{7.6}$$

If $\|\theta\|_{\mathrm{cb}} > 2$, then $\left\|\theta^{(s,r)}\right\| = \|\theta\|_{\mathrm{cb}}$ for some $r, s \ge 2$ with $r + s \le d$.

Proof. Lemma 7.10 shows that $\mathcal{B}$ satisfies von Neumann's inequality. Since $\mathcal{B}$ is commutative and $\Phi_{\theta,\alpha}$ is contractive, Theorem 7.1 gives $\|\theta\| \le 2$. Smith's lemma now shows that θ is completely bounded. Let $J = \ker\theta$, $q : \mathcal{B} \to \mathcal{B}/J$ be the quotient map, and

$$\overline{\theta} : \mathcal{B}/J \longrightarrow M_d(\mathbb{C})$$

be the induced injective homomorphism. At every rectangular level,

$$\left\|\overline{\theta}^{(s,r)}\right\| = \left\|\theta^{(s,r)}\right\|. \tag{7.7}$$

Indeed, one inequality follows from $\theta = \overline{\theta} \circ q$, while the reverse inequality follows by taking lifts whose norms approach the quotient norm. The quotient $\mathcal{B}/J$ is finite-dimensional because it is algebraically isomorphic to the subalgebra $\theta(\mathcal{B})$ of $M_d(\mathbb{C})$. All estimates below use approximate lifts in $\mathcal{B}$; no invariance of J under α is required.

Let $C = \|\theta\|_{\mathrm{cb}}$. There is nothing to prove if $C \le 1$. If $C > 1$, apply Theorem 5.3 to $\overline{\theta}$. There are positive integers r, s and $X \in M_{s,r}(\mathcal{B}/J)$ such that

$$r + s \le d, \qquad \|X\| \le 1, \qquad \left\|\overline{\theta}^{(s,r)}(X)\right\| = C.$$

If $d = 1$, the inequality $r + s \le d$ is impossible, so $C \le 1$. If $d = 2$ or 3, the only possible pairs are $(r, s) = (1, 1)$, $(1, 2)$, and $(2, 1)$ with only the first pair available when $d = 2$. Using lifts of norm at most $1 + \varepsilon$ and Theorem 7.11, every available level is bounded by $2(1 + \varepsilon)$. Letting $\varepsilon \to 0^+$ proves (7.5).

For arbitrary $d \ge 2$, Lemma 7.12 and (7.7) imply

$$C \le 2\sqrt{\min\{r, s\}} \le 2\sqrt{\lfloor d/2 \rfloor},$$

which proves (7.6). Finally, if $C > 2$, the detecting pair cannot have $r = 1$ or $s = 1$, by the scalar, column, and row estimates and the same approximate-lift argument. Hence $r, s \ge 2$. □

The disk-algebra example following Corollary 7.5 also proves sharpness in Theorem 7.11 and in the case $d = 2$ of Theorem 7.13. Taking the direct sum with $f \mapsto f(0)$ gives the same sharp example for $d = 3$.

Unlike Theorem 1.5, the abstract argument above gives the endpoint column and row estimates but not corresponding higher-order positive square-function series. We therefore do not claim an abstract analog of Theorem 1.6.

8. Noncommutative operator algebras

The scalar estimate in Theorem 7.1 uses a commutation hypothesis, and the complete Q-algebra estimates use commutativity of the domain. We now give the counterexample in Theorem 1.3 and prove Theorem 1.4 for matrix algebras containing the diagonal.

For a linear or antilinear map L, we use the entrywise amplification $L^{(k)}([b_{ij}]) = [L(b_{ij})]$ and the norm $\|L\|_{\mathrm{cb}} = \sup_{k\ge1} \left\|L^{(k)}\right\|$. This is the convention in [12, Section 6]. In particular,

$$[\Phi_{\theta,\alpha}(b_{ij})] = \frac{1}{2}\big([\theta(b_{ij})] + [\theta(\alpha(b_{ji}))]^*\big),$$

where the adjoint is taken on the whole block matrix.

We first define the algebra and the two maps, construct completely positive maps proving the required complete contractivity, and then establish the strict norm inequality.

Put

$$P = \begin{pmatrix} 1 & -5/2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad N = E_{12}, \qquad I = I_3, \tag{8.1}$$

and define

$$X(a, c, u) = aP + c(I - P) + uN = \begin{pmatrix} a & u - \frac{5}{2}(a - c) & a - c \\ 0 & c & 0 \\ 0 & 0 & c \end{pmatrix}, \tag{8.2}$$

$$\mathcal{A} = \{X(a, c, u) : a, c, u \in \mathbb{C}\} \subset M_3(\mathbb{C}).$$

The matrix norms on $\mathcal{A}$ are those inherited from $M_3(\mathbb{C})$. The matrices I, P, N have real entries and form a complex basis of $\mathcal{A}$. Define θ by complex-linear extension of

$$\theta(I) = I_2, \qquad \theta(P) = E_{11}, \qquad \theta(N) = \frac{16}{19}E_{12}, \tag{8.3}$$

where the matrix units on the right belong to $M_2(\mathbb{C})$. Define α by antilinear extension of

$$\alpha(I) = I, \qquad \alpha(P) = \frac{7}{17}P + \frac{3}{8}N, \qquad \alpha(N) = \frac{4}{21}(I - P). \tag{8.4}$$

In particular,

$$\theta(X(a,c,u)) = \begin{pmatrix} a & 16u/19 \\ 0 & c \end{pmatrix}, \qquad \alpha(\mathcal{A}) \subset \mathcal{A}.$$

Lemma 8.1. *The space $\mathcal{A}$ is a three-dimensional noncommutative operator algebra with unity, and $\mathcal{A} \cap \mathcal{A}^* = \mathbb{C}I$. The map θ is an injective homomorphism onto T_2, preserves the identity, and satisfies*

$$\|\theta\|_{\mathrm{cb}} \le \frac{8\sqrt{29}}{19}.$$

Proof. Direct multiplication gives

$$P^2 = P, \qquad PN = N, \qquad NP = N^2 = 0,$$

and hence

$$X(a,c,u)X(a',c',u') = X(aa', cc', au' + uc'). \tag{8.5}$$

These identities prove closure under multiplication and multiplicativity of θ. The coordinate formula proves injectivity, surjectivity onto T_2, and preservation of the identity. If $b = X(a,c,u)$ and $b^* \in \mathcal{A}$, the lower triangular entries of b^* force $a = c$ and $u = 0$. Thus $b = cI$.

The plane $\mathcal{M} = \operatorname{span}\{e_1, 2e_2 + 5e_3\}$ is invariant for $\mathcal{A}$. In the orthonormal basis $e_1, (2e_2 + 5e_3)/\sqrt{29}$, restriction to $\mathcal{M}$ gives the completely contractive homomorphism

$$\pi(X(a,c,u)) = \begin{pmatrix} a & 2u/\sqrt{29} \\ 0 & c \end{pmatrix}.$$

With $L = \operatorname{diag}(8\sqrt{29}/19, 1)$, we have $\theta(b) = L\pi(b)L^{-1}$. Since $8\sqrt{29}/19 > 1$, this gives the asserted bound at every matrix level. □

We next construct a completely positive extension of the symmetrized map. Set

$$v = (4, 0, -1)^T, \qquad w = (1, 2, 4)^T, \tag{8.6}$$

and define

$$\Gamma_\Phi(Z) = \begin{pmatrix} v^*Zv/17 & v^*Zw/19 \\ w^*Zv/19 & w^*Zw/21 \end{pmatrix} \qquad (Z \in M_3(\mathbb{C})). \tag{8.7}$$

Lemma 8.2. *The map $\Gamma_\Phi : M_3(\mathbb{C}) \to M_2(\mathbb{C})$ is completely positive, preserves the identity, and satisfies $\Gamma_\Phi|_{\mathcal{A}} = \Phi_{\theta,\alpha}$.*

Proof. The coefficient matrix

$$G = \begin{pmatrix} 1/17 & 1/19 \\ 1/19 & 1/21 \end{pmatrix} \tag{8.8}$$

is positive definite because $19^2 - 17 \cdot 21 = 4 > 0$. If $G = \sum_j g_j g_j^*$, its Schur multiplier is

$$Z \longmapsto \sum_j \operatorname{diag}(g_j) Z \operatorname{diag}(g_j)^*.$$

This is a completely positive map on $M_2(\mathbb{C})$. Its composition with the completely positive map $Z \mapsto (v,w)^* Z (v,w)$ is Γ_Φ. The identities $v^*w = 0$, $\|v\|^2 = 17$, and $\|w\|^2 = 21$ give $\Gamma_\Phi(I) = I_2$.

Evaluation on the basis yields

$$\Gamma_\Phi(P) = \begin{pmatrix} 12/17 & 0 \\ 3/19 & 0 \end{pmatrix}, \qquad \Gamma_\Phi(N) = \begin{pmatrix} 0 & 8/19 \\ 0 & 2/21 \end{pmatrix}. \tag{8.9}$$

On the other hand,

$$\theta(\alpha(P)) = \begin{pmatrix} 7/17 & 6/19 \\ 0 & 0 \end{pmatrix}, \qquad \theta(\alpha(N)) = \begin{pmatrix} 0 & 0 \\ 0 & 4/21 \end{pmatrix}.$$

Together with (8.3), these give

$$\Gamma_\Phi|_{\mathcal{A}} = \Phi_{\theta,\alpha}, \tag{8.10}$$

since both sides are complex-linear and agree on I, P, N. □

To prove complete contractivity of α, we construct a positive Choi matrix. For a complex-linear map $\Gamma : M_3(\mathbb{C}) \to M_m(\mathbb{C})$, write

$$C_\Gamma = [\Gamma(E_{ij})]_{i,j=1}^3.$$

The blocks are indexed by the matrix units of $M_3(\mathbb{C})$. If $C = [C_{ij}]$ is positive with $m \times m$ blocks, the associated map is $\Gamma_C(Z) = \sum_{i,j} z_{ij} C_{ij}$.

Lemma 8.3. *Let $U, V \in M_m(\mathbb{C})$. Suppose that $M \in M_m(\mathbb{C})$ is positive definite and*

$$Q = I - M - V^* M^{-1} V - (U - M)^* M^{-1} (U - M) \geq 0. \tag{8.11}$$

There is a completely positive map $\Gamma : M_3(\mathbb{C}) \to M_m(\mathbb{C})$ preserving the identity such that

$$\Gamma(E_{11} + E_{13}) = U, \qquad \Gamma(E_{12}) = V.$$

Proof. Put $C = U - M$ and define

$$C_\Gamma = \begin{pmatrix} M \\ V^* \\ C^* \end{pmatrix} M^{-1} \begin{pmatrix} M & V & C \end{pmatrix} + \operatorname{diag}(0, 0, Q). \tag{8.12}$$

This matrix is positive. Choi's criterion [11, Theorem 2] gives a completely positive map. The diagonal blocks sum to I, so the map preserves the identity. The $(1,1)$, $(1,2)$, and $(1,3)$ blocks of C_Γ are M, V, C, which prove the interpolation identities.

For completeness, this implication follows from a Gram factorization. Write $C_\Gamma = \sum_\nu z_\nu z_\nu^*$, with $z_\nu = (z_{\nu,1}, z_{\nu,2}, z_{\nu,3})^T$ and $z_{\nu,i} \in \mathbb{C}^m$. Let $K_\nu : \mathbb{C}^m \to \mathbb{C}^3$ have ith row $z_{\nu,i}^*$. Then

$$\Gamma(Z) = \sum_\nu K_\nu^* Z K_\nu, \qquad \sum_\nu K_\nu^* K_\nu = I_m.$$

The resulting isometric column factorization also proves complete contractivity at every matrix level. □

Lemma 8.4. *There is a completely positive map $\Gamma_\alpha : M_3(\mathbb{C}) \to M_3(\mathbb{C})$ preserving the identity such that*

$$\alpha(b) = \Gamma_\alpha(\bar{b}) \qquad (b \in \mathcal{A}), \tag{8.13}$$

where the bar denotes entrywise conjugation. Thus Γ_α extends the complex-linear map $b \mapsto \alpha(\bar{b})$ on $\mathcal{A}$.

Proof. Apply Lemma 8.3 with

$$\begin{aligned} U &= \alpha(P) + \frac{5}{2}\alpha(N) = \frac{7}{17}P + \frac{3}{8}N + \frac{10}{21}(I - P), \\ V &= \alpha(N) = \frac{4}{21}(I - P), \qquad M = \frac{1}{6}\begin{pmatrix} 4 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}. \end{aligned} \tag{8.14}$$

The matrix M is positive definite and

$$M^{-1} = \begin{pmatrix} 12/7 & -6/7 & 0 \\ -6/7 & 24/7 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

For the matrix Q in (8.11), put $D_0 = \operatorname{diag}(1, 2, 1)$. Direct calculation gives

$$D_0 Q D_0 = \frac{1}{1189524}\begin{pmatrix} 237356 & -18081 & -22540 \\ -18081 & 454919 & 374228 \\ -22540 & 374228 & 508268 \end{pmatrix}. \tag{8.15}$$

The diagonal entries of the integer matrix exceed the sums of the absolute off-diagonal entries in their rows by

$$196735, \qquad 62610, \qquad 111500,$$

respectively. For a Hermitian matrix $H = [h_{ij}]$,

$$z^* H z \geq \sum_i \left(h_{ii} - \sum_{j \neq i} |h_{ij}| \right) |z_i|^2.$$

This follows from $2|z_i z_j| \leq |z_i|^2 + |z_j|^2$. It proves positive definiteness in (8.15), and hence $Q > 0$.

The resulting map satisfies

$$\Gamma_\alpha(P) = U - \frac{5}{2}V = \alpha(P), \qquad \Gamma_\alpha(N) = V = \alpha(N), \qquad \Gamma_\alpha(I) = I,$$

because $P + \frac{5}{2}N = E_{11} + E_{13}$. Since I, P, N have real entries, complex linearity of Γ_α and antilinearity of α give (8.13). □

Proof of Theorem 1.3. Use (8.2)–(8.4). Lemma 8.1 proves the algebraic assertions and the complete upper bound. Lemmas 8.2 and 8.4 give identity-preserving completely positive maps. They are complete contractions [2, Proposition 1.2.10]. At matrix level k, conjugating each input block is entrywise conjugation of the whole scalar matrix in M_{3k}, and therefore preserves its norm. Consequently,

$$\|[\alpha(b_{ij})]\| = \left\|\Gamma_\alpha^{(k)}([\overline{b_{ij}}])\right\| \le \|[b_{ij}]\|\,, \qquad \|[\Phi_{\theta,\alpha}(b_{ij})]\| \le \|[b_{ij}]\|\,.$$

Both maps preserve the identity, so their completely bounded norms are equal to one.

For the lower estimate, take the integer matrix

$$B = 2P + 6N - I = \begin{pmatrix} 1 & 1 & 2 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad \theta(B) = \begin{pmatrix} 1 & 96/19 \\ 0 & -1 \end{pmatrix}. \tag{8.16}$$

For $t \ge 0$, set

$$f(t) = \left\| \begin{pmatrix} 1 & t \\ 0 & -1 \end{pmatrix} \right\| = \frac{\sqrt{4+t^2}+t}{2}.$$

A unitary change of basis in the last two coordinates of $\mathbb{C}^3$ gives

$$\|B\| = \frac{3+\sqrt{5}}{2}, \qquad \|\theta(B)\| = \frac{48+\sqrt{2665}}{19}. \tag{8.17}$$

For $k > 1$, the inequality $f(t) > k$ is equivalent to $t > k - k^{-1}$. With $k = 3 + \sqrt{5}$,

$$k - k^{-1} = \frac{9+5\sqrt{5}}{4} < \frac{96}{19},$$

since $213 > 0$ and

$$213^2 - 125 \cdot 19^2 = 244 > 0. \tag{8.18}$$

Thus $\|\theta(B)\| > 2\,\|B\|$. The ratio in (8.17) proves the lower bound in (1.6).

Finally, Theorem 5.3 applies to $\theta : \mathcal{A} \to M_2(\mathbb{C})$. Since $\|\theta\|_{\mathrm{cb}} > 1$, it gives an extremal at a rectangular level with $r, s \ge 1$ and $r + s \le 2$. Thus $r = s = 1$, and $\|\theta\| = \|\theta\|_{\mathrm{cb}}$. □

The algebra is a nonunitary similarity of a triangular algebra. Put

$$S_0 = I + \frac{5}{2}E_{12} - E_{13}, \qquad \mathcal{A}_0 = \left\{ \begin{pmatrix} a & u & 0 \\ 0 & c & 0 \\ 0 & 0 & c \end{pmatrix} : a, c, u \in \mathbb{C} \right\}.$$

Since $(S_0 - I)^2 = 0$, direct multiplication gives

$$\mathcal{A} = S_0 \mathcal{A}_0 S_0^{-1}.$$

The similarity preserves the multiplication but changes the matrix norms. The algebraic identification

$$X(a, c, u) \longmapsto \begin{pmatrix} a & u \\ 0 & c \end{pmatrix}$$

is not isometric, since $\|P\| = \sqrt{33}/2$ whereas the image of P has norm one.

The vectors in (8.6) satisfy $Nv = 0$ and $Pw = 0$. The symmetrization identities then determine (8.4). Replacing 19 by $t > 0$ in the off-diagonal entries of (8.7), and $16/19$ by $16/t$ in (8.3), leaves α unchanged. Complete positivity and the strict norm violation for B hold simultaneously when

$$\sqrt{357} \le t < \frac{384}{9+5\sqrt{5}}.$$

The integer 19 belongs to this interval by $19^2 - 357 = 4$ and (8.18).

The construction was obtained by semidefinite programming. With θ in (8.3) fixed, preservation of the identity, the range condition for α, and the symmetrization identities on I, P, N impose affine constraints over the real scalars on two Choi matrices. For a Hermitian matrix $H = H_1 + iH_2$ with H_1, H_2 real,

$$H \ge 0 \quad \Longleftrightarrow \quad \begin{pmatrix} H_1 & -H_2 \\ H_2 & H_1 \end{pmatrix} \ge 0.$$

The positivity constraints therefore give a semidefinite feasibility problem in the form of [39, Section 1.1]. The displayed rational matrices give an exact verification independent of the numerical calculation.

Since θ is injective, its kernel is invariant under α. Thus kernel invariance alone does not imply the bound. Moreover,

$$[\theta(P), \theta(\alpha(P))] = \begin{pmatrix} 0 & 6/19 \\ 0 & 0 \end{pmatrix} \neq 0, \qquad B^2 = I_3. \tag{8.19}$$

This also shows where the example fails the commutation hypothesis of Theorem 7.1. Thus the norm inequality fails on an involution.

Corollary 8.5. *Among counterexamples to the complete COR conjecture, the smallest possible target matrix order is two and the smallest possible complex dimension of the domain algebra is three.*

Proof. A scalar target, or a domain of complex dimension at most two, gives $\dim \theta(\mathcal{B}) \leq 2$. Corollary 7.5 therefore excludes a counterexample in either case. Theorem 1.3 attains both minimum dimensions. □

We next prove the bound for digraph algebras. The following consequence of Lemma 5.2 gives the correction estimate needed in the proof.

Lemma 8.6. *Let $\mathcal{B}$ be a finite-dimensional operator algebra with unity, and let $D \subset \mathcal{B}$ be a C^*-subalgebra with the same identity. Let $E : \mathcal{B} \to D$ be complex-linear, and let $\theta : \mathcal{B} \to B(\mathcal{H})$ be a homomorphism preserving the identity whose restriction to D is a $*$-homomorphism. If*

$$\Psi = \frac{1}{2}(\theta + \theta E)$$

is completely contractive, then $\|\theta\|_{\mathrm{cb}} \leq 2$.

Proof. Represent $\mathcal{B}$ completely isometrically and preserving the identity as an operator algebra on a Hilbert space. The finite-dimensional argument below uses compactness of the unit balls of $M_m(\mathcal{B})$ and Lemma 5.2; it does not require a finite-dimensional ambient representation.

First suppose that $\mathcal{H}$ is finite-dimensional and nonzero. Fix $m \geq 1$ and put $c = \left\|\theta^{(m)}\right\|$. If $c \leq 1$, there is nothing to prove. Otherwise, finite dimensionality gives a contraction $F \in M_m(\mathcal{B})$ and a unit vector $x \in \mathcal{H}^m$ such that, with

$$R = \theta^{(m)}(F), \qquad K = \theta^{(m)}(E^{(m)}(F)),$$

one has $\|Rx\| = c$. Apply (5.3) with $B = E^{(m)}(F) \in M_m(D)$. It gives $\langle Rx, Kx\rangle = 0$, and hence

$$c^2 + \|Kx\|^2 = \|(R+K)x\|^2 \leq 4,$$

since $(R+K)/2 = \Psi^{(m)}(F)$ is a contraction. Thus $\left\|\theta^{(m)}\right\| \leq 2$ at every matrix level.

Now let $\mathcal{H}$ be arbitrary. For a unit vector $x = (x_1, \ldots, x_m) \in \mathcal{H}^m$, put

$$\mathcal{K} = \operatorname{span}\{\theta(b)x_j : b \in \mathcal{B},\ 1 \leq j \leq m\}.$$

This space is finite-dimensional, contains every x_j, and is invariant under $\theta(\mathcal{B})$ by multiplicativity. Since $\theta(d)^* = \theta(d^*)$ for $d \in D$, it reduces $\theta(D)$. The restriction $\theta_{\mathcal{K}}(b) = \theta(b)|_{\mathcal{K}}$ is therefore a homomorphism preserving the identity, and its restriction to D is a $*$-homomorphism. The map

$$\frac{1}{2}(\theta_{\mathcal{K}} + \theta_{\mathcal{K}} E) = P_{\mathcal{K}} \Psi(\,\cdot\,)|_{\mathcal{K}}$$

is completely contractive. The finite-dimensional case gives $\|\theta_{\mathcal{K}}\|_{\mathrm{cb}} \leq 2$. For every $F \in M_m(\mathcal{B})$, invariance now yields

$$\left\|\theta^{(m)}(F)x\right\| = \left\|\theta_{\mathcal{K}}^{(m)}(F)x\right\| \leq 2\,\|F\|\,.$$

Taking the supremum over unit vectors and matrix levels proves the assertion. The case $\mathcal{H} = \{0\}$ is immediate. □

For the finite-dimensional diagonal, we first use an argument with central projections as in [12, Section 5]. We then use states to treat the noncommutative summands with arbitrary representation multiplicities.

Lemma 8.7. *Let D be a finite-dimensional C^*-algebra with identity. Let $\eta : D \to B(\mathcal{H})$ be a homomorphism preserving the identity, and let $\gamma : D \to D$ be an antilinear contraction preserving the identity. If*

$$\Xi(d) = \frac{1}{2}\big(\eta(d) + \eta(\gamma(d))^*\big)$$

is positive, then η is a $*$-homomorphism.

Proof. The assertion is immediate if $\mathcal{H} = \{0\}$, so assume otherwise. The map η is bounded because D is finite-dimensional. The complex-linear map $d \mapsto \gamma(d^*)$ is a contraction preserving the identity on D. It is therefore positive and preserves adjoints [2, Proposition 1.2.8] (see also [8, Lemma 1.2]). Consequently, γ is positive and preserves adjoints.

First let p be a central projection of D, and set $U = \eta(2p - I)$ and $V = \eta(\gamma(2p - I))$. Then $U^2 = I$, $UV = VU$, and $(U + V^*)/2$ is a self-adjoint contraction. Since $(U + I)\mathcal{H} \subset \ker(U - I)$ and V leaves $\ker(U - I)$ invariant, self-adjointness of $(U + V^*)/2$ gives, relative to this subspace and its orthogonal complement,

$$U = \begin{pmatrix} I & X \\ 0 & -I \end{pmatrix}, \qquad V = \begin{pmatrix} A & X \\ 0 & -E \end{pmatrix}, \qquad A = A^*, \quad E = E^*, \quad AX + XE = 2X.$$

The element $\gamma(2p - I)$ is a self-adjoint contraction in the finite-dimensional algebra D. It satisfies a polynomial all of whose zeros belong to $[-1, 1]$. The same polynomial annihilates V and A, and its reflected polynomial annihilates E. Thus A and E are self-adjoint contractions with finite spectra. Let P_a and Q_e be the spectral projections of A and E corresponding to eigenvalues a and e. The equality $AX + XE = 2X$ gives $(a + e - 2)P_aXQ_e = 0$. Since $a, e \leq 1$, the only possible nonzero term has $a = e = 1$. Thus X maps the eigenspace of E at 1 into the eigenspace of A at 1 and is zero elsewhere. On these two spaces the symmetrization has the compression

$$\begin{pmatrix} I & X/2 \\ X^*/2 & -I \end{pmatrix}.$$

Its square has diagonal blocks $I + XX^*/4$ and $I + X^*X/4$. Contractivity forces $X = 0$. Hence $\eta(p)$ is an orthogonal projection.

With normalized Haar measure on the compact unitary group of D, put

$$B = \int_{\mathcal{U}(D)} \eta(u)^* \eta(u)\, du.$$

Since $\eta(u)^{-1} = \eta(u^*)$, one has $\|\eta\|^{-2} I \leq B \leq \|\eta\|^2 I$. Haar invariance gives $\eta(u)^* B \eta(u) = B$ for every unitary $u \in D$. Thus $\pi(d) = B^{1/2}\eta(d)B^{-1/2}$ is a $*$-representation. Set $L = B^{-1/2}$ and $G = L^2$. Then

$$\eta(d) = L\pi(d)L^{-1}, \qquad L > 0.$$

For a central projection $p \in D$, the orthogonal projection $\eta(p)$ commutes with $\eta(u)$ and $\eta(u)^*$ for every unitary $u \in D$. It therefore commutes with B and L, and $\pi(p) = \eta(p)$. Thus L and G are block diagonal with respect to the represented simple summands of D. Set $\beta(d) = \pi(\gamma(d^*))$, a positive complex-linear map. The symmetrization identity becomes

$$2L\Xi(d)L = G\pi(d) + \beta(d)G.$$

Fix a simple summand of D on which π is nonzero, with target space $\mathbb{C}^r \otimes \mathcal{L}$, on which $\pi(D)$ acts as $M_r(\mathbb{C}) \otimes I$. Let $j : M_r(\mathbb{C}) \to D$ be the inclusion of that input summand. Compression of βj has the form $b(\,\cdot\,) \otimes I$, where b is positive and $b(I) \leq I$. Write L_r and G_r for the corresponding blocks of L and G, and let $\Xi_r(a)$ be the compression of $\Xi(j(a))$ to this target block. For a state ω on $B(\mathcal{L})$, put

$$g = (\mathrm{id} \otimes \omega)(G_r), \qquad \Lambda(a) = (\mathrm{id} \otimes \omega)(L_r\Xi_r(a)L_r).$$

A state is completely positive [37, Theorem 3]. Compression, positive congruence, and the state slice preserve positivity, so Λ is positive. Since G_r is bounded below by a positive scalar multiple of the identity, $g > 0$. Also $0 \leq j(I) \leq I$ and $\Xi(I) = I$, which give $\Lambda(I) \leq g$. The compressed identity becomes

$$2\Lambda(a) = ga + b(a)g.$$

The positive map b and the positive map Λ preserve adjoints. Comparing this identity at a and at a^* therefore gives $[g, a - b(a)] = 0$. If g has unequal eigenvalues λ_i, λ_j, take the matrix unit e_{ij} in an eigenbasis. Then $b(e_{ij})_{ij} = 1$. The map $T(a) = g^{-1/2}\Lambda(a)g^{-1/2}$ is positive and $0 \leq T(I) \leq I$. We claim that $\|T\| \leq 1$. For a unitary $u = \sum_k \lambda_k e_k$, put $A_k = T(e_k)$ and $W\xi = (A_k^{1/2}\xi)_k$. Then $\|W\| \leq 1$ and

$$T(u) = W^* \operatorname{diag}(\lambda_k I) W,$$

so $\|T(u)\| \leq 1$. Every contraction in $M_r(\mathbb{C})$ is a convex combination of unitaries: apply a singular-value decomposition and express its diagonal contraction as a convex combination of diagonal matrices with entries in $\{-1, 1\}$. This proves the claim.

However,

$$T(e_{ij})_{ij} = \frac{1}{2}\left(\sqrt{\lambda_i/\lambda_j} + \sqrt{\lambda_j/\lambda_i}\right) > 1,$$

which is impossible. Hence g is scalar for every state ω. Writing $G_r = [G_{ij}]$ with $G_{ij} \in B(\mathcal{L})$, we obtain $\omega(G_{ij}) = 0$ for $i \neq j$ and $\omega(G_{ii} - G_{jj}) = 0$ for all i, j and all states ω. States separate operators, so $G_r = I_r \otimes H$ for a positive invertible $H \in B(\mathcal{L})$. This holds on every represented simple summand, including arbitrary multiplicities. Therefore $L = G^{1/2}$ commutes with $\pi(D)$ and $\eta = \pi$. □

Proof of Theorem 1.4. We may assume $\mathcal{H} \neq \{0\}$. Let E_{ij} be the standard matrix units in $M_n(\mathbb{C})$. For $b = [b_{ij}] \in \mathcal{B}$,

$$E_{ii} b E_{jj} = b_{ij} E_{ij} \in \mathcal{B}.$$

Thus $\mathcal{B}$ is spanned by the matrix units it contains. Define $i \preceq j$ when $E_{ij} \in \mathcal{B}$. This relation is reflexive, since every E_{ii} belongs to $\mathcal{B}$, and transitive, since $E_{ij}E_{jk} = E_{ik}$. The relation

$$i \sim j \quad \Longleftrightarrow \quad i \preceq j \text{ and } j \preceq i$$

is therefore an equivalence relation. Let $C_1, \dots, C_r$ be its classes, and put $p_k = \sum_{i \in C_k} E_{ii}$. These are mutually orthogonal projections in $\mathcal{B}$ with sum I. Each $p_k \mathcal{B} p_k$ is the full matrix algebra on the coordinate space indexed by C_k.

Suppose that both $p_k \mathcal{B} p_l$ and $p_l \mathcal{B} p_k$ are nonzero. There are $i, i' \in C_k$ and $j, j' \in C_l$ such that $E_{ij}, E_{j'i'} \in \mathcal{B}$. Since $E_{jj'}, E_{i'i} \in \mathcal{B}$, their product gives

$$E_{ji} = E_{jj'} E_{j'i'} E_{i'i} \in \mathcal{B}.$$

Hence $i \sim j$, so $k = l$. Consequently,

$$p_k \mathcal{B} p_l = \{0\} \text{ or } p_l \mathcal{B} p_k = \{0\} \qquad (k \neq l).$$

It follows that

$$D = \mathcal{B} \cap \mathcal{B}^* = \bigoplus_{k=1}^{r} p_k \mathcal{B} p_k.$$

The map $E_D(x) = \sum_k p_k x p_k$ is a completely positive projection from $M_n(\mathbb{C})$ onto D and preserves the identity.

Write $\Phi = \Phi_{\theta,\alpha}$. Since Φ is a complete contraction preserving the identity on $\mathcal{B} \subset M_n(\mathbb{C})$, Arveson's theorem [2, Theorem 1.2.9] gives a completely positive extension $\widetilde{\Phi} : M_n(\mathbb{C}) \to B(\mathcal{H})$ preserving the identity. In particular, $\Phi|_D$ is completely positive. The complex-linear map $d \mapsto \alpha(d^*)$ is a contraction from D to $M_n(\mathbb{C})$ preserving the identity. It therefore preserves adjoints by [2, Proposition 1.2.8]. Thus

$$\alpha(d)^* = \alpha(d^*) \in \mathcal{B} \qquad (d \in D),$$

which gives $\alpha(D) \subset D$. Lemma 8.7 now shows that $\eta = \theta|_D$ is a $*$-homomorphism. Put $P_k = \eta(p_k)$.

By [37, Theorem 1 and Section 3], there are a Hilbert space $\mathcal{L}$, a $*$-representation $\sigma : M_n(\mathbb{C}) \to B(\mathcal{L})$ preserving the identity, and an isometry $V : \mathcal{H} \to \mathcal{L}$ such that

$$\widetilde{\Phi}(x) = V^* \sigma(x) V \qquad (x \in M_n(\mathbb{C})).$$

The identity $\widetilde{\Phi}(I) = I$ gives $V^*V = I$. Integrate over the compact unitary group $\mathcal{U}(D)$ with normalized Haar measure and set

$$Z = \int_{\mathcal{U}(D)} \sigma(u)^* V \eta(u)\, du.$$

The integrand is norm-continuous, so this is an operator-norm integral. Each integrand has norm one, and therefore $\|Z\| \leq 1$. For $v \in \mathcal{U}(D)$, right invariance of Haar measure gives

$$\sigma(v) Z \eta(v)^* = \int_{\mathcal{U}(D)} \sigma(uv^*)^* V \eta(uv^*)\, du = Z.$$

Since the unitaries linearly span D, it follows that $\sigma(d)Z = Z\eta(d)$ for every $d \in D$. Set $b_0 = Z^*Z$ and define on $M_n(\mathbb{C})$ the completely positive map

$$\Omega_0(x) = Z^* \sigma(x) Z.$$

Then $0 \leq b_0 \leq I$, b_0 commutes with $\eta(D)$, and

$$\Omega_0(d_1 x d_2) = \eta(d_1)\Omega_0(x)\eta(d_2), \qquad \Omega_0(d) = \eta(d) b_0$$

for $x \in M_n(\mathbb{C})$ and $d, d_1, d_2 \in D$.

Let $b \in p_k \mathcal{B} p_l$, where $k \neq l$ and this block is nonzero. Then $p_l \mathcal{B} p_k = \{0\}$. For $u, v \in \mathcal{U}(D)$,

$$P_k \eta(u)^* \Phi(ubv^*) \eta(v) P_l = \frac{1}{2}\theta(b).$$

Indeed, the term containing $\theta(ubv^*)$ gives $\theta(b)/2$, since η preserves adjoints. The other term vanishes because

$$P_l\theta(\alpha(ubv^*))P_k = \theta\big(p_l\alpha(ubv^*)p_k\big) = 0.$$

Here we used only $\alpha(ubv^*) \in \mathcal{B}$, not invariance of the individual blocks under α. Expanding the two norm integrals gives

$$\Omega_0(b) = \int_{\mathcal{U}(D)}\int_{\mathcal{U}(D)} \eta(u)^*\Phi(ubv^*)\eta(v)\,du\,dv.$$

The bimodule identity gives $\Omega_0(b) = P_k\Omega_0(b)P_l$. Compressing the last integral to this block therefore yields

$$\Omega_0(b) = \frac{1}{2}\theta(b) \qquad (b \in p_k\mathcal{B}p_l,\ k \neq l).$$

Since ηE_D is completely positive on $M_n(\mathbb{C})$, so is

$$\Omega(x) = \Omega_0(x) + (I-b_0)^{1/2}\eta(E_Dx)(I-b_0)^{1/2}.$$

For $d \in D$, commutation of b_0 with $\eta(D)$ gives

$$\Omega(d) = \eta(d)b_0 + \eta(d)(I-b_0) = \eta(d).$$

In particular, $\Omega(I) = I$. On every off-diagonal block of $\mathcal{B}$, one has $E_D(b) = 0$ and $\Omega(b) = \theta(b)/2$. Summing the blocks gives

$$\Omega|_{\mathcal{B}} = \frac{1}{2}(\theta + \theta E_D).$$

The left side is completely contractive. Lemma 8.6, applied to $E_D|_{\mathcal{B}}$, now gives $\|\theta\|_{\mathrm{cb}} \leq 2$.

For sharpness, take $\mathcal{B} = T_2$, $L = \mathrm{diag}(2,1)$, $\theta(b) = LbL^{-1}$, and $\alpha(b) = \overline{\mathrm{diag}(b)}$. The map α is an antilinear complete contraction preserving the identity, and the symmetrization is the inclusion of T_2 in $M_2(\mathbb{C})$. Since $\|\theta(E_{12})\| = 2$, the constant is best possible. □

Corollary 8.8. *Let $\mathcal{B} \subset B(\mathcal{H}_0)$ be a finite-dimensional operator algebra with unity. Suppose that it contains mutually orthogonal projections $p_1, \ldots, p_r$ with sum I, that each $p_k\mathcal{B}p_k$ is self-adjoint, and that, for $k \neq l$, at least one of $p_k\mathcal{B}p_l$ and $p_l\mathcal{B}p_k$ is zero. Let $\theta : \mathcal{B} \to B(\mathcal{H})$ be a homomorphism preserving the identity and let $\alpha : \mathcal{B} \to \mathcal{B}$ be an antilinear contraction preserving the identity. If*

$$\Phi(b) = \frac{1}{2}\big(\theta(b) + \theta(\alpha(b))^*\big)$$

is completely contractive, then $\|\theta\|_{\mathrm{cb}} \leq 2$. The constant is sharp.

Proof. Put $\mathcal{C} = C^*(\mathcal{B})$ and $D = \bigoplus_k p_k\mathcal{B}p_k$. The absence of reverse off-diagonal corners gives $D = \mathcal{B}\cap\mathcal{B}^*$. The pinching map $E(b) = \sum_k p_kbp_k$ is a complete contraction from $\mathcal{B}$ onto D preserving the identity. A completely positive extension $\widetilde{\Phi} : \mathcal{C} \to B(\mathcal{H})$ preserving the identity exists by Arveson's theorem. The contraction $d \mapsto \alpha(d^*)$ preserves adjoints, so $\alpha(D) \subset D$. Lemma 8.7 gives a $*$-homomorphism $\eta = \theta|_D$.

Write $\widetilde{\Phi}(x) = V^*\sigma(x)V$ in Stinespring form and put

$$Z = \int_{\mathcal{U}(D)} \sigma(u)^*V\eta(u)\,du, \qquad b_0 = Z^*Z, \qquad \Omega_0(x) = Z^*\sigma(x)Z.$$

Then $\|Z\| \leq 1$, $\sigma(d)Z = Z\eta(d)$, and $0 \leq b_0 \leq I$ belongs to $\eta(D)'$. Consequently,

$$\Omega_0(d_1xd_2) = \eta(d_1)\Omega_0(x)\eta(d_2), \qquad \Omega_0(d) = \eta(d)b_0.$$

For an allowed off-diagonal corner $b \in p_k\mathcal{B}p_l$, the opposite corner is zero. With $P_k = \eta(p_k)$, this implies

$$P_k\eta(u)^*\Phi(ubv^*)\eta(v)P_l = \tfrac{1}{2}\theta(b), \qquad u, v \in \mathcal{U}(D),$$

because $p_l\alpha(ubv^*)p_k = 0$. Expanding the two integrals in $Z^*\sigma(b)Z$ and using its (P_k, P_l) corner therefore gives $\Omega_0(b) = \theta(b)/2$.

The map $\eta E : \mathcal{B} \to B(\mathcal{H})$ is a complete contraction preserving the identity. Extend it to a completely positive map $J : \mathcal{C} \to B(\mathcal{H})$ preserving the identity, and set

$$\Omega(x) = \Omega_0(x) + (I-b_0)^{1/2}J(x)(I-b_0)^{1/2}.$$

This map is completely positive and $\Omega(I) = I$. On D it equals η, while on every off-diagonal corner of $\mathcal{B}$ it equals $\theta/2$. Thus $\Omega|_{\mathcal{B}} = (\theta + \theta E)/2$ is completely contractive. Lemma 8.6 gives the conclusion. The example $\mathcal{B} = T_2$ in Theorem 1.4 gives sharpness. □

For example, if $W \subset B(\mathcal{H}_2, \mathcal{H}_1)$ is a finite-dimensional operator space, this applies to

$$\left\{ \begin{pmatrix} aI_{\mathcal{H}_1} & w \\ 0 & cI_{\mathcal{H}_2} \end{pmatrix} : a, c \in \mathbb{C},\ w \in W \right\}.$$

9. Similarity transfer and scaled q-numerical ranges

Let $D \subset \mathbb{C}$ be a bounded convex domain and let $A \in M_d(\mathbb{C})$ satisfy $\sigma(A) \subset D$. Define the optimal complete spectral constant of D for A by

$$M_{D,\mathrm{cb}}(A) = \sup_{m \geq 1} \sup \left\{ \|F(A)\| : F \in M_m(\mathcal{A}(\overline{D})),\ \|F\|_{\overline{D}} \leq 1 \right\}. \tag{9.1}$$

The same value is obtained with rectangular matrix-valued functions.

The following proposition applies Theorem 5.5 to the functional calculus on a fixed domain.

Proposition 9.1. *Suppose that $M_{D,\mathrm{cb}}(A) > 1$. For every $\gamma \geq 1$, there is a positive invertible matrix S_γ such that*

$$\mathrm{cond}(S_\gamma) = \gamma, \qquad M_{D,\mathrm{cb}}(S_\gamma^{-1} A S_\gamma) = \gamma M_{D,\mathrm{cb}}(A). \tag{9.2}$$

Proof. Let

$$\mathfrak{A}_{A,D} = \mathcal{A}(\overline{D}) / \ker\{f \mapsto f(A)\},$$

with its quotient matrix norms, and let $\theta_{A,D} : \mathfrak{A}_{A,D} \to M_d(\mathbb{C})$ be the induced homomorphism. The proof of Proposition 2.1 applies without change and gives

$$\|\theta_{A,D}\|_{\mathrm{cb}} = M_{D,\mathrm{cb}}(A).$$

The quotient is finite-dimensional because its image is $\mathrm{alg}(A)$. Apply Theorem 5.5. Since

$$f(S^{-1}AS) = S^{-1} f(A) S,$$

the amplified homomorphism is the functional calculus of $S^{-1}AS$ on the same domain D. This proves (9.2). □

The following theorem is the completely bounded form of the transfer principle of Aouichaoui and O'Loughlin [1, Theorem 3.3].

Theorem 9.2. *For each $B \in M_d(\mathbb{C})$, let $K(B) \subset \mathbb{C}$ be a compact complete C-spectral set for B, where $C \geq 1$ is independent of B. Let $A \in M_d(\mathbb{C})$ and let D be a bounded convex domain containing $\sigma(A)$. Suppose that $\gamma > 1$ and*

$$K(S^{-1}AS) \Subset D \qquad \text{whenever } S \text{ is invertible and } \mathrm{cond}(S) < \gamma. \tag{9.3}$$

Then

$$M_{D,\mathrm{cb}}(A) \leq \max\left\{1, \frac{C}{\gamma}\right\}. \tag{9.4}$$

Proof. The conclusion is immediate if $M_{D,\mathrm{cb}}(A) = 1$. Otherwise, fix $1 < \lambda < \gamma$. Proposition 9.1 gives a positive invertible S_λ such that

$$\mathrm{cond}(S_\lambda) = \lambda, \qquad M_{D,\mathrm{cb}}(S_\lambda^{-1} A S_\lambda) = \lambda M_{D,\mathrm{cb}}(A).$$

By (9.3), $K(S_\lambda^{-1} A S_\lambda) \Subset D$. Every matrix-valued function in $\mathcal{A}(\overline{D})$ is holomorphic on a neighborhood of this compact set. By Runge's theorem, applied entrywise, the complete C-spectral estimate extends from rational functions to such functions. Hence

$$M_{D,\mathrm{cb}}(S_\lambda^{-1} A S_\lambda) \leq C.$$

Thus $\lambda M_{D,\mathrm{cb}}(A) \leq C$. Letting $\lambda \to \gamma^-$ proves (9.4). □

In the following discussion of q-numerical ranges, assume that $d \geq 2$. For $0 < |q| \leq 1$, define

$$W_q(A) = \{\langle Ax, y\rangle : \|x\| = \|y\| = 1,\ \langle x, y\rangle = q\}, \qquad \Omega_q(A) = q^{-1} W_q(A).$$

The set $\Omega_q(A)$ depends only on $|q|$. It is compact and convex, and it contains the spectrum of A; see [1, 29, 38]. The scaled range and its spectral constants were studied in [33]. For $0 < r \leq 1$, put

$$\chi(r) = \frac{1 + \sqrt{1 - r^2}}{r}. \tag{9.5}$$

Aouichaoui and O'Loughlin proved the exact similarity formula

$$\Omega_r(A) = \bigcup_{\mathrm{cond}(S) \leq \chi(r)} W(S^{-1}AS). \tag{9.6}$$

If A is not scalar, $0 < r < 1$, and $\operatorname{cond}(S) < \chi(r)$, then

$$W(S^{-1}AS) \Subset \Omega_r(A)^\circ \tag{9.7}$$

[1, Theorem 2.2 and Corollary 2.3].

For $d \geq 2$, let $C_{q,d}^{\mathrm{cb}}$ be the least $C \geq 1$ such that

$$\|F(A)\| \leq C \max_{z \in \Omega_q(A)} \|F(z)\| \tag{9.8}$$

for every $A \in M_d(\mathbb{C})$ and every square matrix-valued polynomial F. Since $\Omega_q(A)$ is convex, the same constant is obtained for matrix-valued functions holomorphic on a neighborhood of $\Omega_q(A)$.

Theorem 9.3. *For every $d \geq 2$ and $0 < |q| \leq 1$,*

$$\max\left\{1, \frac{2}{\chi(|q|)}\right\} \leq C_{q,d}^{\mathrm{cb}} \leq \max\left\{1, \frac{Q_{\mathrm{cb}}(d)}{\chi(|q|)}\right\}. \tag{9.9}$$

In particular,

$$C_{q,d}^{\mathrm{cb}} \leq \max\left\{1, \frac{(1+\sqrt{2})|q|}{1+\sqrt{1-|q|^2}}\right\}. \tag{9.10}$$

Consequently, $\Omega_q(A)$ is a complete spectral set for every matrix A of every order at least two when $|q| \leq 1/\sqrt{2}$. For $d = 2, 3$,

$$C_{q,d}^{\mathrm{cb}} = \max\left\{1, \frac{2|q|}{1+\sqrt{1-|q|^2}}\right\}. \tag{9.11}$$

Thus, in these two dimensions, $\Omega_q(A)$ is a complete spectral set for every A if and only if $|q| \leq 4/5$.

Proof. If $|q| = 1$, then $\Omega_q(A) = W(A)$ up to the harmless reduction to $|q|$, and the upper bound in (9.9) is the definition of $Q_{\mathrm{cb}}(d)$. Suppose that $0 < |q| < 1$. Scalar matrices have complete spectral constant one on $\Omega_q(A)$. If A is not scalar, apply Theorem 9.2 with

$$K(B) = W(B), \qquad C = Q_{\mathrm{cb}}(d), \qquad D = \Omega_{|q|}(A)^\circ, \qquad \gamma = \chi(|q|).$$

The hypothesis follows from (9.7). This proves the upper bound.

For the lower bound, let $N_d = 2e_1e_2^* \in M_d(\mathbb{C})$. The rank-one q-numerical-radius formula [21, Theorem 2.1] gives

$$\max_{z \in \Omega_q(N_d)} |z| = \chi(|q|).$$

Taking the scalar polynomial $p(z) = z$ gives the lower bound $2/\chi(|q|)$, while constant functions give the lower bound one. This proves (9.9).

The complete Crouzeix–Palencia inequality [18, 19] gives $Q_{\mathrm{cb}}(d) \leq 1 + \sqrt{2}$, which proves (9.10). Since $\chi(1/\sqrt{2}) = 1 + \sqrt{2}$, the dimension-free complete spectral-set assertion follows. Theorem 1.1 gives $Q_{\mathrm{cb}}(2) = Q_{\mathrm{cb}}(3) = 2$, so the two bounds coincide for $d = 2, 3$. Finally, $\chi(r) \geq 2$ exactly when $r \leq 4/5$. □

Remark 9.4. Theorems 5.5 and 9.2 hold in every finite dimension. They give complete estimates on domains larger than the numerical range, but they do not control the rectangular levels left by Theorem 5.3. In particular, at $q = 1$ Theorem 9.3 returns $Q_{\mathrm{cb}}(d)$ itself. For $d = 4$, the unresolved level is the 2×2 matrix level.

10. Equality and quantitative estimates

At $\tau = 1/2$, both square-function estimates are needed to identify the reducing subspace associated with the singular value 2. The results in this section are first stated for bounded operators. For an operator T satisfying these estimates, put

$$E_T = \ker(4I - T^*T), \qquad F_T = \frac{1}{2}TE_T.$$

Theorem 10.1. *Let $T \in \mathcal{B}(H)$ satisfy*

$$\sum_{k=0}^{\infty} 4^{-k} T^{*k} T^k \preceq 2I, \qquad \sum_{k=0}^{\infty} 4^{-k} T^k T^{*k} \preceq 2I.$$

Then $E_T \perp F_T$, the subspace $E_T \oplus F_T$ reduces T, and, after identifying E_T with F_T by $x \mapsto Tx/2$,

$$T \simeq_u \begin{pmatrix} 0 & 0 \\ 2I_{E_T} & 0 \end{pmatrix} \oplus R.$$

If $\|T\| = 2$ is attained, then $E_T \neq \{0\}$. If H is finite-dimensional and $\|T\| = 2$, then $\|R\| < 2$, and T is not diagonalizable.

Proof. Put $E = E_T$. The map $U = T|_E/2$ is an isometry from E onto $F = F_T$. Let $x \in E$. The first square-function estimate gives

$$\begin{aligned} 2\|x\|^2 &\geq \sum_{k=0}^{\infty} 4^{-k} \left\|T^k x\right\|^2 \\ &= 2\|x\|^2 + \sum_{k=2}^{\infty} 4^{-k} \left\|T^k x\right\|^2. \end{aligned}$$

Thus $T^2 x = 0$. If $y = Ux$, then

$$Ty = 0, \qquad T^* y = 2x.$$

The second square-function estimate applied to y gives $(T^*)^2 y = 0$, and therefore $T^* x = 0$.

If $x, z \in E$, then

$$\langle x, Uz \rangle = \frac{1}{2} \langle T^* x, z \rangle = 0.$$

Hence $E \perp F$, and

$$TE = 2F, \qquad TF = 0, \qquad T^* F = 2E, \qquad T^* E = 0.$$

Thus $E \oplus F$ reduces T and gives the stated block.

If $\|T\| = 2$ is attained at a unit vector x, then $T^*Tx = 4x$, so $x \in E$. Suppose now that H is finite-dimensional and $\|T\| = 2$. Then $E \neq \{0\}$. If the restriction R to $(E \oplus F)^\perp$ had norm 2, a unit vector at which its norm is attained would belong to $E \cap (E \oplus F)^\perp$, a contradiction. Hence $\|R\| < 2$. The nonzero nilpotent summand shows that T is not diagonalizable. □

Remark 10.2. As a consequence, we obtain another proof that $\psi(A) < 2$ if $A = V\Lambda V^{-1}$ is diagonalizable. Indeed, $\psi(A)$ is attained, say at a scalar-valued function f with $\|f\|_{W(A)} = 1$. If $\psi(A) = 2$, then $T = f(A)$ satisfies $\|T\| = 2$, and Theorem 1.5 with $\tau = 1/2$ allows us to apply the preceding theorem to conclude that T is not diagonalizable. On the other hand, $T = f(A) = Vf(\Lambda)V^{-1}$, so T is diagonalizable, which is a contradiction.

When the norm is not attained, E_T may be trivial. The next theorem replaces it by two-dimensional subspaces on which T approaches the nilpotent equality block.

Theorem 10.3. *Let $T \in \mathcal{B}(H)$ satisfy the two square-function estimates in Theorem 10.1, and assume that $\|T\| = 2$. Then there are orthogonal decompositions*

$$H = \mathcal{M}_j \oplus \mathcal{M}_j^\perp, \qquad \dim \mathcal{M}_j = 2,$$

and operators $D_j \in \mathcal{B}(\mathcal{M}_j^\perp)$ such that, in a suitable orthonormal basis of $\mathcal{M}_j$,

$$\left\| T - \left(\begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix} \oplus D_j \right) \right\| \longrightarrow 0.$$

Proof. Since $4 = \|T^*T\|$ belongs to the spectrum of the positive operator T^*T, there are unit vectors x_j such that

$$\|(T^*T - 4I)x_j\| \longrightarrow 0.$$

After discarding finitely many indices, put

$$s_j = \|Tx_j\| > 0, \qquad y_j = \frac{Tx_j}{s_j}.$$

Then $s_j \to 2$ and

$$\|T^* y_j - 2x_j\| \longrightarrow 0.$$

The first square-function estimate gives $\left\|T^2 x_j\right\| \to 0$, and hence $\|Ty_j\| \to 0$. Since $\|T^* y_j\| \to 2$, the second square-function estimate gives $\left\|(T^*)^2 y_j\right\| \to 0$. Indeed,

$$2T^* x_j = (T^*)^2 y_j - T^*(T^* y_j - 2x_j) \longrightarrow 0,$$

and

$$|\langle y_j, x_j \rangle| = \frac{|\langle Tx_j, x_j \rangle|}{s_j} = \frac{|\langle x_j, T^* x_j \rangle|}{s_j} \longrightarrow 0.$$

Write

$$\beta_j = \langle y_j, x_j \rangle, \qquad \rho_j = (1 - |\beta_j|^2)^{1/2}, \qquad e_j = \frac{y_j - \beta_j x_j}{\rho_j},$$

after discarding finitely many further indices. Put $\mathcal{M}_j = \operatorname{span}\{x_j, e_j\}$. Then $\{x_j, e_j\}$ is orthonormal and

$$Tx_j - 2e_j \longrightarrow 0, \qquad Te_j \longrightarrow 0, \qquad T^*x_j \longrightarrow 0, \qquad T^*e_j - 2x_j \longrightarrow 0.$$

Let P_j be the orthogonal projection onto $\mathcal{M}_j$, let $N_j x_j = 2e_j$ and $N_j e_j = 0$, and set

$$D_j = (I - P_j)T|_{\mathcal{M}_j^\perp}.$$

The four preceding limits give

$$\|P_j T P_j - N_j\| \longrightarrow 0, \qquad \|(I - P_j)TP_j\| \longrightarrow 0, \qquad \|P_j T(I - P_j)\| \longrightarrow 0.$$

These are the nonzero blocks of $T - (N_j \oplus D_j)$, and the result follows. □

We next solve the optimization problem used in the quantitative estimates. For $0 < s \le 2$, let τ_s be the unique number in $[1/2, 3/4)$ satisfying

$$2s^2\tau_s(1 - \tau_s)^2 + 4\tau_s - 3 = 0, \tag{10.1}$$

and put

$$\eta(s) = \left(\frac{2\tau_s - 1}{2\tau_s^3(1 - \tau_s)^2}\right)^{1/2}.$$

Thus $\eta(2) = 0$. The functions L_τ and L were defined in (4.5).

Lemma 10.4. *For $0 < s \le 2$,*

$$\eta(s) = \min_{0<\tau<1} \frac{1}{\tau^2}\left(\frac{\tau}{1 - \tau} - s^2\tau^2\right)^{1/2}. \tag{10.2}$$

Moreover,

$$\eta(s) \le 2\sqrt{4 - s^2}, \tag{10.3}$$

with strict inequality when $s < 2$. For $0 < s < 2$,

$$L(s) = \max_{0<\tau<1} L_\tau(s) > L_{1/2}(s) = \frac{s^4}{4(4 - s^2)}. \tag{10.4}$$

The functions $s \mapsto \tau_s$ and $s \mapsto \eta(s)$ are continuous on $(0, 2]$, and L is continuous on $(0, 2)$.

Proof. Put

$$q_s(\tau) = 2s^2\tau(1 - \tau)^2 + 4\tau - 3.$$

Its derivative satisfies

$$q_s'(\tau) = 2s^2(1 - \tau)(1 - 3\tau) + 4 \ge 4 - \frac{2s^2}{3} > 0.$$

Also, $q_s(0) = -3$, $q_s(1) = 1$, $q_s(1/2) = s^2/4 - 1 \le 0$, and $q_s(3/4) = 3s^2/32 > 0$. Thus (10.1) has the asserted unique solution.

The square of the expression minimized in (10.2) is

$$d_s(\tau) = \frac{1}{\tau^3(1 - \tau)} - \frac{s^2}{\tau^2}.$$

A direct calculation gives

$$d_s'(\tau) = \frac{q_s(\tau)}{\tau^4(1 - \tau)^2}.$$

Since $d_s(\tau)$ tends to infinity at both endpoints, its unique minimum occurs at τ_s. Substitution of (10.1) gives

$$d_s(\tau_s) = \frac{2\tau_s - 1}{2\tau_s^3(1 - \tau_s)^2}.$$

This proves (10.2). Choosing $\tau = 1/2$ gives (10.3). The inequality is strict for $s < 2$ because then $\tau_s > 1/2$.

For $0 < s < 2$, the denominator in $L_\tau(s)$ is positive. Differentiation gives

$$\frac{\partial}{\partial\tau} L_\tau(s) = -\frac{s^4\tau^2 q_s(\tau)}{\{1 - s^2\tau(1 - \tau)\}^2}.$$

Hence the unique maximum occurs at τ_s. Since $\tau_s > 1/2$, (10.4) follows. Continuity follows from the implicit function theorem applied to (10.1). □

Put

$$\vartheta(s) = \min\{s^2, \eta(s)\}, \qquad 0 < s \le 2.$$

Theorem 10.5. *Let $T_1, \dots, T_q \in \mathcal{B}(H)$ satisfy, for every $0 < \tau < 1$,*

$$\sum_{k=0}^{\infty} \tau^{2k} \sum_{\alpha \in \mathcal{W}_k} T_\alpha^* T_\alpha \preceq \frac{1}{1-\tau} I. \tag{10.5}$$

Let

$$s = \left\| (T_1, \dots, T_q)^{\mathsf{T}} \right\| > 0,$$

and suppose that this norm is attained at a unit vector x. Then $s \le 2$ and

$$\left(\sum_{i,j=1}^{q} \|T_i T_j x\|^2 \right)^{1/2} \le \vartheta(s). \tag{10.6}$$

The function ϑ cannot be replaced by a smaller function of s, since the scalar case $q = 1$ is sharp.

Proof. Keeping the terms of degrees zero and one in (10.5), and taking $\tau = 1/2$, gives $s \le 2$. Applying the quadratic form at x and retaining the terms through degree two gives

$$\tau^4 \sum_{i,j=1}^{q} \|T_i T_j x\|^2 \le \frac{\tau}{1-\tau} - \tau^2 s^2.$$

Lemma 10.4 gives the bound by $\eta(s)$. On the other hand, the definition of the column norm gives

$$\sum_{i,j=1}^{q} \|T_i T_j x\|^2 \le s^2 \sum_{j=1}^{q} \|T_j x\|^2 = s^4.$$

Taking the smaller of these two bounds proves (10.6). The sharp scalar examples are given in Theorem 10.7. □

The same optimization gives a lower bound on every eigenvector condition number that realizes a diagonalization in dimension three.

Corollary 10.6. *Let $A \in M_3(\mathbb{C})$ be diagonalizable, write $A = V \Lambda V^{-1}$, and put*

$$C = \psi_{\mathrm{cb}}(A), \qquad \kappa = \mathrm{cond}(V).$$

Then

$$\kappa^4 \ge L(C) > \frac{C^4}{4(4 - C^2)}. \tag{10.7}$$

In particular, if $A_j = V_j \Lambda_j V_j^{-1} \in M_3(\mathbb{C})$ are diagonalizable and $\psi_{\mathrm{cb}}(A_j) \to 2$, then $\mathrm{cond}(V_j) \to \infty$ for every choice of eigenvector matrices V_j.

Proof. If $C = 1$, apply Theorem 4.4 to $I = V I V^{-1}$. We may therefore assume that $C > 1$. Choose a minimal scalar, column, or row extremal as in the proof of Theorem 1.7. Apply (4.8) to that scalar, column, or adjoint row tuple. This gives (10.7). The final assertion follows. □

Theorem 10.7. *Let $T \in \mathcal{B}(H)$ and assume that, for every $0 < \tau < 1$,*

$$\sum_{k=0}^{\infty} \tau^{2k} T^{*k} T^k \preceq \frac{1}{1-\tau} I, \qquad \sum_{k=0}^{\infty} \tau^{2k} T^k T^{*k} \preceq \frac{1}{1-\tau} I.$$

Assume that $s = \|T\| > 0$ is attained, and choose unit singular vectors x, y such that

$$Tx = sy, \qquad T^* y = sx.$$

Then $s \le 2$ and

$$\left\| T^2 x \right\|, \ \left\| (T^*)^2 y \right\| \le \vartheta(s).$$

The estimate for $\left\| T^2 x \right\|$ is sharp for every $0 < s \le 2$, already among three-dimensional nilpotent weighted shifts. Taking adjoints gives sharpness for $\left\| (T^*)^2 y \right\|$. More precisely,

$$\vartheta(s) = \begin{cases} s^2, & 0 < s \le 3/2, \\ \eta(s), & 3/2 \le s \le 2. \end{cases}$$

Proof. Theorem 10.5, applied with $q = 1$, gives $s \le 2$ and $\|T^2 x\| \le \vartheta(s)$. Applying the same theorem to T^* and y gives $\|(T^*)^2 y\| \le \vartheta(s)$.

For $s > 0$ and $a \ge 0$, let

$$W_{s,a} = \begin{pmatrix} 0 & 0 & 0 \\ s & 0 & 0 \\ 0 & a & 0 \end{pmatrix}.$$

Let e_1, e_2, e_3 be the standard basis vectors of $\mathbb{C}^3$. Then $W_{s,a}e_1 = se_2$, $W_{s,a}e_2 = ae_3$, and

$$\sum_{k=0}^{\infty} \tau^{2k} W_{s,a}^{*k} W_{s,a}^{k} = \operatorname{diag}\bigl(1 + s^2\tau^2 + s^2a^2\tau^4, 1 + a^2\tau^2, 1\bigr),$$
$$\sum_{k=0}^{\infty} \tau^{2k} W_{s,a}^{k} W_{s,a}^{*k} = \operatorname{diag}\bigl(1, 1 + s^2\tau^2, 1 + a^2\tau^2 + s^2a^2\tau^4\bigr).$$

Suppose first that $0 < s \le 3/2$ and take $a = s$. Put $b = s^2$ and

$$R_b(\tau) = 1 - b\tau + b\tau^2 - b^2\tau^3 + b^2\tau^4.$$

Since

$$\frac{1}{1-\tau} - (1 + b\tau^2 + b^2\tau^4) = \frac{\tau R_b(\tau)}{1-\tau}$$

and

$$\frac{\partial R_b(\tau)}{\partial b} = -\tau(1-\tau)(1 + 2b\tau^2) \le 0,$$

one has

$$R_b(\tau) \ge R_{9/4}(\tau) = \frac{(3\tau - 2)^2(9\tau^2 + 3\tau + 4)}{16} \ge 0.$$

Thus $W_{s,s}$ satisfies both square-function estimates. Its norm is s and $\|W_{s,s}^2 e_1\| = s^2$.

Now suppose that $3/2 \le s \le 2$. Evaluating (10.2) at $\tau = 2/3$ gives

$$\eta(s)^2 \le \frac{81}{8} - \frac{9}{4}s^2 \le s^4.$$

Set $a = \eta(s)/s$. Then $0 \le a \le s$, and the definition of $\eta(s)$ gives

$$1 + s^2\tau^2 + \eta(s)^2\tau^4 \le \frac{1}{1-\tau} \qquad (0 < \tau < 1).$$

Each remaining diagonal entry of either square function is bounded by $1 + s^2\tau^2 + \eta(s)^2\tau^4$ or by $1 + s^2\tau^2$. Moreover, $1 + s^2\tau^2 \le (1-\tau)^{-1}$ because $s \le 2$. Hence $W_{s,a}$ satisfies both estimates, $\|W_{s,a}\| = s$, and

$$\|W_{s,a}^2 e_1\| = sa = \eta(s).$$

This proves sharpness and the displayed formula for ϑ. □

We next obtain an operator-norm estimate. More precisely, we choose a two-dimensional subspace $\mathcal{M}$ and compare T with $\left(\begin{smallmatrix} 0 & 0 \\ 2 & 0 \end{smallmatrix}\right) \oplus D$ relative to the decomposition $H = \mathcal{M} \oplus \mathcal{M}^{\perp}$.

Theorem 10.8. *Let $T \in \mathcal{B}(H)$ satisfy*

$$\sum_{k=0}^{\infty} 4^{-k} T^{*k} T^k \preceq 2I, \qquad \sum_{k=0}^{\infty} 4^{-k} T^k T^{*k} \preceq 2I.$$

Put

$$s = \|T\|, \qquad \delta = 1 - \frac{s^2}{4}.$$

If $0 \le \delta \le 1/16$, and either $\delta > 0$ or $\|T\|$ is attained, then there are an orthogonal decomposition $H = \mathcal{M} \oplus \mathcal{M}^{\perp}$, with $\dim \mathcal{M} = 2$, and an operator $D \in \mathcal{B}(\mathcal{M}^{\perp})$ such that

$$\left\| T - \left(\begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix} \oplus D \right) \right\| \le 8\sqrt{\delta}.$$

The displayed 2×2 matrix is written in a suitable orthonormal basis of $\mathcal{M}$.

Proof. First suppose that $\|T\|$ is attained. Choose unit singular vectors x, y such that

$$Tx = sy, \qquad T^*y = sx.$$

Since $s \geq \sqrt{15}/2 > \sqrt{2}$, the vectors x and y are linearly independent. Indeed, if y were a constant of modulus one times x, then the first square-function series on x would be a geometric series with ratio $s^2/4 > 1/2$, and its sum would be larger than 2. Put

$$\beta = \langle y, x\rangle, \qquad \rho = (1 - |\beta|^2)^{1/2}, \qquad e = \frac{y - \beta x}{\rho}, \qquad \mathcal{M} = \operatorname{span}\{x, e\}.$$

Then $\{x, e\}$ is an orthonormal basis of $\mathcal{M}$ and $y = \beta x + \rho e$. Let P be the orthogonal projection onto $\mathcal{M}$. Define N on $\mathcal{M}$ by

$$Nx = 2e, \qquad Ne = 0,$$

and put $D = (I - P)T|_{\mathcal{M}^\perp}$.

First,

$$(PTP - N)x = s\beta x + (s\rho - 2)e,$$

while

$$\|(PTP - N)e\| \leq \frac{\|Ty\| + s\,|\beta|}{\rho}.$$

Using the Hilbert–Schmidt norm of $(PTP - N)|_{\mathcal{M}}$ gives

$$\|PTP - N\| \leq \left(s^2\,|\beta|^2 + |s\rho - 2|^2 + \frac{(\|Ty\| + s\,|\beta|)^2}{\rho^2} \right)^{1/2}. \tag{10.8}$$

Moreover,

$$\|(I - P)TP\| \leq \frac{\|Ty\|}{\rho}. \tag{10.9}$$

Indeed, $(I - P)TPx = 0$, and the estimate on e follows from $e = (y - \beta x)/\rho$. If $w = (I - P)T^*x$, then

$$(I - P)T^*Px = w, \qquad (I - P)T^*Pe = -\frac{\beta}{\rho} w,$$

because $T^*y = sx$. Hence

$$\|PT(I - P)\| \leq \frac{\|T^*x\|}{\rho}. \tag{10.10}$$

The norm of $PT(I - P) + (I - P)TP$ is at most the larger of the two bounds in (10.9) and (10.10).

The two square-function estimates give

$$\left\|T^2x\right\|, \ \left\|(T^*)^2y\right\| \leq 4\sqrt{\delta}.$$

Since $T^2x = sTy$, $(T^*)^2y = sT^*x$, and $s^2 = 4(1 - \delta)$, one obtains

$$\|Ty\|, \ \|T^*x\| \leq \frac{11}{5}\sqrt{\delta}, \qquad |\beta| \leq \frac{\sqrt{\delta}}{1 - \delta} \leq \frac{4}{15}.$$

Thus $\rho \geq 15/16$, $s\,|\beta| \leq (11/5)\sqrt{\delta}$, and

$$2 - s\rho \leq 2\delta + 2\,|\beta|^2 \leq \frac{11}{10}\sqrt{\delta}.$$

It follows from (10.8) that

$$\|PTP - N\| \leq \frac{28}{5}\sqrt{\delta},$$

and $PT(I-P)+(I-P)TP$ has norm at most $(12/5)\sqrt{\delta}$. The triangle inequality gives the stated bound.

Now suppose that $0 < \delta \leq 1/16$, without assuming norm attainment. Choose unit vectors x_j with $s_j = \|Tx_j\| \to s$, and put $y_j = Tx_j/s_j$. Since $s^2I - T^*T \succeq 0$, $\left\|(T^*T - s^2I)x_j\right\| \to 0$, whence

$$r_j = T^*y_j - sx_j \longrightarrow 0.$$

Applying the column estimate through degree two to x_j gives $\limsup_j \left\|T^2x_j\right\| \leq 4\sqrt{\delta}$. Applying the row estimate to y_j and using $T^*y_j - sx_j \to 0$ gives the analogous bound for $(T^*)^2y_j$. It follows that

$$\limsup_j \|Ty_j\|, \quad \limsup_j \|T^*x_j\| \leq \frac{4\sqrt{\delta}}{s}, \qquad \limsup_j |\langle y_j, x_j\rangle| \leq \frac{\sqrt{\delta}}{1 - \delta}.$$

Define $\beta_j, \rho_j, e_j, \mathcal{M}_j, P_j, N_j, D_j$ as above. These are well-defined for large j, since $\limsup_j |\beta_j| \le 4/15 < 1$. The compression and lower-block calculations apply with s_j in place of s. For the upper block,

$$(I - P_j)T^* e_j = \frac{-\beta_j (I - P_j)T^* x_j + (I - P_j) r_j}{\rho_j},$$

so its norm is at most $\|T^* x_j\| / \rho_j + o(1)$. Using $4/s \le 11/5$, $\liminf_j \rho_j \ge 15/16$, and the same compression estimate, we obtain

$$\limsup_j \|T - (N_j \oplus D_j)\| \le \frac{\sqrt{631741} + 352}{150} \sqrt{\delta} < 8\sqrt{\delta}.$$

The strict inequality follows from $631741 < 848^2$. Choosing a sufficiently large j proves the assertion. □

Corollary 10.9. *Let $A \in M_n(\mathbb{C})$ and let $f \in \mathcal{A}(W(A))$ satisfy $\|f\|_{W(A)} = 1$. If*

$$\|f(A)\| \ge 2 - \varepsilon, \qquad 0 \le \varepsilon \le 2 - \frac{\sqrt{15}}{2},$$

then there are an orthogonal decomposition $\mathbb{C}^n = \mathcal{M} \oplus \mathcal{M}^\perp$, with $\dim \mathcal{M} = 2$, and a matrix D on $\mathcal{M}^\perp$ such that

$$\left\| f(A) - \left(\begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix} \oplus D \right) \right\| \le 8\sqrt{\varepsilon}.$$

The displayed block is written in a suitable orthonormal basis of $\mathcal{M}$.

Proof. Put $T = f(A)$. The bound on ε gives $\|T\| \ge \sqrt{15}/2$, so Theorem 10.8 applies. Since

$$1 - \frac{\|T\|^2}{4} \le 2 - \|T\| \le \varepsilon,$$

the result follows. □

11. Representing measures for extremal functions

We now apply the square-function estimates to extremal functions on a fixed bounded Jordan domain. Let Ω be such a domain, let $A \in M_n(\mathbb{C})$, and assume

$$W(A) \subset \overline{\Omega}, \qquad \sigma(A) \subset \Omega.$$

We write $\mathcal{A}(\overline{\Omega})$ for the functions that are continuous on $\overline{\Omega}$ and holomorphic in Ω. For $h \in H^\infty(\Omega)$, the matrix $h(A)$ is defined by the holomorphic functional calculus. Put

$$\psi_\Omega(A) = \sup\{\|h(A)\| : h \in H^\infty(\Omega),\ \|h\|_{\infty,\Omega} \le 1\}.$$

We call f a normalized extremal function for (A, Ω) if

$$\|f\|_{\infty,\Omega} = 1, \qquad \|f(A)\| = \psi_\Omega(A).$$

Proposition 11.1. *Under the preceding assumptions, $\psi_\Omega(A) \le 2$. The supremum is attained, and every normalized extremal function has the form*

$$f = B \circ \varphi,$$

where $\varphi : \Omega \to \mathbb{D}$ is a conformal bijection and B is a finite Blaschke product, that is, a unimodular constant times a possibly empty finite product of automorphisms of $\mathbb{D}$, of degree at most $d_A - 1$, where $d_A = \deg m_A$. In particular,

$$|f| = 1 \qquad \text{on } \partial\Omega. \tag{11.1}$$

If $\Omega = W(A)^\circ$, then

$$\psi_\Omega(A) = \psi(A).$$

Proof. Let $\varphi : \Omega \to \mathbb{D}$ be a conformal bijection. Since $\partial\Omega$ is a Jordan curve, φ and φ^{-1} extend continuously to the closures [23, Chapter 2]. For $h \in H^\infty(\Omega)$ and $0 < r < 1$, put

$$h_r(z) = h\big(\varphi^{-1}(r\varphi(z))\big).$$

Then $h_r \in \mathcal{A}(\overline{\Omega})$, $\|h_r\|_{\overline{\Omega}} \le \|h\|_{\infty,\Omega}$, and $h_r(A) \to h(A)$ as $r \to 1^-$. By Mergelyan's theorem, h_r can be approximated uniformly on $\overline{\Omega}$ by polynomials. Since $W(A) \subset \overline{\Omega}$, Theorem 1.5 gives

$$\|h_r(A)\| \le 2\,\|h\|_{\infty,\Omega}\,.$$

Letting $r \to 1^-$ proves $\psi_\Omega(A) \le 2$.

To apply Crouzeix's extremal theorem on a disk, put $T = \varphi(A)$. For $h \in H^\infty(\Omega)$ and $g = h \circ \varphi^{-1}$, the composition rule gives $h(A) = g(T)$, and their supremum norms agree. Thus $\psi_\Omega(A) = \psi_{\mathbb{D}}(T)$, and normalized extremals correspond in both directions. The disk satisfies the hypotheses of the cited theorem. The same transfer applies to every invariant restriction of A.

Crouzeix's finite-dimensional extremal theorem gives attainment [13, Theorem 2.1]. Let f be any normalized extremal and choose a unit vector x with $\|f(A)x\| = \psi_\Omega(A)$. The cyclic subspace $E = \operatorname{span}\{A^j x : j \geq 0\}$ is invariant under A and has dimension at most d_A. Holomorphic functional calculus respects this restriction. Hence $\psi_\Omega(A|_E) \leq \psi_\Omega(A)$, while the vector x gives the reverse inequality for f. Thus f is also a normalized extremal for $A|_E$. Applying the same theorem to $A|_E$ gives the stated Blaschke form with degree at most $\dim E - 1 \leq d_A - 1$.

If $\Omega = W(A)^\circ$, polynomials give $\psi_\Omega(A) \geq \psi(A)$. Repeating the radial approximation above and using the definition of $\psi(A)$ in place of the constant 2 gives the reverse inequality. □

Put

$$K = \psi_\Omega(A),$$

and assume $K > 1$. Choose a normalized extremal function f as in Proposition 11.1 and a unit vector x such that

$$\|f(A)x\| = K.$$

Write

$$F = f(A), \qquad y = \frac{1}{K} Fx.$$

Then

$$Fx = Ky, \qquad F^* y = Kx.$$

Apply Lemma 5.2 at the scalar level to the homomorphism $h \mapsto h(A)$ and to the extremal function f. This gives

$$\langle Fx, x \rangle = 0, \qquad x \perp y. \tag{11.2}$$

This is the usual extremal orthogonality relation. See also [7, Section 4.1]. The proof of [7, Theorem 4.5] also applies to the present Jordan domain, without smoothness of the boundary. We include it because the uniqueness statement is used below.

Lemma 11.2. *There is a unique probability measure μ on $\partial\Omega$ such that*

$$\langle h(A)x, x \rangle = \int_{\partial\Omega} h \, d\mu \qquad (h \in \mathcal{A}(\overline{\Omega})). \tag{11.3}$$

Moreover,

$$\int_{\partial\Omega} f \, d\mu = 0. \tag{11.4}$$

Proof. Define

$$\ell(h) = \langle h(A)x, x \rangle \qquad (h \in \mathcal{A}(\overline{\Omega})).$$

Let $h \in \mathcal{A}(\overline{\Omega})$ and suppose that $\operatorname{Re} h \geq 0$ on Ω. For $t \geq 0$,

$$\left\| e^{-th} f \right\|_{\overline{\Omega}} \leq 1.$$

Extremality therefore gives

$$\left\| e^{-th(A)} Fx \right\| \leq K.$$

The function $t \mapsto \left\| e^{-th(A)} Fx \right\|^2$ has a maximum at the endpoint $t = 0$ of $[0, \infty)$. Since F commutes with $h(A)$ and $F^* Fx = K^2 x$, its right derivative at 0 is nonpositive. This gives

$$\operatorname{Re} \ell(h) \geq 0.$$

Let $\mathcal{R}$ be the real parts on $\partial\Omega$ of functions in $\mathcal{A}(\overline{\Omega})$, and define

$$\mathcal{L}(\operatorname{Re} h) = \operatorname{Re} \ell(h).$$

This is well defined. If $\operatorname{Re} h = 0$ on $\partial\Omega$, then h is a purely imaginary constant. If $\operatorname{Re} h \geq 0$ on $\partial\Omega$, the minimum principle gives $\operatorname{Re} h \geq 0$ in Ω. The preceding argument and $\mathcal{L}(1) = 1$ therefore show that $\mathcal{L}$ is a positive functional of norm one on $\mathcal{R}$. Let $\varphi : \Omega \to \mathbb{D}$ be a conformal bijection, extended continuously to the closures. The real and imaginary parts of φ^k, restricted to $\partial\Omega$, belong to $\mathcal{R}$ and, when $\varphi(\zeta) = e^{it}$, equal $\cos(kt)$ and $\sin(kt)$. Since the real trigonometric polynomials are uniformly dense in $C(\mathbb{T}, \mathbb{R})$, the

space $\mathcal{R}$ is uniformly dense in $C(\partial\Omega,\mathbb{R})$. Hence $\mathcal{L}$ extends uniquely to a positive functional of norm one on that space. The Riesz representation theorem gives a unique probability measure μ such that

$$\operatorname{Re}\ell(h)=\int_{\partial\Omega}\operatorname{Re}h\,d\mu.$$

Applying this identity to $-ih$ also gives the imaginary parts and proves (11.3). Finally, (11.2) gives (11.4). □

The following two identities are the only matrix coefficients needed below.

Proposition 11.3. *For every $h\in\mathcal{A}(\overline{\Omega})$,*

$$\langle h(A)y,y\rangle=\int_{\partial\Omega}h\,d\mu, \tag{11.5}$$

$$K\,\langle h(A)y,x\rangle=\int_{\partial\Omega}hf\,d\mu. \tag{11.6}$$

Proof. Since F commutes with $h(A)$,

$$\begin{aligned}K^2\,\langle h(A)y,y\rangle&=\langle h(A)Fx,Fx\rangle\\&=\langle h(A)x,F^*Fx\rangle\\&=K^2\,\langle h(A)x,x\rangle\,.\end{aligned}$$

This proves (11.5). Using $F^*y=Kx$ and then (11.5),

$$\begin{aligned}K\,\langle h(A)y,x\rangle&=\langle h(A)y,F^*y\rangle\\&=\langle Fh(A)y,y\rangle\\&=\int_{\partial\Omega}hf\,d\mu.\end{aligned}$$

□

Put

$$m=\|f\|_{W(A)}\,,\qquad s=\frac{K}{m}.$$

Since $m\le 1$ and the scalar Crouzeix inequality applied to f on $W(A)$ gives $K\le 2m$, one has $m>0$ and $1<s\le 2$. For a Borel map g and a measure λ, the pushforward measure $g_*\lambda$ is defined by $g_*\lambda(E)=\lambda(g^{-1}(E))$ for every Borel set E. We write $m_{\mathbb{T}}$ for normalized arclength measure on $\mathbb{T}$.

Theorem 11.4. *Set*

$$\varepsilon_{K,m}=m^2\vartheta(s),\qquad s=\frac{K}{m}.$$

Then, for every $h\in\mathcal{A}(\overline{\Omega})$,

$$\left|\int_{\partial\Omega}fh\,d\mu\right|\le\varepsilon_{K,m}\,\|h\|_{\overline{\Omega}}\,. \tag{11.7}$$

Moreover,

$$0\le\varepsilon_{K,m}\le 2m\sqrt{4m^2-K^2}\le 2\sqrt{4-K^2}. \tag{11.8}$$

If $\nu=f_*\mu$, then ν is a probability measure on $\mathbb{T}$ and

$$\left|\int_{\mathbb{T}}\zeta^k\,d\nu(\zeta)\right|\le\varepsilon_{K,m}\qquad(k\in\mathbb{Z}\setminus\{0\}). \tag{11.9}$$

Let (A_j,Ω_j,f_j,x_j) be a sequence satisfying the assumptions above, and define K_j,m_j,s_j,μ_j, and $\nu_j=(f_j)_*\mu_j$ as above. If $s_j=K_j/m_j\to 2$, then ν_j converges weakly to $m_{\mathbb{T}}$.

Proof. The normalized matrix $T=F/m$ satisfies the two square-function estimates for every $0<\tau<1$, has norm s, and attains its norm at x. Theorem 10.7 gives

$$\left\|T^2x\right\|\le\vartheta(s).$$

Since $Fy=F^2x/K$,

$$\|Fy\|\le\frac{m^2\vartheta(s)}{K}. \tag{11.10}$$

By (11.6) and commutativity,

$$\begin{aligned}\int_{\partial\Omega} fh\,d\mu &= K\,\langle h(A)y, x\rangle \\ &= \langle Fh(A)y, y\rangle \\ &= \langle h(A)Fy, y\rangle\,.\end{aligned}$$

The definition of K gives $\|h(A)\| \le K\,\|h\|_{\overline{\Omega}}$. Together with (11.10), this proves (11.7).

Since $\vartheta(s) \le \eta(s)$, Lemma 10.4 gives

$$\varepsilon_{K,m} \le 2m^2\sqrt{4 - \frac{K^2}{m^2}} = 2m\sqrt{4m^2 - K^2}.$$

Since $m \le 1$ and $K \le 2m$, both factors on the right in

$$4 - K^2 - m^2(4m^2 - K^2) = (1 - m^2)\big(4(1 + m^2) - K^2\big)$$

are nonnegative. This proves the second inequality in (11.8).

By (11.1), ν is supported on $\mathbb{T}$. Taking $h = f^{k-1}$ in (11.7) gives (11.9) for $k \ge 1$. The estimates for negative k follow by conjugation. For a sequence in the final assertion, $\vartheta(s_j) \to 0$ and $m_j \le 1$, so $\varepsilon_{K_j,m_j} \to 0$. Thus every nonzero Fourier coefficient of ν_j tends to zero. Weak convergence follows from the uniform density of the trigonometric polynomials in $C(\mathbb{T})$. □

Corollary 11.5. *If*

$$K = 2m, \qquad m = \|f\|_{W(A)}\,,$$

then

$$\int_{\partial\Omega} fh\,d\mu = 0 \qquad (h \in \mathcal{A}(\overline{\Omega})),$$

and

$$f_*\mu = m_{\mathbb{T}}.$$

Proof. The assumption is $s = 2$, so $\vartheta(s) = 0$. The first assertion follows from Theorem 11.4. Taking $h = f^{k-1}$ gives vanishing of all positive Fourier coefficients of $f_*\mu$. The negative coefficients are their complex conjugates, so $f_*\mu = m_{\mathbb{T}}$. □

12. Polynomial equality and equality in dimension three

We first characterise equality for scalar polynomials in every finite dimension. We then combine the scalar rigidity theorem with the complete bound to classify equality in dimension three.

We write $A \simeq B$ for unitary similarity. For a matrix A, put $w(A) = \max_{z\in W(A)} |z|$. For $m \ge 1$, let $C_m \in M_{m+1}(\mathbb{C})$ have nonzero entries only on the first superdiagonal. For $m = 1$, its only nonzero entry is 2. For $m \ge 2$, its first and last superdiagonal entries are $\sqrt{2}$, and all intervening entries are 1. These are the Crabb–Choi–Crouzeix matrices. They satisfy

$$W(C_m) = \overline{\mathbb{D}}, \qquad \|C_m^m\| = 2$$

[26, Section 4].

Proposition 12.1. *Let $A \in M_d(\mathbb{C})$ satisfy $W(A) \subset \overline{\mathbb{D}}$, and let p be a nonzero polynomial. Then*

$$\|p(A)\| = 2\max_{|z|\le 1} |p(z)|$$

if and only if there are $c \ne 0$ and an integer $1 \le m < d$ such that $p(z) = cz^m$ and A is unitarily similar to $C_m \oplus B$, where $w(B) \le 1$. The block B is omitted when $m + 1 = d$.

Proof. Divide p by its supremum norm on $\overline{\mathbb{D}}$. Thus $\|p\|_{\overline{\mathbb{D}}} = 1$ and $\|p(A)\| = 2$. Suppose that $Av = \lambda v$, $\|v\| = 1$, and $|\lambda| = 1$. Since

$$I - \mathrm{Re}(\overline{\lambda}A) \ge 0, \qquad \big\langle (I - \mathrm{Re}(\overline{\lambda}A))v, v\big\rangle = 0,$$

positivity gives $\mathrm{Re}(\overline{\lambda}A)v = v$. It follows that $A^*v = \overline{\lambda}v$, so $\mathbb{C}v$ reduces A. Removing these reducing subspaces successively gives

$$A \simeq A_0 \oplus D, \qquad A_0 \in M_{d_0}(\mathbb{C}), \qquad \sigma(A_0) \subset \mathbb{D},$$

where D is diagonal and $\sigma(D) \subset \mathbb{T}$. Either block may be absent before equality is used. Since $\|p(D)\| \le 1$, equality implies that A_0 is present and $\|p(A_0)\| = 2$.

Apply Proposition 11.1 to $(A_0, \mathbb{D})$. Its bound gives $\psi_{\mathbb{D}}(A_0) \le 2$, and p attains equality. Thus p is a normalized extremal function. By the same proposition, which uses Crouzeix's finite-dimensional extremal theorem [13, Theorem 2.1], p is a finite Blaschke product of degree at most $d_0 - 1$. A polynomial finite Blaschke product is a unimodular monomial. Indeed, any nonzero zero $\alpha \in \mathbb{D}$ in its Blaschke factorisation would produce a pole at $1/\overline{\alpha}$, which cannot cancel against a numerator zero in $\mathbb{D}$. Consequently $p(z) = \eta z^m$, where $|\eta| = 1$. A constant polynomial cannot have norm 2 at A_0, so $1 \le m < d$.

Now $\|A^m\| = 2$. The numerical-radius power inequality gives

$$2 = \|A^m\| \le 2w(A^m) \le 2w(A)^m \le 2,$$

so $w(A) = 1$. The equality theorem for powers, proved from Crabb's theorem in [26, Theorems 12 and 14], gives a unitary decomposition $A \simeq C_m \oplus B$, with $w(B) \le 1$. This proves necessity and allows us to remove the normalization of p.

Conversely, if $A \simeq C_m \oplus B$ and $w(B) \le 1$, then $W(A) = \overline{\mathbb{D}}$. The numerical-radius power inequality gives $\|B^m\| \le 2$, while $\|C_m^m\| = 2$. Hence $\|p(A)\| = 2|c| = 2\,\|p\|_{\overline{\mathbb{D}}}$ for $p(z) = cz^m$. □

Jin's rigidity theorem states that $\psi(A) = 2$ forces $W(A)$ to be a nondegenerate disk in every finite dimension [28, Theorem 1]. Together with the preceding proposition, it gives the following classification. The matrix decomposition for monomials is the theorem of Hnětynková and Tichý cited above. The additional conclusion is that, after an affine change of variable, every polynomial attaining equality must be a monomial.

Theorem 12.2. *Let $A \in M_d(\mathbb{C})$ and let p be a polynomial with $p(A) \neq 0$. Then*

$$\|p(A)\| = 2 \max_{z \in W(A)} |p(z)|$$

if and only if there are $a \in \mathbb{C}$, $r > 0$, $c \neq 0$, and an integer $1 \le m < d$ such that

$$p(z) = c(z-a)^m, \qquad A \simeq aI + r(C_m \oplus B), \qquad w(B) \le 1.$$

The block B is omitted when $m + 1 = d$.

Proof. Equality and the scalar bound give $\psi(A) = 2$. Jin's rigidity theorem [28, Theorem 1] gives $W(A) = a + r\overline{\mathbb{D}}$, with $r > 0$. Put $T = (A - aI)/r$ and $q(z) = p(a + rz)$. Proposition 12.1 gives $q(z) = c_0 z^m$ and $T \simeq C_m \oplus B$, with $1 \le m < d$ and $w(B) \le 1$. Therefore $p(z) = c_0 r^{-m}(z-a)^m$, as required. The converse follows from the converse in Proposition 12.1 by the same affine change of variable. □

The next consequence answers the degree question in [24, Section 8.4].

Corollary 12.3. *Let $A \in M_d(\mathbb{C})$ and let p be a polynomial with $\deg p \ge d$ and $p(A) \neq 0$. Then*

$$\|p(A)\| < 2 \max_{z \in W(A)} |p(z)|.$$

Proof. The scalar bound gives the non-strict inequality. Equality would give $\deg p < d$ by Theorem 12.2. □

In dimension three, Theorem 1.6 transfers Jin's rigidity theorem to the complete constant.

Corollary 12.4. *Let $A \in M_d(\mathbb{C})$, where $d \le 3$. If $W(A)$ is not a nondegenerate disk, then $\psi_{\mathrm{cb}}(A) < 2$.*

Proof. Theorem 1.1 gives $\psi_{\mathrm{cb}}(A) \le 2$. Equality would give $\psi(A) = 2$ by Theorem 1.6, and then $W(A)$ would be a nondegenerate disk by [28, Theorem 1]. □

Crouzeix classified the matrices of order three with unit-disk numerical range and scalar constant two [16, Lemma 10.2]. The preceding results remove the disk assumption and give the complete equality classification.

Corollary 12.5. *For $A \in M_3(\mathbb{C})$, the following statements are equivalent.*

(i) $\psi(A) = 2$.
(ii) $\psi_{\mathrm{cb}}(A) = 2$.
(iii) There are $a \in \mathbb{C}$ and $b \in \mathbb{C} \setminus \{0\}$ such that A is unitarily similar to $aI + bB_\xi$ or $aI + bM_s$, where

$$B_\xi = \begin{pmatrix} 0 & 0 & 2 \\ 0 & \xi & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad |\xi| \le 1,$$

and

$$M_s = \begin{pmatrix} 0 & \sqrt{2(1-s^2)} & 2s \\ 0 & -s & \sqrt{2(1-s^2)} \\ 0 & 0 & 0 \end{pmatrix}, \qquad 0 \le s \le 1.$$

Proof. Statements (i) and (ii) are equivalent by Theorem 1.6. If (i) holds, Jin's rigidity theorem gives $W(A) = a + r\overline{\mathbb{D}}$, with $r > 0$. Apply [16, Lemma 10.2] to $(A - aI)/r$. Its two families are precisely B_ξ and $e^{i\vartheta}M_s$, with $s = \sin\phi$, $0 \le \phi \le \pi/2$. Absorbing the unimodular factor into b gives (iii). Conversely, the same lemma gives $\psi(B_\xi) = \psi(M_s) = 2$. Unitary similarity and a nonconstant affine change preserve ψ, so (iii) implies (i). □

We finish with algebraic consequences of equality that apply in arbitrary finite dimension. Write $r_A(\lambda)$ for the size of the largest Jordan block of A at λ, and put $D(z_0, r) = \{z \in \mathbb{C} : |z - z_0| < r\}$.

Lemma 12.6. *Let A be a matrix and let f be holomorphic on a neighborhood of $\sigma(A)$. Then*

$$\operatorname{alg}(f(A)) = \operatorname{alg}(A)$$

if and only if

(i) f is injective on $\sigma(A)$.
(ii) $f'(\lambda) \neq 0$ whenever $r_A(\lambda) \ge 2$.

Proof. The inclusion $\operatorname{alg}(f(A)) \subset \operatorname{alg}(A)$ is automatic. Suppose first that $A = q(f(A))$ for a polynomial q. If $\lambda, \nu \in \sigma(A)$, then

$$q(f(\lambda)) = \lambda, \qquad q(f(\nu)) = \nu.$$

Thus f is injective on $\sigma(A)$. If $r_A(\lambda) \ge 2$, comparison of first derivatives on a Jordan block at λ gives $q'(f(\lambda))f'(\lambda) = 1$.

Conversely, assume (i) and (ii). For each $\lambda \in \sigma(A)$, the function f has a local holomorphic inverse at $f(\lambda)$ when $r_A(\lambda) \ge 2$. When $r_A(\lambda) = 1$, only the value of an inverse is needed. Since the points $f(\lambda)$ are distinct, Hermite interpolation gives a polynomial q whose derivatives through order $r_A(\lambda) - 1$ agree at $f(\lambda)$ with those of the corresponding local inverse. It follows that $q(f(A)) = A$. □

Theorem 12.7. *Let $A \in M_n(\mathbb{C})$ and let Ω be a bounded Jordan domain. Assume*

$$W(A) \subset \overline{\Omega}, \qquad \sigma(A) \subset \Omega.$$

Let $f \in \mathcal{A}(\overline{\Omega})$ satisfy

$$\|f\|_{\overline{\Omega}} = 1, \qquad \|f(A)\| = 2.$$

If $\operatorname{alg}(f(A)) = \operatorname{alg}(A)$, then there are $z_0 \in \Omega$ and $c \neq 0$ such that

$$\Omega = z_0 + c\mathbb{D}, \qquad f(z_0 + c\zeta) = \zeta \quad (\zeta \in \mathbb{D}).$$

Proof. Put $F = f(A)$. Theorem 1.5 gives

$$2 = \|F\| \le 2\,\|f\|_{W(A)} \le 2.$$

Thus F satisfies the two square-function estimates at $\tau = 1/2$. By Theorem 10.1, there is a two-dimensional reducing subspace V_0 such that, in a suitable orthonormal basis,

$$F|_{V_0} = N, \qquad N = \begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix}.$$

Choose a polynomial q such that $A = q(F)$. Then V_0 also reduces A, and

$$A|_{V_0} = q(N) = z_0 I + cN, \qquad z_0 = q(0), \qquad c = q'(0).$$

The identity $f(A|_{V_0}) = N$ gives $f(z_0) = 0$ and $cf'(z_0) = 1$. Hence $c \neq 0$. The number z_0 is the eigenvalue of $A|_{V_0}$, so $z_0 \in \sigma(A) \subset \Omega$.

Since $W(N) = \overline{\mathbb{D}}$,

$$\{z : |z - z_0| \le |c|\} = W(A|_{V_0}) \subset W(A) \subset \overline{\Omega}.$$

Since the interior of the closure of a Jordan domain is the domain itself, $D(z_0, |c|) \subset \Omega$. The function $g(\zeta) = f(z_0 + c\zeta)$ satisfies $|g| \le 1$ on $\mathbb{D}$. Since $g'(0) = 1$, it is nonconstant, and the maximum principle shows that it maps $\mathbb{D}$ into $\mathbb{D}$. Moreover, $g(0) = 0$ and $g'(0) = 1$. Equality in Schwarz's lemma gives $g(\zeta) = \zeta$. The identity theorem gives $f(z) = (z - z_0)/c$ on Ω. The bound $\|f\|_{\overline{\Omega}} = 1$ then gives $\Omega \subset z_0 + c\mathbb{D}$. The reverse inclusion was proved above, so $\Omega = z_0 + c\mathbb{D}$. □

Corollary 12.8. *Under the assumptions of Theorem 12.7, suppose that Ω is not a disk. Then either*

(i) $f(\lambda) = f(\nu)$ for two distinct eigenvalues $\lambda, \nu \in \sigma(A)$, or

(ii) $f'(\lambda) = 0$ *for some* $\lambda \in \sigma(A)$ *with* $r_A(\lambda) \geq 2$.

Proof. Otherwise Lemma 12.6 would give $\operatorname{alg}(f(A)) = \operatorname{alg}(A)$, and Theorem 12.7 would imply that Ω is a disk. □

Proposition 12.9. *Let* A *be a matrix, and let* $f \in \mathcal{A}(W(A))$ *be holomorphic on a neighborhood of* $\sigma(A)$. *Assume*

$$M = \max_{z \in W(A)} |f(z)| > 0, \qquad \|f(A)\| = 2M.$$

Then there are an eigenvalue λ *of* A, *a Jordan block of size* r *at* λ, *and an integer* $m \geq 1$ *such that* f *has a zero of order* m *at* λ *and*

$$m < r < 3m. \tag{12.1}$$

Proof. After replacing f by f/M, Theorem 10.1 shows that the Jordan form of $f(A)$ contains a block $J_2(0)$. Applying f to the Jordan blocks of A, there must be a block $J_r(\lambda)$ for which the Jordan form of $f(J_r(\lambda))$ contains $J_2(0)$. Hence $f(\lambda) = 0$. If the zero has order m, write

$$f(J_r(\lambda)) = N^m g(N), \qquad N = J_r(\lambda) - \lambda I,$$

where $g(N)$ is invertible and commutes with N. For every $k \geq 1$,

$$(N^m g(N))^k = N^{mk} g(N)^k.$$

The matrix $g(N)^k$ is invertible and commutes with N. Hence $(N^m g(N))^k$ and N^{mk} have kernels of the same dimension for every k, so $f(J_r(\lambda))$ and N^m have the same Jordan block sizes.

Write $r = qm + s$, where $q \geq 0$ and $0 \leq s < m$. The Jordan blocks of N^m have sizes $q+1$, repeated s times, and q, repeated $m - s$ times. Blocks of size zero are omitted. A block of size two occurs exactly when either $q = 1$ and $s > 0$, or $q = 2$ and $s < m$. These conditions are equivalent to (12.1). □

13. A family with a line segment on the boundary

We apply the preceding results to

$$A_{\mu,r} = \begin{pmatrix} i\mu & 0 & ir \\ 0 & -i\mu & ir \\ ir & ir & -1 \end{pmatrix}, \qquad \mu > 0, \qquad r \geq 0. \tag{13.1}$$

For a unit vector $x = (x_1, x_2, x_3)^{\mathsf{T}}$,

$$\operatorname{Re}(x^* A_{\mu,r} x) = -|x_3|^2.$$

Hence the line $\operatorname{Re} z = 0$ supports the numerical range, and the vectors with $x_3 = 0$ give

$$W(A_{\mu,r}) \cap i\mathbb{R} = [-i\mu, i\mu]. \tag{13.2}$$

Thus the displayed line segment is contained in the boundary of the numerical range. If $r = 0$, then

$$A_{\mu,0} = \operatorname{diag}(i\mu, -i\mu, -1)$$

is normal, and therefore

$$\psi(A_{\mu,0}) = \psi_{\mathrm{cb}}(A_{\mu,0}) = 1. \tag{13.3}$$

Corollary 13.1. *For every* $\mu > 0$ *and* $r \geq 0$,

$$\psi_{\mathrm{cb}}(A_{\mu,r}) < 2.$$

Proof. The nontrivial boundary segment in (13.2) excludes a nondegenerate disk. Apply Corollary 12.4. □

The remainder of the section concerns $r > 0$.

Lemma 13.2. *If* $r > 0$, *then* $A_{\mu,r}$ *is unitarily irreducible.*

Proof. Let X commute with $A_{\mu,r}$ and $A_{\mu,r}^*$. It commutes with $\operatorname{Re} A_{\mu,r} = \operatorname{diag}(0, 0, -1)$, so $X = B \oplus [c]$ with $B \in M_2(\mathbb{C})$. It also commutes with

$$\frac{A_{\mu,r} - A_{\mu,r}^*}{2i} = \begin{pmatrix} \mu & 0 & r \\ 0 & -\mu & r \\ r & r & 0 \end{pmatrix}.$$

The upper-left block shows that B is diagonal. The upper-right block gives $B(r, r)^{\mathsf{T}} = c(r, r)^{\mathsf{T}}$. Since $r > 0$, one has $B = cI_2$. Thus the joint commutant of $A_{\mu,r}$ and $A_{\mu,r}^*$ consists of scalar matrices. An orthogonal projection onto a reducing subspace belongs to this joint commutant. □

The family is unitarily equivalent, up to sign, to a real family used by Crouzeix [16, Section 9, p. 436]. For $a > 0$ and $b \geq 0$, put

$$C_{a,b} = \begin{pmatrix} 0 & a & b \\ -a & 0 & b \\ -b & -b & 1 \end{pmatrix}.$$

Proposition 13.3. *The matrix $A_{a,b}$ is unitarily equivalent to $-C_{a,b}$. Hence*

$$\psi(A_{a,b}) = \psi(C_{a,b}), \qquad \psi_{\mathrm{cb}}(A_{a,b}) = \psi_{\mathrm{cb}}(C_{a,b}).$$

Proof. Let

$$u_1 = \frac{e^{i\pi/4}}{\sqrt{2}} \begin{pmatrix} i \\ 1 \\ 0 \end{pmatrix}, \quad u_2 = \frac{e^{3i\pi/4}}{\sqrt{2}} \begin{pmatrix} -i \\ 1 \\ 0 \end{pmatrix}, \quad u_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

and put $U = (u_1, u_2, u_3)$. Direct multiplication gives $U^*(-C_{a,b})U = A_{a,b}$. The two ratios are invariant under unitary equivalence and under multiplication of the matrix by a nonzero scalar. □

Put

$$h = \frac{r}{\mu}, \qquad q = \mu^2, \qquad y = h^2.$$

With respect to the orthonormal basis

$$\frac{e_1 - e_2}{\sqrt{2}}, \qquad \frac{e_1 + e_2}{\sqrt{2}}, \qquad e_3,$$

the matrix $A_{\mu,r}$ is unitarily equivalent to

$$\widetilde{A}_{\mu,h} = \begin{pmatrix} 0 & i\mu & 0 \\ i\mu & 0 & i\sqrt{2}h\mu \\ 0 & i\sqrt{2}h\mu & -1 \end{pmatrix}. \tag{13.4}$$

Its characteristic polynomial is

$$p_{q,y}(\lambda) = \lambda^3 + \lambda^2 + (1 + 2y)q\lambda + q, \tag{13.5}$$

and its discriminant is

$$\begin{aligned} \Delta(q, y) = -4q\{&8q^2y^3 + (12q^2 - q)y^2 \\ &+ (6q^2 - 10q)y + (q + 1)^2\}. \end{aligned} \tag{13.6}$$

If $h > 0$, neither 0 nor -1 is a root of $p_{q,y}$. For every root λ,

$$x(\lambda) = \left(\frac{i\mu}{\lambda},\ 1,\ \frac{i\sqrt{2}h\mu}{1 + \lambda} \right)^{\mathsf{T}} \tag{13.7}$$

is an eigenvector of $\widetilde{A}_{\mu,h}$. Every eigenspace is one-dimensional. Consequently, $A_{\mu,r}$ is diagonalizable exactly when $\Delta(q, y) \neq 0$.

When the roots $\lambda_1, \lambda_2, \lambda_3$ are simple, let $v_j = x(\lambda_j)/\left\|x(\lambda_j)\right\|$ and put $V = (v_1, v_2, v_3)$. Then

$$\widetilde{A}_{\mu,h} = V \operatorname{diag}(\lambda_1, \lambda_2, \lambda_3) V^{-1}.$$

Theorem 1.7 gives the following complete estimate.

Corollary 13.4. *Assume $r > 0$ and $\Delta(q, y) \neq 0$. With V as above and $\kappa = \operatorname{cond}(V)$,*

$$\psi_{\mathrm{cb}}(A_{\mu,r}) \leq \min\{\kappa, \Phi(\kappa)\} < 2. \tag{13.8}$$

Lemma 13.5. *Let $A \in M_n(\mathbb{C})$, let $\lambda \in \sigma(A) \cap \partial W(A)$, and let $u \neq 0$ satisfy $Au = \lambda u$. Then $A^*u = \overline{\lambda}u$. In particular, if A is unitarily irreducible and $n \geq 2$, then $\sigma(A) \subset W(A)^\circ$.*

Proof. After translation and rotation, suppose that $\lambda = 0$ and that the numerical range lies in the closed right half-plane. Then $\operatorname{Re} A \succeq 0$. If $Au = 0$, positivity and $\langle (\operatorname{Re} A)u, u \rangle = 0$ give $(\operatorname{Re} A)u = 0$, and hence $A^*u = 0$. Thus the span of u reduces A. □

For $r > 0$, Lemmas 13.2 and 13.5 give

$$\sigma(A_{\mu,r}) \subset W(A_{\mu,r})^\circ.$$

Proposition 13.6. *Assume $r > 0$ and $\Delta(q, y) = 0$. There is a unique $t \in (0, 1/2)$ such that*

$$q = t^2(1 - 2t), \qquad y = \frac{(1-t)^2}{t(1-2t)}. \tag{13.9}$$

Equivalently,

$$\mu = t\sqrt{1-2t}, \qquad r = (1-t)\sqrt{t}. \tag{13.10}$$

The repeated eigenvalue is $-t$ and the remaining eigenvalue is $2t - 1$. Moreover,

$$q \le \frac{1}{27}, \qquad y \ge 4, \tag{13.11}$$

with equality in either inequality exactly when $t = 1/3$. If $t \neq 1/3$, the Jordan form is $J_2(-t) \oplus [2t - 1]$. If $t = 1/3$, the Jordan form is $J_3(-1/3)$.

Proof. Let λ be a repeated root of $p_{q,y}$. Since the polynomial has real coefficients and degree three, λ is real. The equations $p_{q,y}(\lambda) = p'_{q,y}(\lambda) = 0$ give

$$q = \lambda^2(2\lambda + 1), \qquad y = -\frac{(\lambda+1)^2}{\lambda(2\lambda+1)}.$$

Since $q > 0$ and $y > 0$, one has $-1/2 < \lambda < 0$. Put $t = -\lambda$. Then

$$p_{q,y}(z) = (z + t)^2(z - 2t + 1),$$

and (13.9) follows. The formulas determine t uniquely. Furthermore,

$$\frac{1}{27} - q = \frac{(1-3t)^2(1+6t)}{27}, \qquad y - 4 = \frac{(1-3t)^2}{t(1-2t)},$$

which proves (13.11). The eigenvector calculation in (13.7) shows that every eigenspace is one-dimensional. This gives the two Jordan forms. □

At the triple value, the matrix is affinely equivalent to the 3×3 Kac–Murdock–Szegő matrix considered in [17].

Proposition 13.7. *Let*

$$\mu_* = \frac{1}{3\sqrt{3}}, \qquad r_* = \frac{2}{3\sqrt{3}}, \qquad K_3 = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Then

$$A_{\mu_*, r_*} + \frac{1}{3} I \simeq_u -\frac{2}{3} K_3.$$

Consequently,

$$\psi(A_{\mu_*,r_*}) = \psi(K_3), \qquad \psi_{\mathrm{cb}}(A_{\mu_*,r_*}) = \psi_{\mathrm{cb}}(K_3).$$

Proof. Set

$$U = \frac{1}{\sqrt{3}} \begin{pmatrix} -\frac{1}{2} - \frac{\sqrt{3}}{2} i & 1 & -\frac{1}{2} + \frac{\sqrt{3}}{2} i \\ \frac{1}{2} - \frac{\sqrt{3}}{2} i & -1 & \frac{1}{2} + \frac{\sqrt{3}}{2} i \\ 1 & 1 & 1 \end{pmatrix}.$$

The columns of U are orthonormal, and direct multiplication gives

$$U^* \left(A_{\mu_*,r_*} + \frac{1}{3} I \right) U = -\frac{2}{3} K_3.$$

The two ratios are invariant under unitary equivalence and under $A \mapsto \alpha A + \beta I$, where $\alpha \neq 0$. □

Lemma 13.8. *Let*

$$f_1(z) = \frac{\sum_{j=1}^{7} c_j z^j}{1 + \sum_{j=1}^{7} d_j z^j},$$

where

$$10^5(c_1, \dots, c_7) = (73400, 49736, 7268, -521, 13, 61, -251),$$
$$10^5(d_1, \dots, d_7) = (32564, -3291, 1000, -400, 84, -242, 28).$$

Then f_1 is holomorphic on a neighborhood of $\overline{\mathbb{D}}$ and

$$f_1(\overline{\mathbb{D}}) \subset W(K_3). \tag{13.12}$$

Proof. Write

$$d_0(z) = 1 + \sum_{j=1}^{7} d_j z^j.$$

For $|z| \le 1$, the coefficients give

$$|d_0(z)| \ge 1 - \sum_{j=1}^{7} |d_j| > 0.$$

Thus d_0 has no zero in $\overline{\mathbb{D}}$, and f_1 is holomorphic on a neighborhood of $\overline{\mathbb{D}}$.

We verify (13.12) in exact rational arithmetic. Parametrize the upper semicircle by

$$\zeta(u) = \frac{1+iu}{1-iu}, \qquad 0 \le u < \infty,$$

and put $w(u) = f_1(\zeta(u))$. The common denominator of $\operatorname{Re} w(u)$ and $\operatorname{Im} w(u)$ is

$$\begin{aligned} D(u) = {} & 3892636881u^{14} + 53940184519u^{12} + 168968707589u^{10} \\ & + 416744879107u^8 + 436966909139u^6 + 280138656021u^4 \\ & + 111953512087u^2 + 16833246049, \end{aligned}$$

which is positive on $[0,\infty)$. Let $C, S_1, S_2, S_3 \in \mathbb{Z}[u]$ be the primitive numerators with positive leading coefficients, after cancelling common factors in the rational expressions obtained by substituting $w = w(u)$ into

$$\begin{gathered} 27\,|w|^4 - 18\,|w|^2 - 8\operatorname{Re} w - 1, \\ \operatorname{Re} w + \frac{1}{2}, \qquad -\operatorname{Re} w, \qquad \frac{1}{12} - (\operatorname{Im} w)^2, \end{gathered}$$

respectively. These expressions are C/D^2, $S_1/(2D)$, $2S_2/D$, and $S_3/(12D^2)$. Their signs therefore agree with those of the corresponding polynomials. We use Sturm sequences in exact rational arithmetic [6, Chapter 2]. The following table gives the resulting variations. The entry $+\infty$ means that the signs of the leading coefficients are used.

polynomial	degree	interval	left variation	right variation	sign
C	28	$[0, 5/2]$	14	14	$C(0) < 0$
S_1	14	$[12/5, +\infty)$	7	7	$S_1(12/5) > 0$
S_2	14	$[12/5, +\infty)$	6	6	$S_2(12/5) > 0$
S_3	28	$[12/5, +\infty)$	12	12	$S_3(12/5) > 0$

Thus $w(u)$ belongs to

$$\mathcal{C} = \{z : 27\,|z|^4 - 18\,|z|^2 - 8\operatorname{Re} z - 1 \le 0\}$$

for $0 \le u \le 5/2$, and it belongs to

$$\mathcal{R} = \left\{ z : -\frac{1}{2} \le \operatorname{Re} z \le 0,\ |\operatorname{Im} z| \le \frac{\sqrt{3}}{6} \right\}$$

for $u \ge 12/5$. The intervals overlap. Letting $u \to \infty$ and using the continuity of f_1 and the closedness of $\mathcal{R}$, we obtain $f_1(-1) \in \mathcal{R}$. Both $\mathcal{C}$ and $\mathcal{R}$ are contained in $W(K_3)$ [17, Section 2]. The coefficients of f_1 are real, and $W(K_3)$ is symmetric with respect to the real axis. Hence $f_1(\mathbb{T}) \subset W(K_3)$. Applying the maximum principle to every supporting half-plane of the convex set $W(K_3)$ proves (13.12). □

The next proof gives an independent certificate of strictness at the triple point, using Lemma 13.8. It also gives a completely bounded counterpart to Crouzeix's numerical results [15].

Theorem 13.9. *At the triple point,*

$$\psi_{\mathrm{cb}}(A_{\mu_*, r_*}) < 2. \tag{13.13}$$

Moreover, there is an $\varepsilon > 0$ such that

$$|\mu - \mu_*| + |r - r_*| < \varepsilon \quad \Longrightarrow \quad \psi_{\mathrm{cb}}(A_{\mu,r}) < 2.$$

Proof. To prove (13.13), by Proposition 13.7, it suffices to show that $\psi_{\mathrm{cb}}(K_3) < 2$. Let $\Omega = W(K_3)^\circ$. By Proposition 13.7 and (13.2), $W(K_3)$ has a line segment on its boundary and hence is not a disk. Trivially, K_3 is unitarily irreducible, its Jordan form is $J_3(0)$, and $\sigma(K_3) = \{0\}$. Moreover, Lemma 13.5 implies that $0 \in \Omega$.

Suppose that $\psi_{\mathrm{cb}}(K_3) = 2$. By Theorem 1.6, $\psi(K_3) = 2$. Proposition 11.1 gives a normalized extremal function f of Blaschke degree at most two. Since the Jordan form of K_3 is $J_3(0)$, Proposition 12.9 forces f to have a double zero at zero. Choose a conformal map $\varphi : \Omega \to \mathbb{D}$ with $\varphi(0) = 0$. It follows that $f = \eta\varphi^2$, where $|\eta| = 1$. As $K_3^3 = 0$,

$$f(K_3) = \eta\varphi'(0)^2 K_3^2.$$

Using $\|f(K_3)\| = 2$ and $\|K_3^2\| = 1$, we obtain

$$\mathrm{crad}_\Omega(0)^2 := \frac{1}{|\varphi'(0)|^2} = \frac{1}{2}.$$

On the other hand, Lemma 13.8 gives a nonconstant holomorphic function f_1 such that

$$f_1(\overline{\mathbb{D}}) \subset W(K_3), \qquad f_1(0) = 0, \qquad f_1'(0) = \frac{367}{500}.$$

By the open mapping theorem, $f_1(\mathbb{D}) \subset \Omega$. Let $\varphi : \Omega \to \mathbb{D}$ be conformal with $\varphi(0) = 0$. Schwarz's lemma applied to $\varphi \circ f_1$ implies that

$$|\varphi'(0)|\,\frac{367}{500} \le 1,$$

and hence

$$\mathrm{crad}_\Omega(0) = \frac{1}{|\varphi'(0)|} \ge \frac{367}{500} > \frac{1}{\sqrt{2}},$$

which is a contradiction.

Every eigenvalue of A_{μ_*,r_*} lies in the interior of its numerical range. The complete Crouzeix ratio is continuous at such matrices [16, Theorem 2.1]. Since $(\mu, r) \mapsto A_{\mu,r}$ is continuous, the strict inequality persists in a neighborhood of (μ_*, r_*). □

For $0 < t < 1/2$, put

$$A_t = A_{\mu(t),r(t)}, \qquad \mu(t) = t\sqrt{1-2t}, \qquad r(t) = (1-t)\sqrt{t}. \tag{13.14}$$

Proposition 13.10. *One has*

$$\lim_{t\to 1/2^-} \psi(A_t) = 2, \qquad \lim_{t\to 1/2^-} \psi_{\mathrm{cb}}(A_t) = 2. \tag{13.15}$$

In particular,

$$\sup_{0<t<1/2} \psi_{\mathrm{cb}}(A_t) = 2.$$

Proof. As $t \to 1/2^-$, the matrices A_t converge to

$$A_* = \begin{pmatrix} 0 & 0 & i/(2\sqrt{2}) \\ 0 & 0 & i/(2\sqrt{2}) \\ i/(2\sqrt{2}) & i/(2\sqrt{2}) & -1 \end{pmatrix}.$$

With respect to the orthonormal basis

$$\frac{1}{\sqrt{2}}(1,-1,0)^{\mathsf{T}}, \qquad \frac{1}{\sqrt{2}}(1,1,0)^{\mathsf{T}}, \qquad (0,0,1)^{\mathsf{T}},$$

the matrix A_* is unitarily equivalent to

$$[0] \oplus B, \qquad B = \begin{pmatrix} 0 & i/2 \\ i/2 & -1 \end{pmatrix}.$$

The matrix

$$N = B + \frac{1}{2}I = \frac{1}{2}\begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}$$

satisfies $N^2 = 0$ and $\|N\| = 1$. Hence $W(B)$ is the closed disk with center $-1/2$ and radius $1/2$. This disk contains zero, so it is also $W(A_*)$. For $p(z) = 2z + 1$,

$$\max_{z\in W(A_*)} |p(z)| = 1, \qquad \|p(A_*)\| = 2.$$

The numerical range satisfies $d_{\mathrm{H}}(W(X), W(Y)) \le \|X - Y\|$. Therefore

$$\|p(A_t)\| \longrightarrow 2, \qquad \max_{z\in W(A_t)} |p(z)| \longrightarrow 1.$$

It follows that $\liminf_{t\to 1/2^-} \psi(A_t) \geq 2$. The scalar Crouzeix inequality gives the reverse bound. Finally,

$$\psi(A_t) \leq \psi_{\mathrm{cb}}(A_t) \leq 2$$

by Theorem 1.1, which proves the second limit. □

Corollary 13.11. *For t sufficiently close to $1/2$, let g_t be a normalized scalar extremal function for $(A_t, W(A_t)^\circ)$, and choose a unit vector x_t such that*

$$\|g_t(A_t)x_t\| = \psi(A_t).$$

Let μ_t be the representing measure associated with g_t and x_t, and put $\nu_t = (g_t)_*\mu_t$. There are a unitary matrix U_t and a scalar d_t such that

$$\left\| U_t^* g_t(A_t) U_t - \left(\begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix} \oplus [d_t] \right) \right\| \leq 8\sqrt{2 - \psi(A_t)}.$$

Moreover,

$$\left| \int_{\mathbb{T}} \zeta^k \, d\nu_t(\zeta) \right| \leq \vartheta(\psi(A_t)) \qquad (k \in \mathbb{Z} \setminus \{0\}).$$

Consequently, the matrix estimate tends to zero and ν_t converges weakly to $m_{\mathbb{T}}$ as $t \to 1/2^-$.

Proof. For $0 < t < 1/2$, the spectrum of A_t lies in the interior of its numerical range. Proposition 11.1 gives g_t in the form $B_t \circ \varphi_t$, where B_t is a finite Blaschke product and $\varphi_t : W(A_t)^\circ \to \mathbb{D}$ is conformal. Hence $\|g_t\|_{W(A_t)} = 1$. Proposition 13.10 and Corollary 10.9 give the matrix estimate. Theorem 11.4, with $K = s = \psi(A_t)$ and $m = 1$, gives the Fourier estimate and weak convergence. □

Corollary 13.1 gives $\psi_{\mathrm{cb}}(A_t) < 2$ for every $0 < t < 1/2$, while Proposition 13.10 shows that these constants tend to two as $t \to 1/2^-$. Thus every point of the defective curve satisfies a strict bound, but no constant smaller than two works for the whole curve.

Acknowledgments

We are deeply grateful to our friend Antti Perälä for his invaluable discussions and insightful comments on this paper.

The first-named author was partially funded by the Swedish Research Council under grant agreement no. 2025-05053 and by the Swedish Energy Agency under project no. P2025-04323. The second-named author was also partially funded by the National Science Centre, Poland, under the Weave-UNISONO programme, grant no. UMO-2025/07/Y/ST1/00146. The third-named author was supported in part by the Engineering and Physical Sciences Research Council grant EP/Y008375/1 and the Research Council of Finland grant 375631.

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